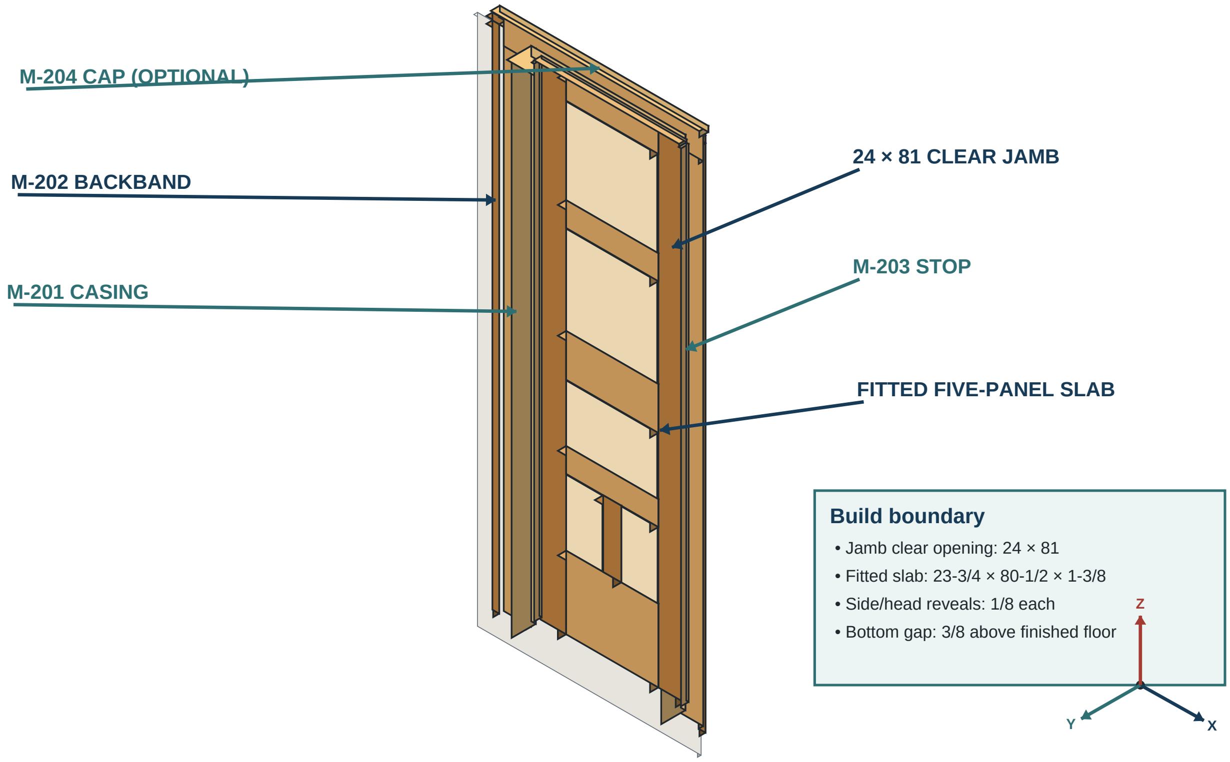
Cabinetmaker’s Door & Millwork Builder’s Guide

Five-panel quarter-sawn white-oak interior door, concealed DF 500 joinery, completed-jamb fitting, and matching custom moulding.

1 Complete door and millwork system

The slab, jamb, stop, casing, backband, and optional cap are separate fitted layers.

SYSTEM



DC-1916-001 | Rev. G | dimension-controlled isometric | written dimensions control

DOOR BUILD	MILLWORK BUILD	CONTROL RULE
13 wood parts, 12 joints, 30 final factory tenons, five floating panels.	Stops, casing, backband, and only evidence-supported cap or plinth details.	Written dimensions and signed field values control every rendering.

SCOPE GATE: The existing completed jamb is not fabricated. Before cutting, identify the opening, handing, swing, trim-face count, physical hardware, builder, date, and Rev. G traveler.

Revision and governing records

The checked-in specification, signed field records, physical samples, and released registers govern every production decision.

Control	Authority	Bench use	Hold condition
Specification	Fixed geometry	Dimensions and tolerances	Any unsynchronized change
Field record	Completed opening	Final fit and trim formulas	Unsigned or conflicting value
Test piece	Actual setup	Cutter, fit, and web proof	Breakout, bind, or mismatch
Rendering	Sequence only	Orientation and relationships	Never used as a cutting template

BLUE	AMBER	RED
Fixed shop control from the project specification.	Field value, sample, or physical hardware required.	Stop; correct and revalidate before continuing.

1. Mark every printed sheet and shop traveler **Rev. G / LS-05 / RL-02**. LS-05 supersedes LS-04; RL-02 supersedes RL-01. Remove or conspicuously obsolete every earlier story stick, rail schedule, panel cut sheet, stock map, and rough-length list so two dimensional authorities cannot coexist.
2. Use this priority when information conflicts: signed field-release value; approved project change; physical hardware or approved sample; Rev. G specification, LS-05 stock map, and RL-02 rough-length schedule; rendering. A rendering never overrides a written dimension.
3. Rev. G retains the confirmed 24 x 81 jamb, 23-7/8 x 80-5/8 prefit assembly, and maximum 23-3/4 x 80-1/2 fitted slab. It changes the two equal stiles to 4-1/4 inches, every rail shoulder to 15-1/4 inches, the bottom bays to 6-1/8 inches, and the panel widths accordingly. LS-05 retains the balanced three-layer frame—5/16 + 7/8 + 5/16 before pressing and 1/4 + 7/8 + 1/4 after symmetric surfacing—while eliminating resawing.
4. Enter each approved departure in a change log with date, reason, affected part IDs, old value, new value, author, and approver. Revise every downstream story stick, schedule, setup card, and inspection record before work resumes.
5. Give each physical sample a traceable ID: groove/tongue TP-G, full-width spacer sample TP-SP-FW, lower-panel spacer sample TP-SP-L, standard Domino TP-J, muntin sections TP-11 and TP-12, profile sample TP-M, and finish sample TP-F.
6. At each shift start, compare the traveler revision, story-stick revision, and setup-card revision. Initial the comparison; do not rely on memory.
7. Quarantine previously cut 79- or 81-inch finished stiles, former D-101H/J/N/K/L panels, 14-3/4- or 15-inch rails, every pre-Rev. G story stick, every pre-LS-05 stock map, and any solid-frame blank made under LS-02. No obsolete dimension, end-spliced stile layer, resawn face, or unrecorded seam may transfer into Rev. G.

Release gate: All active documents and samples show the same revision and opening ID.

STOP: Quarantine work when a dimension differs among controlling records.

REMAKE: A machined feature cannot be “accepted by note” if it violates the current structural or movement requirement.

How to use gates and dispositions

Read in order. Each operation defines setup, action, acceptance, hold, correction, and remake conditions.

1. Work page by page. Complete the numbered steps, enter measured results, and sign the page gate before beginning the next dependent operation.
2. Use four dispositions: **ACCEPT** meets every stated criterion; **HOLD** awaits evidence or cure; **CORRECT** is recoverable without weakening or hiding a defect; **REMAKE** cannot be brought into tolerance without compromising the part.
3. Measure with calibrated tools of suitable resolution. Record actual numbers—never “OK”—for moisture, thickness, width, height, groove size, mortise centers, web, diagonals, twist, and installed reveals.
4. Keep setup samples beside the machine until every production part from that setup is checked. Mark each part immediately after inspection to prevent an unchecked part joining the accepted stack.
5. A STOP instruction freezes only the affected operation, but also freezes every operation that depends on it. Protect and label held parts; identify the root cause before changing a setup.
6. A REMAKE decision is permanent for the named use. Cross-mark the rejected feature, remove the component ID, and retain the piece only as a clearly labeled setup blank if it remains safe.

Release gate: Builder and checker understand the next operation, its datum, measuring method, and disposition limits.

STOP: Do not continue because adhesive, machine time, or an installer is waiting.

REMAKE: Never bury a failed gate inside an assembly.

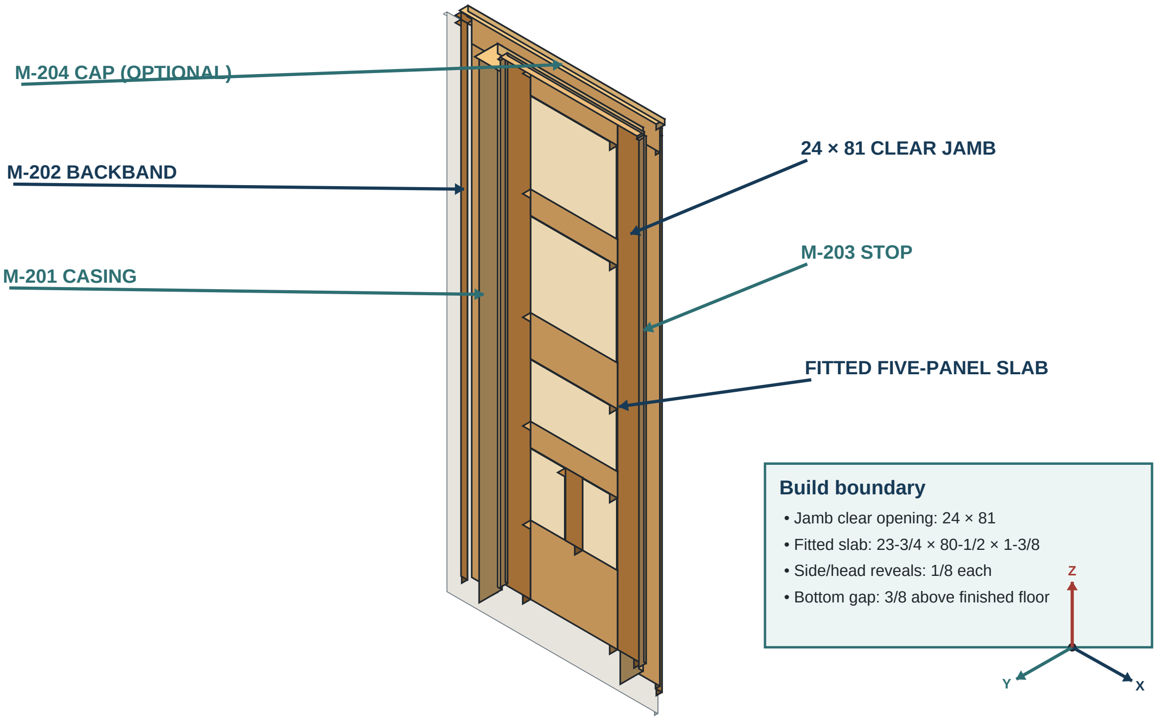
The completed opening

Treat slab, completed jamb, stop, casing, backband, and optional details as separate fitted layers.

1 Complete door and millwork system

The slab, jamb, stop, casing, backband, and optional cap are separate fitted layers.

SYSTEM



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1. Identify the layers from the opening outward: slab; hinges and lock; completed jamb; one M-203 stop set; 3/16-inch casing reveal; M-201 casing; M-202 backband; wall; optional cap, plinth, and baseboard transition.
2. Mark the hinge jamb, strike jamb, head jamb, finished-floor datum, room Face A, optional Face B, door show face, swing direction, and pull/push sides on a field sketch.
3. Distinguish the **operating door clearances** from the **3/16-inch casing reveal**. The casing reveal locates trim on the jamb and is not a prescribed gap around the slab.
4. Confirm whether the existing stop is removed, reused, or replaced. This package provides one three-piece stop set regardless of whether one or two room faces receive casing.
5. Probe the trim footprint for sound fastening substrate and map wall/jamb offsets. Do not assume the visible wall surface is parallel to the jamb or contains framing.
6. Trace the full door swing, including hardware projection, floor changes, nearby fixtures, and the latch path into the strike.

Inspection: The field sketch names every datum and layer and agrees with photographs.

STOP: Unresolved handing, stop ownership, floor datum, or face count blocks fabrication.

REMAKE: A casing or stop cut from the slab dimension instead of its own field layer is not reusable unless it independently fits.

Master workflow and release gates

No operation advances until its incoming measurements, tooling, samples, and records are accepted.

Gate	Incoming evidence	Release test	Retained record
1 Opening	Jamb, floor, handing, trim	Geometry and scope signed	Field worksheet
2 Hardware	Physical hinges, lock, strike	Every dependent dimension known	Hardware sheet
3 Material	Moisture, defects, grain pairs	Board allocation accepted	Cut map
4 Machine	Cutter, guard, fence, stops	Labeled sample passes	Setup sample
5 Dry fit	All parts and relieved tenons	Square, flat, free panels	Dry-fit sheet
6 Glue-up	Timed rehearsal and clamps	Inside open time; geometry held	Glue record
7 Hung slab	Final fit and hardware	Swing, reveal, latch pass	Install record
8 Millwork	Face formulas and dry fit	Joints and bearing pass	Face register
9 Final	Finish and full system	Every acceptance item passes	Build record

1. Release historic evidence, completed-jamb measurements, opening geometry, floor, handing, swing, and trim faces.
2. Obtain and measure hinges, mortise lock, strike, spindle, escutcheons, and screws; clear the lock envelope before J-06.
3. Allocate A-O to the door under LS-05/RL-02, use M as the primary D-101H panel source, keep A intact as panel reserve, and keep N/O full as short-member core reserve pending compatible-white-oak and condition verification. Release continuous stile layers and short-member nesting, thickness-plane whole face mother billets without resawing, press and section same-layup coupons, laminate the balanced frame blanks, then final mill the frame, panels, and separately purchased moulding.
4. Machine and inspect grooves by the prescribed dado/router split; glue panel blanks, size them, form tongues and corner reliefs, prefinish panel edges, and compression-test the specified 3/4-inch-long foam spacers.
5. Prove every Domino joint on production-thickness samples; complete the first full-door dry fit while Stage A remains dry.
6. Perform timed Stage A glue-up and cure inspection; then complete the second full-door dry fit with the cured lower module.
7. Perform timed Stage B glue-up, correct square/twist while wet, cure, and surface only through P100.
8. Fit the slab to the completed jamb; machine hinges, lock, spindle/key holes, and strike; temporarily hang and release the operating door.
9. Dry-fit all stop, moulding, cap, plinth, and return layers. Final-sand only released surfaces to P120, rehang with setup hardware and repeat the operating release, then finish and fully cure. Install brass, then permanent trim in the released plinth/casing/backband-return/cap/stop-last order; complete the final cycle and field release.

STOP: Skipping a dry fit, hardware gate, cure, or operating-door release invalidates every downstream gate.

REMAKE: Do not use finish, filler, clamp pressure, or enlarged hardware gains to conceal failed geometry.

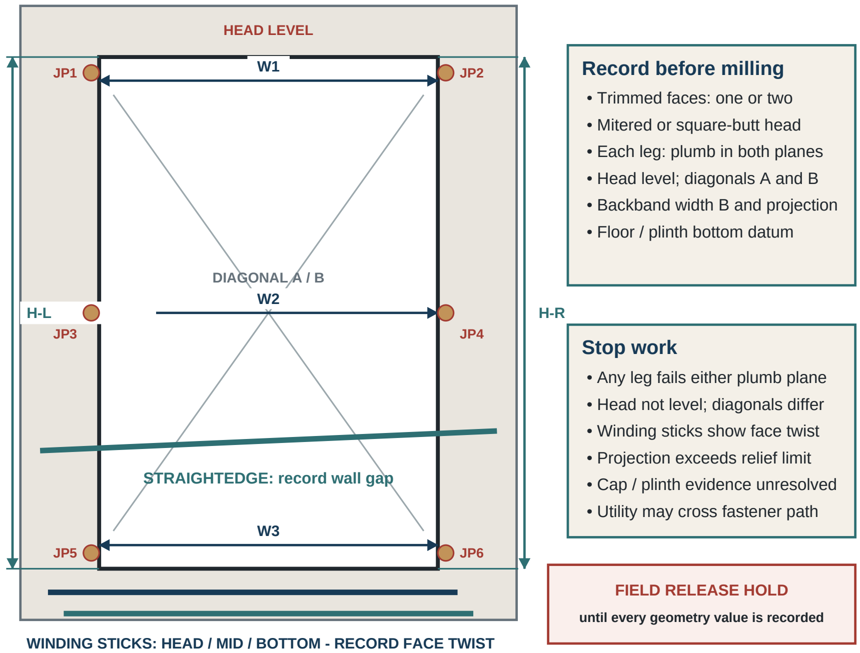
Existing conditions and historic fit

Document both faces, adjacent original millwork, floor, wall, hardware, and finish before choosing trim details.

2 Field measurement gate

Measure the completed jamb and trim footprint before any production blank is sized.

HOLD



DC-1916-001 | Rev. G | quarter-sawn white oak | written dimensions control

1. Photograph the complete opening square-on and at raking angles from both rooms. Include adjacent doors, casing corners, head treatment, stops, plinths, baseboard, floor transitions, hardware, and surviving finish.
2. Measure the adjacent original casing, backband, stop, cap, and plinth profiles at several undamaged locations. Record whether head joints are mitered or square-butt on each face.
3. Make rubbings, contour traces, or full-size profile blocks where edges are not simple rectangles. Label room face, orientation, date, and measurement location.
4. Inspect paint and finish stratigraphy only by an approved safe method. Treat unknown historic coatings as potentially hazardous; do not sand or heat them without the required assessment.
5. Decide separately for Face A and Face B whether to use casing, backband, cap, and plinth. Record baseboard termination and the intended self-return or scribe at every exposed end.
6. Approve one full-size moulding profile sample and one finish sample against the surviving work under normal and raking light.

Release gate: Each visible choice is supported by field evidence or is explicitly recorded as a new compatible intervention.

STOP: Conflicting adjacent examples require a written design decision.

REMAKE: Reject profiles whose overall size, projection, head language, or visible easing differs from the approved sample.

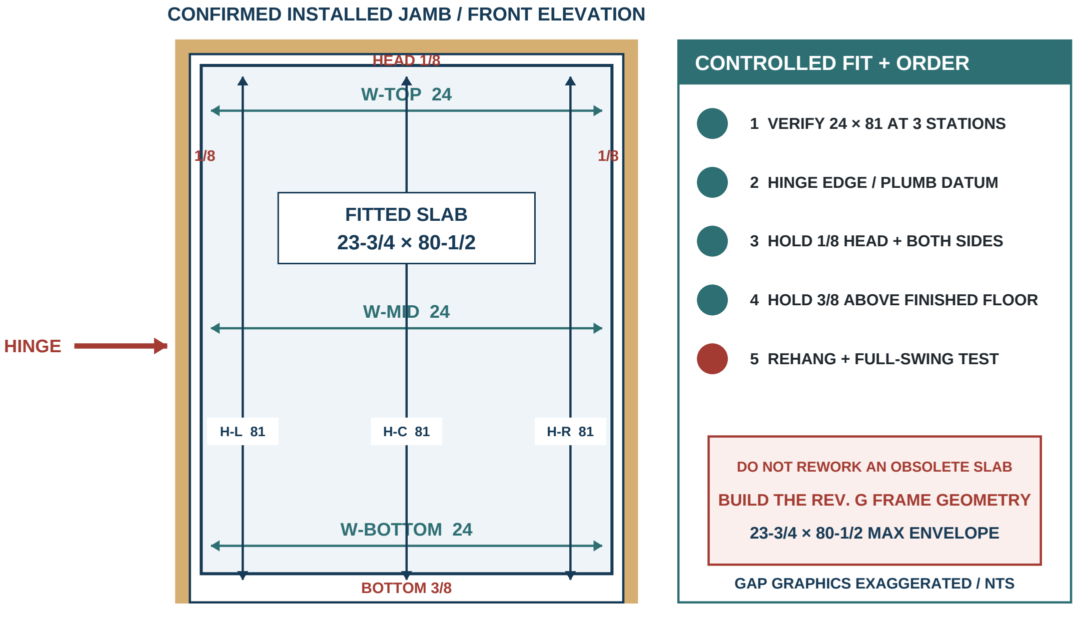
Measure the completed jamb

Record limiting widths and heights, plumb, square, twist, projection, floor datum, handing, and swing.

9A Confirmed jamb and fitted-slab gap map

The 24 x 81 clear jamb is fixed; the 23-3/4 x 80-1/2 slab resolves the written operating gaps.

CONTROLLED FIT



FINISHED FLOOR DATUM | 3/8 BOTTOM GAP | WRITTEN DIMENSIONS CONTROL

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Completed-jamb field	Top	Middle	Bottom / left / right
Width	________	________	________
Height	Left ________	Center ________	Right ________
Door clearance	Head ________	Hinge ________	Lock ________ / floor ________
Geometry	Plumb ________	Square ________	Twist / projection ________

1. Remove only loose obstructions authorized for survey. Use a rigid rule, story pole, level, winding sticks, feeler gauges, and two diagonal tapes; verify the tools against a known standard.
2. Measure clear jamb width at the head, lock-rail elevation, and near the floor. Record each value and its exact elevation, not merely the smallest result.
3. Measure from the selected finished-floor datum to the underside of the head jamb at the hinge side, center, and lock side. Record rugs, thresholds, seasonal floor movement, and required undercut conditions.
4. Check each jamb leg for plumb in and out of the opening and along the opening. Check the head for level, both diagonals for square, and the jamb faces for twist with winding sticks.
5. Measure jamb projection/recession from each wall face at both corners, mid-height, and the floor. Map wall humps across the full 4-1/2-inch casing and 1-1/8-inch backband footprint.
6. Record existing hinge-gain locations, depths, screw condition, stop line, strike location, and all damaged or unsound substrate.
7. Repeat the critical width and height measurements independently. Resolve any difference greater than 1/32 inch before signing.

STOP: Moisture, moving jambs, loose plaster, unknown floor datum, or incompatible diagonals requires correction or redesign.

REMAKE: A field story stick that cannot be reconciled to the written measurements must be remade before transfer.

Jamb, prefit, and fitted-slab states

Keep the confirmed 24 x 81 jamb, 23-7/8 x 80-5/8 prefit assembly, and 23-3/4 x 80-1/2 fitted slab distinct.

CONFIRMED JAMB	PREFIT ASSEMBLY	FINISHED FITTED SLAB
24 x 81 inches clear and reported consistent throughout the installed finished jamb.	23-7/8 x 80-5/8 x 1-3/8 inches after cure and surfacing; retain 1/16 on every perimeter edge.	23-3/4 x 80-1/2 x 1-3/8 inches: 1/8 at both sides and head, 3/8 at bottom; any latch-edge bevel stays inside that maximum rectangle.

1. Keep three states separate on every record: **confirmed finished jamb 24 x 81**, **cured surfaced prefit assembly 23-7/8 x 80-5/8 x 1-3/8**, and **finished fitted slab no larger than 23-3/4 x 80-1/2 x 1-3/8** before any handing-specific latch-edge bevel.
2. Carry 1/16 inch removable stock outside each final perimeter edge of the prefit assembly. Scribe the 23-3/4 x 80-1/2 final rectangle from the designated final-top and hinge-edge datums; do not “center” the final rectangle by eye.
3. Derive final width from the confirmed jamb as **24 - 1/8 hinge side - 1/8 lock side = 23-3/4 inches**. Derive final height as **81 - 1/8 head - 3/8 bottom = 80-1/2 inches**.
4. Preserve the maximum final rectangle while accounting for jamb bow, twist, hinge axis, lock-edge bevel, and the full swing path. Any latch-edge bevel removes material inside that maximum envelope and may not increase it.
5. Limit planned fitting removal to the carried 1/16 inch at each prefit perimeter edge. More removal changes stile width, rail relationships, hardware cover, or the specified gaps and requires a revised geometric review.
6. Record actual hung width at top/middle/bottom, height at both sides, both 1/8-inch side reveals, the 1/8-inch head reveal, and the 3/8-inch bottom gap after fitting.

STOP: If the jamb recheck does not confirm the recorded 24 x 81 opening or the prefit assembly cannot yield the final rectangle within its 1/16-inch edge allowances, stop before rail crosscutting.

REMAKE: A slab beyond the 23-3/4 x 80-1/2 maximum is not made acceptable by changing the specified reveals or bottom gap.

Operation-specific safety and PPE

Match guarding, extraction, support, workholding, and emergency controls to each operation.

Operation	Primary setup	Required control	Stop if
Through rail grooves	1/4-in dado stack	Correct cartridge/insert, guard strategy, push blocks, full support	Feed or support is uncertain
Stopped/segmented grooves	Plunge router + edge guide	Clamped stock and positive start/stop blocks	A drop cut or freehand start is proposed
Domino	DF 500 only	Clamped stock, fixed fence, extraction	Machine rocks or slot breaks out
Trim profile	Saw / plunge router / planes	Wide mother blank and stable support	Default profile needs a dado
Hardware	Guided router / chisels	Physical part and steel setup screws	Template differs from hardware
Nailing	Sequential-actuation nailer	Verified substrate and utility path	Fastener path is uncertain
Finish	Rubio system	Ventilation and safe rag disposal	Sample or disposal plan is absent

1. Read the current manufacturer instructions for the saw, dado stack, jointer, planer, plunge router, DF 500, nailers, dust extractor, adhesive, and finish. Record any control that is stricter than this guide.
2. Wear eye and hearing protection for machining; use effective dust extraction and the respiratory protection required by the exposure assessment. Quarter-sawn white-oak dust is not controlled by opening a door.
3. Inspect guards, riving knife where applicable, brake, switches, cords, hoses, and emergency access before each setup. Disconnect power before cutter changes, measurements near cutters, or jam clearing.
4. Support long or wide parts independently. Keep hands out of projected cutter and fastener paths; use push blocks and workholding that cannot rotate or lift.
5. Machine small moulding from a wide mother blank whenever practical. Never run a narrow, short, warped, or unsupported member against a cutter because its final size appears simple.
6. Use sequential-fire mode for nailers. Verify substrate and concealed utilities before fastening.
7. Lay oil-wet cloths flat outdoors to cure, or submerge them in water inside a sealed metal container for lawful disposal. Follow current Rubio ventilation and waste instructions.

STOP: Missing guard, extraction, stable workholding, or safe feed path cancels the operation.

REMAKE: A part caught, kicked back, burned deeply, or struck by a fastener is quarantined and structurally assessed before any reuse.

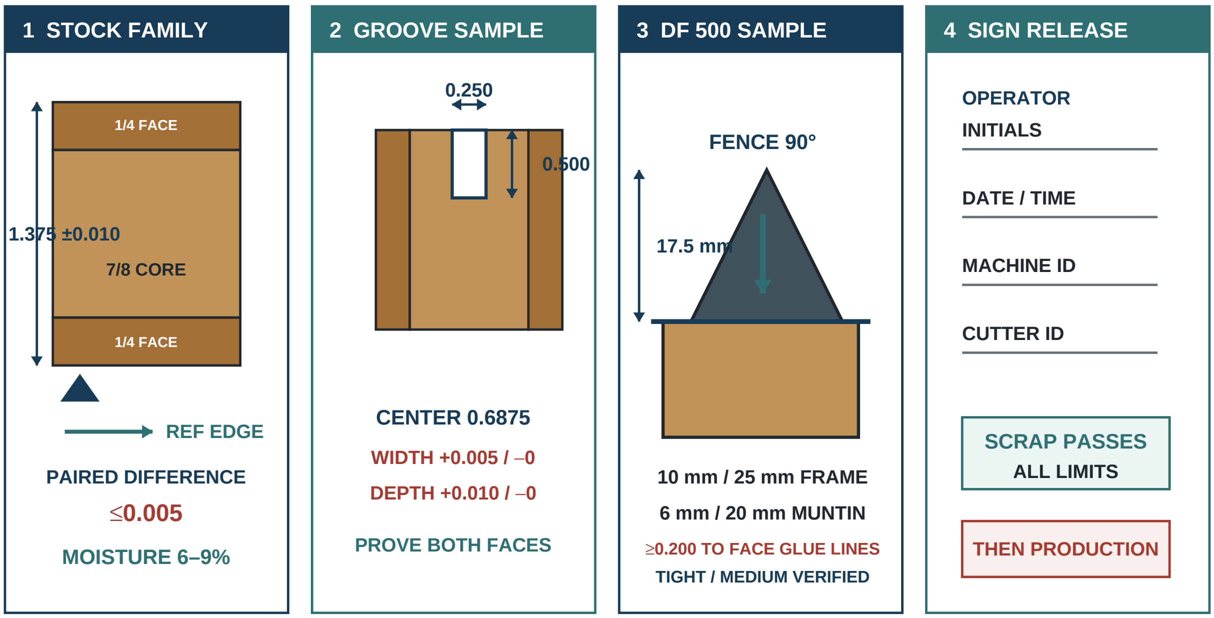
Calibrate before touching project stock

Prove saw, jointer, planer, router, DF 500, squares, and assembly platform on scrap.

3A Machine calibration and matched-scrap release

Calibrate thickness, reference geometry, groove, and DF 500 settings before a production member.

SETUP GATE



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1. Clean and inspect the jointer tables/fence, planer bed and rollers, table-saw arbor/flange/fence, crosscut fence, plunge-router collet/base/edge guide, and DF 500 fence/pins.
2. Verify jointer tables coplanar, fence square, planer cut parallel, rip fence parallel to the blade without dangerous toe-in, and crosscut fence square using a five-cut or equivalent test.
3. Check blade and cutter condition. Install the 1/4-inch dado stack only for the supported through-rail operations on Page 20; install a sharp 1/4-inch plunge-router bit for Pages 21–22.
4. Verify the router depth stop and edge-guide offset on scrap. Verify DF 500 cutter diameter, 17.5 mm fence height, 90-degree fence, and plunge stops with a measured test mortise.
5. Check calipers, combination squares, straightedges, winding sticks, moisture meter, and tapes against shop standards. Record tool ID and date.
6. Plane one full-width face mother billet through the staged no-resaw sequence, including its intermediate rest and a flat carrier where required. Then press one full-thickness 5/16 + 7/8 + 5/16 calibration blank through the production sequence. After symmetric surfacing, check the nominal 1/4 + 7/8 + 1/4 section, thickness at corners/center, glue-line position, edge squareness, groove location, and DF cutter clearance.

Release gate: Calibration results are recorded and remain with the setup.

STOP: Drift, vibration, burning, snipe inside a final cut line, or repeatability worse than the feature tolerance requires repair and retest.

REMAKE: Never “average” inconsistent machine settings across production parts.

Materials, test stock, and consumables

Reserve A-O for the door; use finished M for the upper panel, keep damaged A intact, keep 74-inch N/O full as short-core reserve, and release continuous layers, coupons, and seam maps before replacement stock.

Family	Controlled specification	Planning quantity	Selection rule
Door lumber	A now 72-3/4 in; B-F/H-K rough QS oak; G confirmed S4S at 15/16 x 4-1/2 x 75; L now 6 in wide; balanced frame layup	No purchase currently released	LS-05 / RL-02; no resaw; G thickness/width/length pass; prove B/C faces, K/I lanes, J's 6-1/2-in F-core envelope, revised J/L map, coupons, and D/F seams
Panels	STOCK-M/G/H/D primary; A is first upper-panel fallback	G thickness/width/length fit four J staves; CHK-P-01 still controls; all A-O milling-gated	CHK-RS-01 + CHK-P-01; vertical grain and matched lower pair
Casing + stop	S4S QS white oak, 4/4 x nominal 6 in x 8 ft	7 boards; about 28 dealer BF	Four sides, both heads, full-length spare; mill stop from a wide mother blank
Backband	S4S QS white oak, 6/4 x nominal 6 in x 8 ft	2 boards; about 12 dealer BF	All backband; optional caps only when field-supported
Order control	Correct notation is 4/4, not 4 x 4	About 40 dealer BF total	No plinth stock; no A-O stock credited to moulding
10 x 50 tenons	Factory beech	60: 26 final + 26 test + 8 spare	Segregate and label each group
6 x 40 tenons	Factory beech	12: 4 final + 4 test + 4 spare	Segregate and label each group
Panel spacers	3/4-in-long oil/silicone-free foam	30: 20 installed + 10 test/spare	PB265 is candidate only; release H/J/N and K/L separately
Hardware	Brass visible set	Physical project set	Measure before machining
Adhesive + shop kit	Fresh Type II PVA; timer, cauls, clamps, P120 abrasives	One controlled production set	Same-layup coupon and timed rehearsal must pass
Finish	Rubio Oil Plus 2C Pure	Coverage + sample allowance	Room-approved sample controls

STOP: Substitute tenons, spacers, hardware, adhesive, or finish require a new sample/release.

REMAKE: Moisture-damaged tenons, expired adhesive, or contaminated finish cannot enter production.

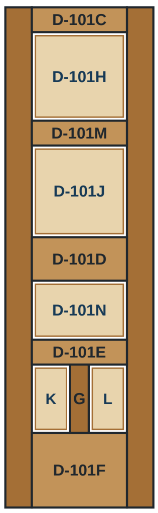
Door frame members

Build the eight structural members as balanced three-layer families while preserving every finished dimension.

1 Lay out every cut part

Confirm identifiers, orientation, and count before mortising.

DRY FIT / NO GLUE



Frame parts

- D-101A/B: 4-1/4 x 80-1/2 equal stiles
- D-101C/M/E: 4 x 15-1/4 rails
- D-101D: 7 x 15-1/4 lock rail
- D-101F: 12 x 15-1/4 bottom rail
- D-101G: 3 x 11 muntin
- Frame section: 1/4 + 7/8 + 1/4

Floating panels

- H: 15-7/8 W x 15-1/16 L
- J: 15-7/8 W x 15-9/16 L
- N: 15-7/8 W x 10-5/16 L
- K/L: 6-7/8 W x 11-13/16 L
- All panel grain runs vertically

Mark before assembly

- Show face on every frame member
- Reference face and reference edge
- Top and bottom of both stiles
- Top/reference edge of every rail
- Joint IDs at each rail end
- Panel ID on the back face
- Pair D-101K and D-101L visually
- Keep all show faces upward on bench

If one label is missing, stop and remark the part.

DC-1916-001 | Rev. G | show face up | drawings not cutting templates

ID	Member	Qty	Glue blank T x W x L	Finished T x W x L
D-101A	Hinge stile	1	1.5 x 4.375 x 81.125	1.375 x 4.25 x 80.5
D-101B	Lock stile	1	1.5 x 4.375 x 81.125	1.375 x 4.25 x 80.5
D-101C	Top rail	1	1.5 x 4.25 x 16.25	1.375 x 4 x 15.25
D-101M	Upper intermediate rail	1	1.5 x 4.25 x 16.25	1.375 x 4 x 15.25
D-101D	Lock rail	1	1.5 x 7.25 x 16.25	1.375 x 7 x 15.25
D-101E	Lower intermediate rail	1	1.5 x 4.25 x 16.25	1.375 x 4 x 15.25
D-101F	Bottom rail	1	1.5 x 12.25 x 16.25	1.375 x 12 x 15.25
D-101G	Center muntin	1	1.5 x 3.25 x 12	1.375 x 3 x 11

SECTION	STILES	HOLD
Preglue 5/16 + 7/8 + 5/16; finish 1/4 + 7/8 + 1/4. Plane faces in staged passes; never resaw.	D-101A B/K/K; D-101B C/I/I (show/core/back). I/K ordinary-ripped into one core and one back-face lane; no end joints.	Coupons and D/F seam map pass before joinery; perimeter fitting waits for the jamb.

STOP: Do not cut final stile length until the dimensional-state gate passes.

REMAKE: Reject a finished frame blank with pith, active check, unstable bow/twist, or insufficient clean wood at a joint.

Panels and field-fit moulding

Panel sizes include 7/16-inch tongue projection; moulding lengths remain scheme- and field-dependent.

Floating-panel shop schedule

Panel	Qty	Rough T x W x L	Finished T x W x L
D-101H	1	0.625 x 16.375 x 15.5625	0.5 x 15.875 x 15.0625
D-101J	1	0.625 x 16.375 x 16.0625	0.5 x 15.875 x 15.5625
D-101N	1	0.625 x 16.375 x 10.8125	0.5 x 15.875 x 10.3125
D-101K	1	0.625 x 7.375 x 12.3125	0.5 x 6.875 x 11.8125
D-101L	1	0.625 x 7.375 x 12.3125	0.5 x 6.875 x 11.8125

Field-fit moulding schedule

Moulding	Qty	Finished section	Length control
M-201A/B	2 per trimmed face	0.75 x 4.5	field transfer
M-201C	1 per trimmed face	0.75 x 4.5	field transfer
M-202A/B	2 per trimmed face	1 x 1.125	field transfer
M-202C	1 per trimmed face	1 x 1.125	field transfer
M-203A-C	3 total per opening	0.375 x 1.25	field transfer from operating door
M-204	1 per treated face	0.875 x 1.5	field transfer
M-205A/B	2 per treated face	0.875 x field verify	field verify

1. Treat panel dimensions as overall finished blank dimensions before tongue formation; do not add tongue projection to them.
2. Field formulas, not the slab size, control moulding lengths. Add 2 inches rough length only after calculating the correct layer.

STOP: Do not release a cut list carrying the former panel sizes or a generic moulding length.

REMAKE: A panel cut to an obsolete width/length or a moulding member cut from nominal door dimensions cannot be patched longer.

Board allocation, moisture, and grain

Follow the A-O map in order: prove continuous stile layers first, then release panels, cores, and the short-face pool.

BOARD A-O: WHAT EACH BOARD BECOMES

Do not start panel or short-member cuts until the continuous B/C/I/K stile-layer gates pass.

A — **KEEP FULL**
72-3/4 reserve: panel fallback, setup, repair

F — **FP-05**
two clear 16-1/4 face billets; map knots first

K — **D-101A**
hinge-stile core + continuous back face

B — **D-101A**
continuous hinge-stile show face

G — **D-101J**
four center-panel staves; follow controlled G-01 cuts

L — **D/E/F**
lock-rail, lower-rail, and bottom-rail core staves

C — **D-101B**
continuous lock-stile show face + eligible width offcuts

H — **D-101N**
three lower-center panel staves; tail conditional

M — **D-101H**
three upper-panel staves; tail may enter FP-05

D — **K + L**
four lower-panel staves; tail may enter FP-05

I — **D-101B**
lock-stile core + continuous back face

N — **KEEP FULL**
finished white-oak short-core reserve

E — **FP-05**
three clear 16-1/4 face billets; map knots first

J — **C/M/F/G**
top and upper rail cores, wide F core, muntin core

O — **KEEP FULL**
finished white-oak short-core and repair reserve

CUT ORDER: prove B/C/I/K -> map M/G/H/D panels -> map J/L cores -> release E/F and eligible offcuts.

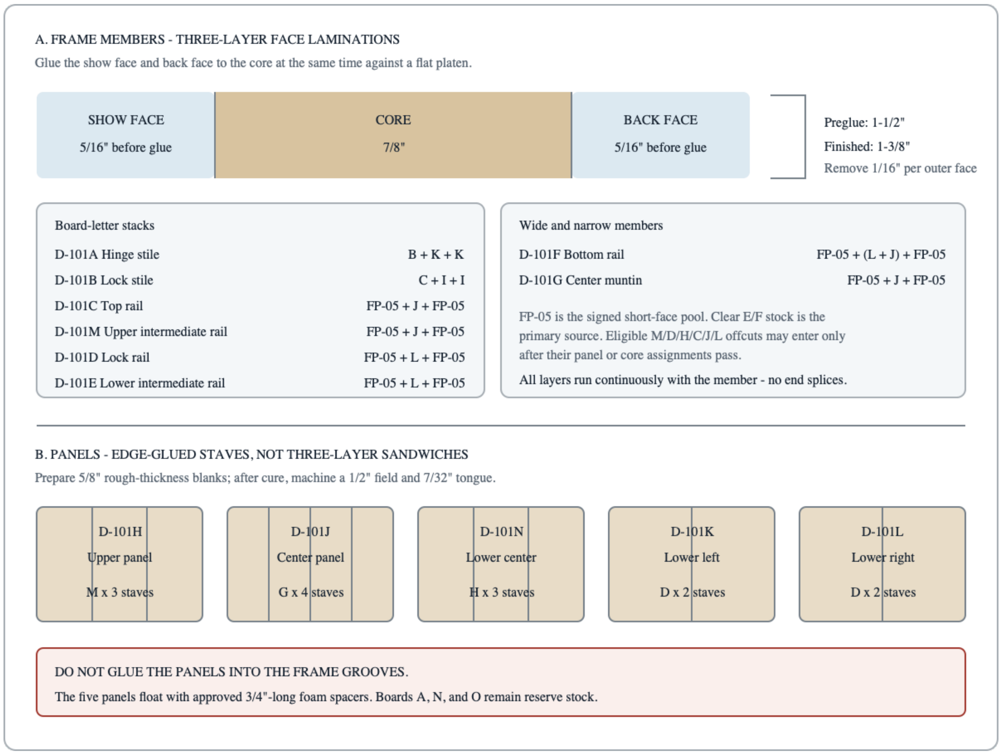
Step	Action	Release check
1	Keep A full; keep N/O at 1-3/4 x 11 x 74.	Reserve labels remain attached.
2	Prove B/C show faces and I/K two-lane yield.	All four continuous stile layers pass.
3	Map M/G/H/D panel billets.	Five panel blanks have clear, stable yield.
4	Map J/L cores.	C, M, D, E, F, and G cores are accounted for.
5	Release FP-05 face billets from E/F and eligible offcuts.	Every face is clear, traceable, and long enough.

STOP: Stock outside moisture range or without clean joint/hardware zones remains on hold.

REMAKE: A mismatched replacement visible within a paired set requires reallocation of the set, not isolated acceptance.

Plane, rest, press, and prove the balanced layup

Thickness-plane whole mother billets to 5/16 faces without resawing, retain a 7/8 core, and distinguish the eight frame laminations from the five edge-glued floating panels.



1. Do not rough-crosscut a critical B/C/I/K board. First flatten and joint only enough to prove the complete continuous stile-layer envelope and sign CHK-ST-01/02. The long show/core/back stacks are B/K/K for D-101A and C/I/I for D-101B.
2. Mill every core to a glue-ready 7/8-inch thickness. Make each 5/16-inch face by thickness-planing the entire mother billet in staged, alternating light passes; sticker and rest at an intermediate thickness, then take the final passes only after the billet remains flat. Use a straight, flat carrier/sled when the planer manufacturer’s minimum safe thickness or stable registration requires it. Do not resaw. Keep grain parallel with member length; no cross-grain layer, butt joint, scarf, or hidden short block is allowed.
3. Edge-glue D-101D/F core and face sheets separately at the released seam locations. Cure, flatten, and verify every seam before the three-layer press.
4. Prepare a full-thickness same-layup coupon for each pressing family. Apply the production adhesive spread and pressure, press both faces simultaneously against a flat platen with rigid registered cauls, and prevent layer slip.
5. After cure, section the coupon through representative groove and DF operations. Accept only void-free glue lines, wood failure rather than adhesive-line failure, and at least 0.200 inch actual core wood from each cutter envelope to a face glue line.
6. Plane panel laminations to a common glue-up thickness above 1/2 inch. Joint glue edges with the show-face orientation marked so adjacent pieces can be checked for a closed, square seam.
7. Rough-mill each separately purchased moulding family. Plane future M-203 stop and other narrow parts as wide mother blanks; do not rip them to hazardous narrow widths yet.
8. Sticker every cured blank on aligned, equal-thickness sticks over a flat support with air to both faces. Record date, moisture, bow, twist, press family, adhesive lot, and coupon ID.

Inspection: RF bears without rocking; RE is square within 0.003 inch per inch; no final dimension has been consumed.

STOP: New checks, cupping, or stress movement require longer rest or reallocation.

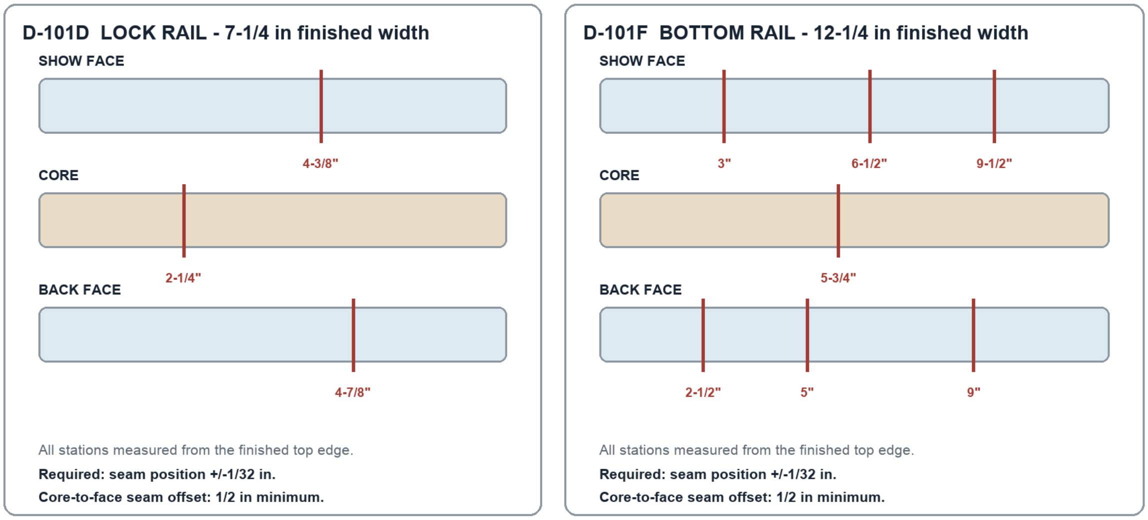
REMAKE: A blank that cannot clean up within finished size plus allowance is rejected now.

Rest, surface symmetrically, and verify seams

Remove 1/16 from each outer face, retain a 1/4 + 7/8 + 1/4 section, and mark the D-101D/F seams exactly as shown.

WIDE RAIL SEAM MAPS: D-101D AND D-101F

Mark every seam on both ends after final width and before any Domino mortise.



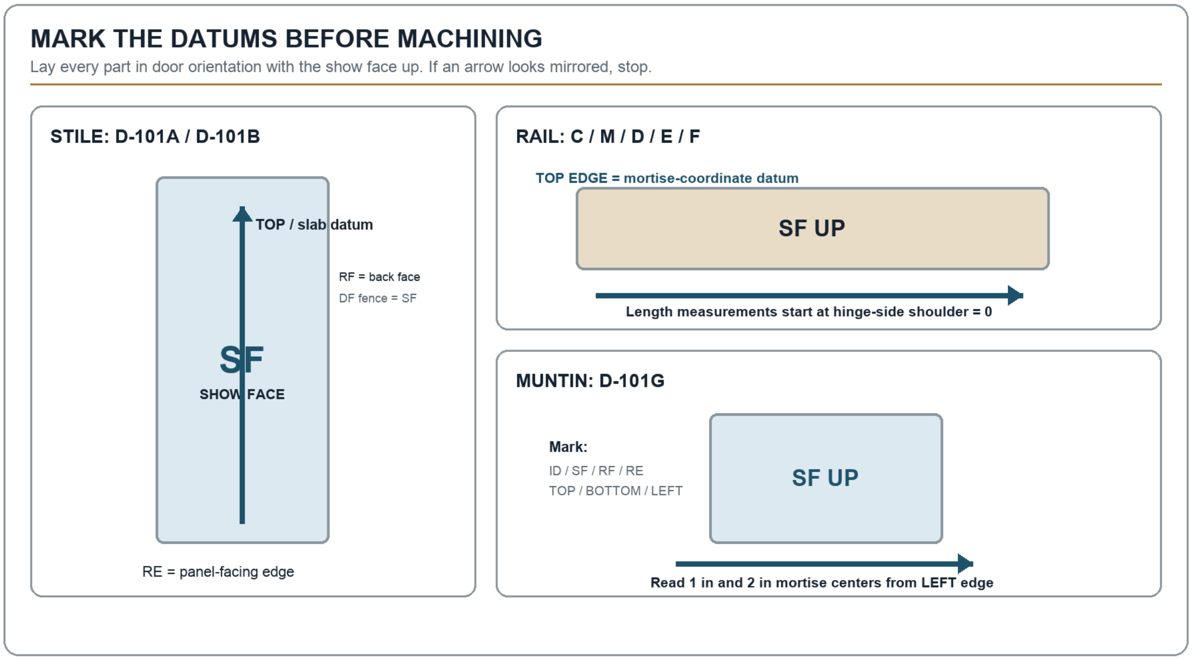
Step	Action	Check
1	Rest every cured lamination.	Moisture and shape are stable.
2	Plane 1/16 from each outside face in alternating passes.	Finished section is 1/4 + 7/8 + 1/4.
3	Mark D-101D core/show/back seams and D-101F seams.	Measure from finished top; every station is within +/-1/32.
4	Verify seam offsets before Domino machining.	Core-to-face offset is at least 1/2.

STOP: Do not machine grooves in a part that rocks, lacks a continuous datum, or differs from its mating member beyond tolerance.

REMAKE: Any member below finished thickness/width, or with irrecoverable twist in a show surface, is replaced.

Component marks and face discipline

Mark ID, SF, RF, RE, top, bottom, and direction before any groove or Domino operation.



Step	Action	Check
1	Lay the part in door orientation with SF up.	ID and arrows read normally.
2	Mark ID, SF, RF, RE, top, and bottom.	Marks are permanent or on blue tape.
3	Mark the hinge-side shoulder as rail zero.	All length coordinates use the same end.
4	Transfer from the story stick and knife mating lines.	Never measure from an unfinished or rounded edge.

Inspection: Place the parts in door orientation with SF up; every arrow and ID must read coherently without mental mirroring.

STOP: Ambiguous face, edge, top, or handing requires re-establishing the datum from the grain map and field record.

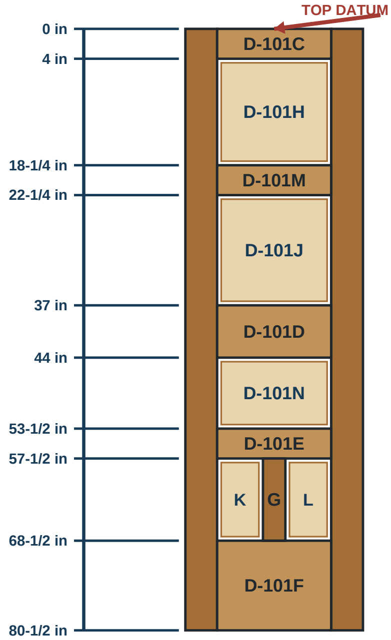
REMAKE: A feature machined from the wrong face or mirrored datum is rejected unless a documented section proves full compliance and no visual consequence.

Story stick and rail stack

Transfer every shoulder and stile mortise center from the slab-top master datum.

0 Story-stick rail stack

All vertical locations originate at the top edge of the finished slab.



Use one master story stick

- Hook the top of D-101A/B
- Knife every shoulder location
- Transfer to both stiles together
- Do not leapfrog a tape measure
- Stack must total exactly 80-1/2 inches

Opening checks

- Upper: 14-1/4 inches
- Center: 14-3/4 inches
- Lower-center: 9-1/2 inches
- Bottom: 11 inches
- Bottom pair: 6-1/8 inches each

DC-1916-001 | Rev. G | show face up | drawings not cutting templates

1. Prepare one stable, full-length story stick hooked from slab top zero. Mark both edges so one edge carries rail shoulders and panel openings while the other carries absolute stile mortise centers.
2. Transfer the vertical stack from the scribed final-top datum: D-101C 0–4; opening H 4–18-1/4; D-101M 18-1/4–22-1/4; opening J 22-1/4–37; D-101D 37–44.
3. Continue: opening N 44–53-1/2; D-101E 53-1/2–57-1/2; bottom opening 57-1/2–68-1/2; D-101F 68-1/2–80-1/2. Sum the marked bands independently to confirm 80-1/2 inches. Independently verify the clear opening heights: H 14-1/4, J 14-3/4, N 9-1/2, and the bottom field 11 inches. The full-width clear field is 15-1/4 inches; D-101G divides the bottom field into two 6-1/8-inch clear bays.
4. Mark absolute stile Domino centers down from the final top: C 1.250/2.750; M 19.500/21.000; D 38.250/40.500/42.750; E 54.750/56.250; F 69.750/72.500/76.000/79.250.
5. On a separate 15-1/4-inch rail-shoulder stick, mark the D-101G footprint from 6-1/8 to 9-1/8 inches and its receiver mortise centers at 7-1/8 and 8-1/8 inches from the hinge-side shoulder.
6. Clamp the master stick—not a tape—to D-101A and D-101B with tops flush. Knife every shoulder, opening, and mortise center, then compare both stiles face-to-face.

Inspection: The two stile transfers coincide within 0.010 inch and the complete stack returns exactly to 80-1/2. The lock center remains 40 inches above the slab bottom, or 40-1/2 inches down from the final top.

STOP: Do not “split” a transfer error among openings.

REMAKE: A damaged, shortened, or cumulatively laid-out story stick is replaced and all transfers are rechecked.

Geometry and machining split

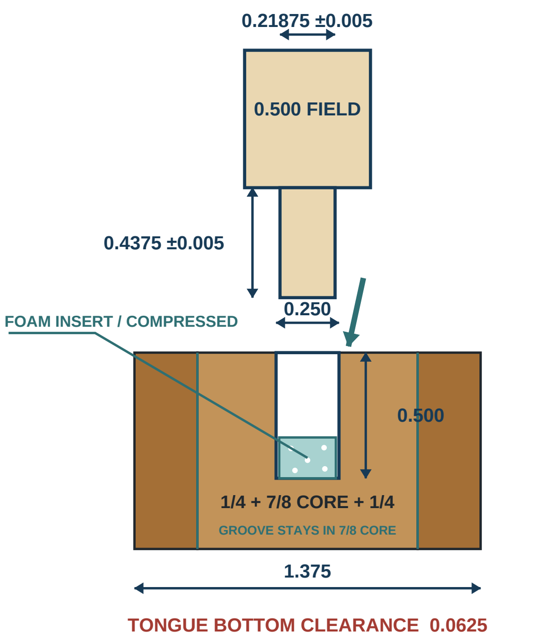
Use the dado only for supported through rail grooves; route stopped, segmented, and muntin grooves.

4 Panel groove and floating tongue cutaway

The 7/32 +/-0.005-inch tongue enters a centered 1/4 x 1/2 groove without glue.

CUTAWAY

EXACT CROSS-SECTION / SHOW FACE →



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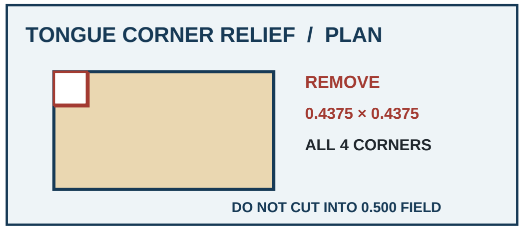
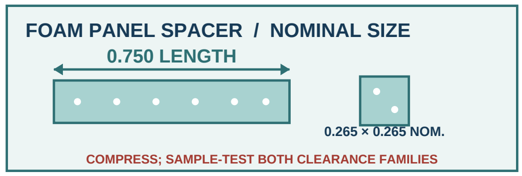




Movement, spacer, and glue boundary

- Full panels: 0.250 total side clearance
- Lower pair: 0.125 total side clearance
- Parallel clearance: 0.0625 total
- Prefinish edges; panel tongues remain dry
- Spacer does not set panel cut size

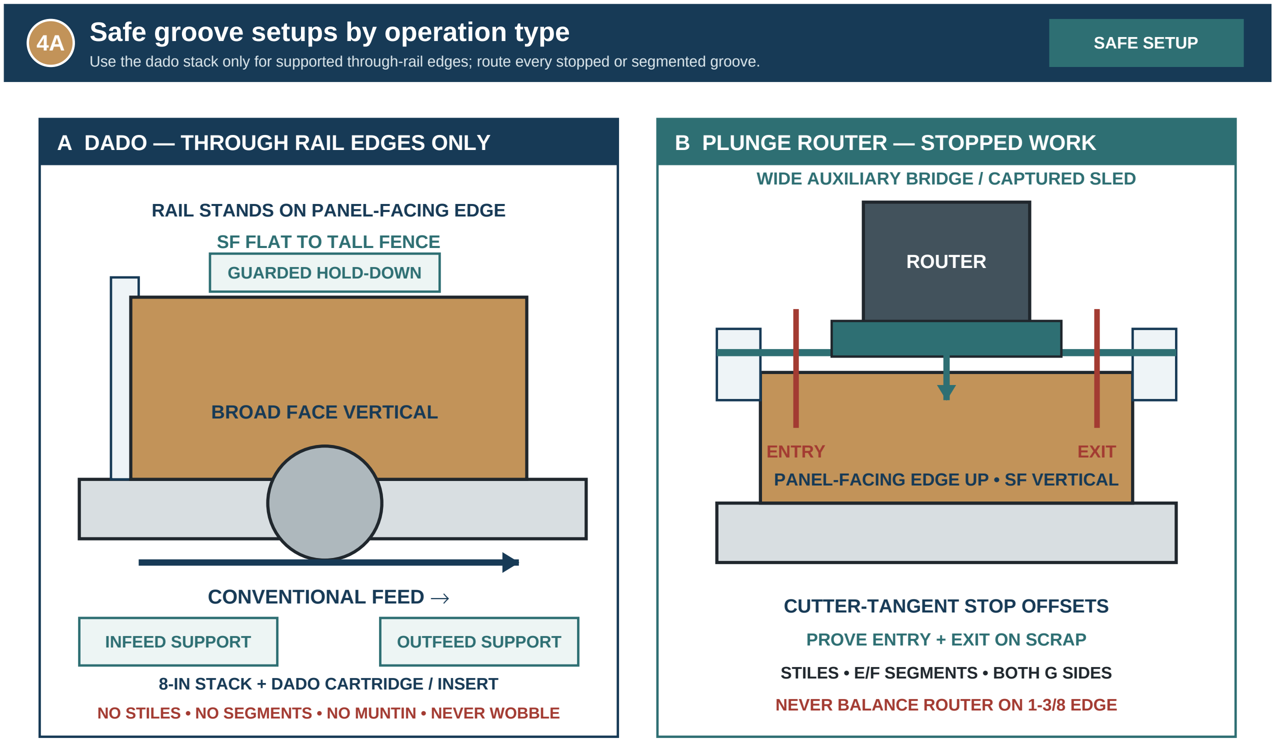
1. Center every panel groove 11/16 inch from SF in the 1-3/8-inch frame thickness. Use RF/SF against the machine datum for every member so opposed grooves align.
2. Machine grooves to **1/4 inch wide +.005/-.000 x 1/2 inch deep +.010/-.000**. Prove width, depth, and centerline on production-thickness TP-G before production.
3. Verify the complete groove-width envelope remains inside the 7/8-inch core between the two face glue lines. A groove that enters a face lamella or glue line rejects the laminated member.
4. Use the guarded dado stack only for conventionally fed, supported, uninterrupted rail-edge grooves: C bottom; M both edges; D both edges; E top.
5. Use the plunge router only in an edge-up captured cradle with a wide, roll-resistant bridge and positive stops for all opening-only stile grooves, E-bottom and F-top segments, and both long edges of G.
6. Keep groove bottoms clean and continuous. Opposed centerlines must align within 0.010 inch; the approved 7/32-inch tongue must slide without rocking or clamp pressure.
7. Preserve measured wood between **every relevant standard-rail mortise**—tight and medium—and its adjacent groove at **0.150 inch minimum**. At D-101G preserve **0.100 inch minimum groove-to-mortise web** and **0.220 inch minimum inter-mortise web**; between the tight/medium mortises in each E/F rail receiver preserve **0.100 inch minimum inter-mortise web**.

STOP: A dado stack is prohibited for stopped, segmented, or muntin grooves because a blind plunge/withdrawal on the saw cannot be positively controlled.

REMAKE: Wrong-face grooves, overrun into a joint footprint, width/depth below target, or web below minimum require a new structural member.

Supported through rail grooves

Prove the SawStop-compatible dado setup and conventionally feed only stable, fully supported rail edges.



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1. Disconnect the saw. On the identified SawStop, use only a compatible **8-inch stacked dado**—never a wobble dado—with the dedicated dado brake cartridge and compatible zero-clearance insert required by the exact saw manual. Build a sharp 1/4-inch stack, verify full arbor/nut engagement, and install every specified non-through-cut control before power-up.
2. Stand TP-G on the narrow panel-facing edge with SF flat against a tall, rigid auxiliary fence. Set the dado center 11/16 inch from SF and cutter height to 1/2 inch; prove both dimensions with a sectioned sample rather than the elevation scale.
3. Provide continuous infeed/outfeed support at table height. Use horizontal feather pressure against the tall fence, a guarded overhead hold-down, and trailing push blocks so the rail cannot tip, lift, or rotate; keep hands outside the marked cutter plane.
4. Mark each approved through edge: C bottom; M top/bottom; D top/bottom; E top. Confirm the opposing edge mark before every pass.
5. Feed conventionally from leading shoulder to trailing shoulder at a steady rate. Maintain fence contact until the member has fully cleared; never reverse over the running cutter.
6. Brush out chips and gauge each groove at both ends and center before stacking the part. Record width, depth, centerline, and edge ID.

Accept: Width .250–.255, depth .500–.510, clean bottom, no burn or wandering centerline.

STOP: Lift, chatter, fence separation, burning, or unsupported stock requires shutdown and setup correction.

REMAKE: A through groove on E bottom or F top, or any groove entering the 6-1/8–9-1/8-inch muntin footprint, destroys required joint wood.

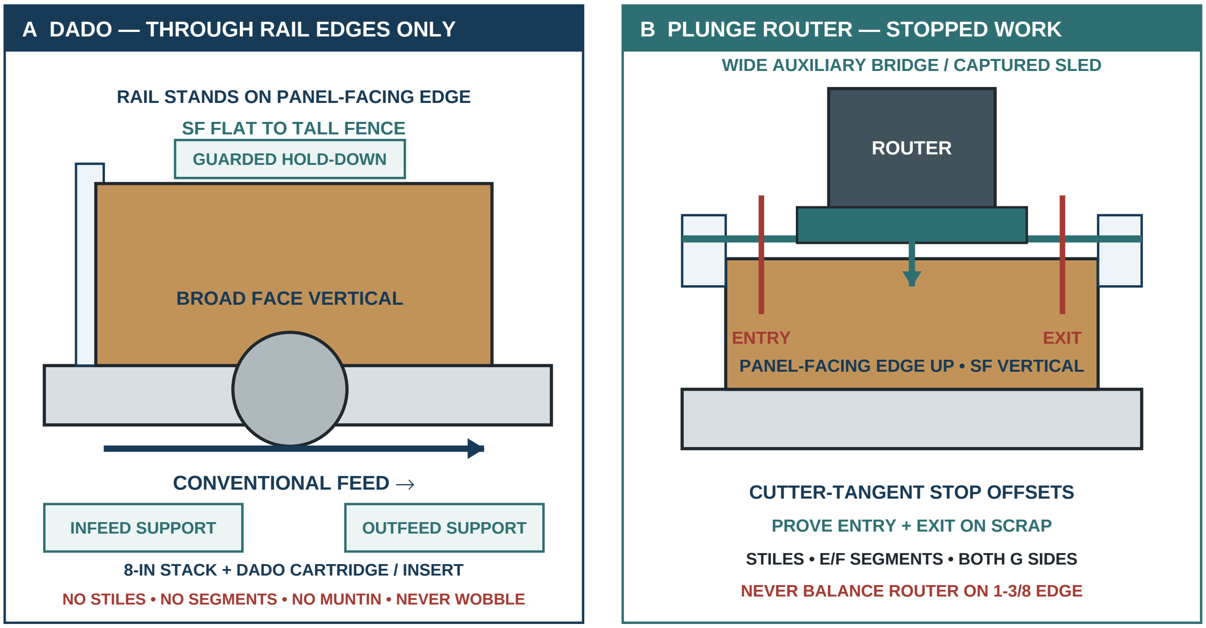
Opening-only stile grooves

Use a clamped plunge router, edge guide, and positive start/stop blocks—never a freehand drop cut.

4A **Safe groove setups by operation type**

Use the dado stack only for supported through-rail edges; route every stopped or segmented groove.

SAFE SETUP



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1. Stand D-101A with its narrow panel-facing edge horizontal and facing up; SF is vertical against the fixed datum side of a full-length captured cradle. Support both broad faces so the stile cannot bow or roll, and clamp below the router bridge and outside every travel zone. Repeat independently for D-101B.
2. Fit a sharp 1/4-inch bit with verified collet engagement and extraction. Use a wide auxiliary router base or captured bridge that bears on two coplanar rails spanning both sides of the 1-3/8-inch edge; the router may not balance on the stile edge alone.
3. On production-thickness TP-G, set the cutter center 11/16 inch from SF. Prove the conventional-feed arrow, bridge stability, .250–.255 width, and .500–.510 final depth before setting production stops.
4. Mark the required cutter-tangent starts and ends at 4–18-1/4, 22-1/4–37, 44–53-1/2, and 57-1/2–68-1/2 inches from the final slab top. On scrap, measure the separate entry and exit offsets from the cutter tangent to the bridge's stop-contact faces; place the blocks by those offsets, label them ENTRY/EXIT, and verify the resulting cut—not the router-base position.
5. Make two controlled conventional-feed passes, first approximately 1/4 inch deep and then to final depth. Start with the carriage against ENTRY, bring the bit to full speed, plunge, feed to EXIT, stop travel, retract fully above the edge, and only then switch off.
6. Gauge each segment and probe both terminations. The cutter may not enter a rail-joint footprint; the full 7/16 panel-corner relief must clear the radiused termination.

STOP: Never freehand the start or stop and never pull the live bit backward to “clean” an end.

REMAKE: Any overrun across a rail shoulder/joint zone, visible offset between segments, or groove outside tolerance requires a new stile.

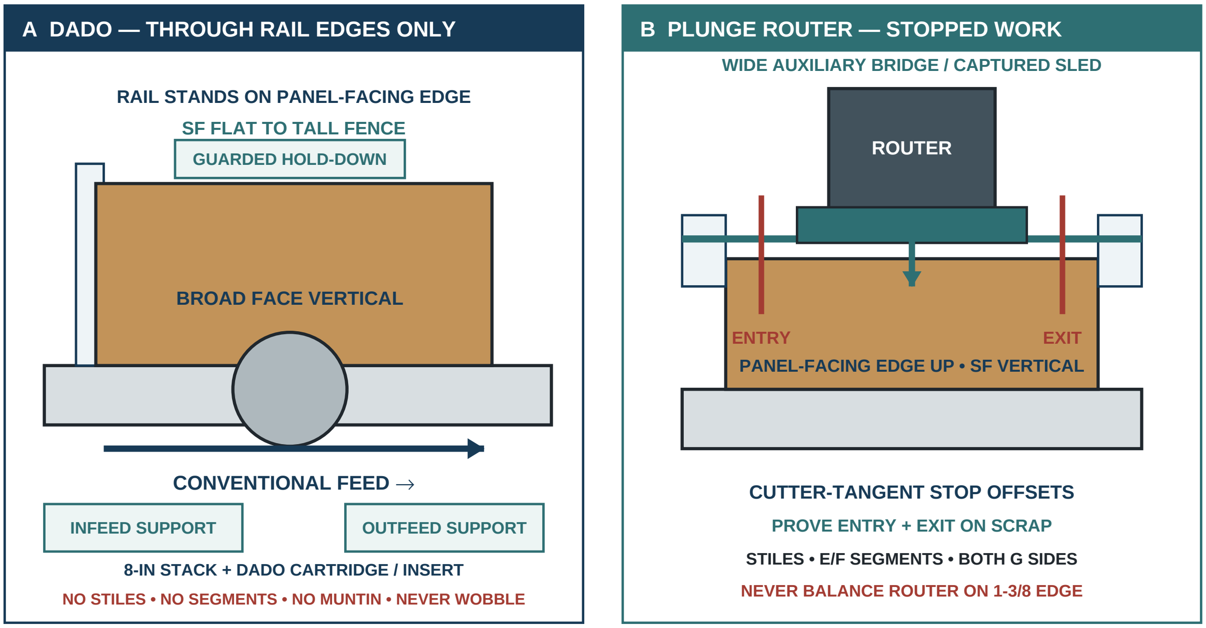
Segmented lower rails and muntin

Preserve the full 6-1/8-to-9-1/8-inch joint footprint and machine D-101G only in a captured support fixture.

4A Safe groove setups by operation type

Use the dado stack only for supported through-rail edges; route every stopped or segmented groove.

SAFE SETUP



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1. Stand D-101E edge-up in the captured cradle with its bottom panel-facing edge horizontal/up, SF vertical against the datum side, and the hinge-side shoulder at the recorded zero. Apply the proven cutter-tangent ENTRY/EXIT offsets to segments **0–6-1/8** and **9-1/8–15-1/4** inches.
2. Route each E-bottom segment in the two proven conventional-feed depth passes. Retract fully at each stop; never connect the cuts across the solid 6-1/8–9-1/8-inch D-101G footprint.
3. Repeat the same edge-up setup on D-101F's top panel-facing edge. Confirm SF and the hinge-side zero before applying the separate tangent offsets to 0–6-1/8 and 9-1/8–15-1/4.
4. Cradle D-101G with one 11-inch panel-facing edge horizontal/up and SF against the same datum. Route that edge with sacrificial end support, then re-cradle the opposite edge without changing the SF centerline reference; never balance the router directly on the 1-3/8-inch edge.
5. Measure the remaining 6-1/8–9-1/8-inch solid rail islands and confirm they are uncut, full thickness, and exactly aligned when E/G/F are placed in assembly orientation.
6. After G mortise samples are made, section TP-11/TP-12 through both joint halves. Measure D-101G groove-to-mortise webs at **0.100 inch minimum**, D-101G inter-mortise webs at **0.220 inch minimum**, and E/F rail-receiver inter-mortise webs at **0.100 inch minimum**.

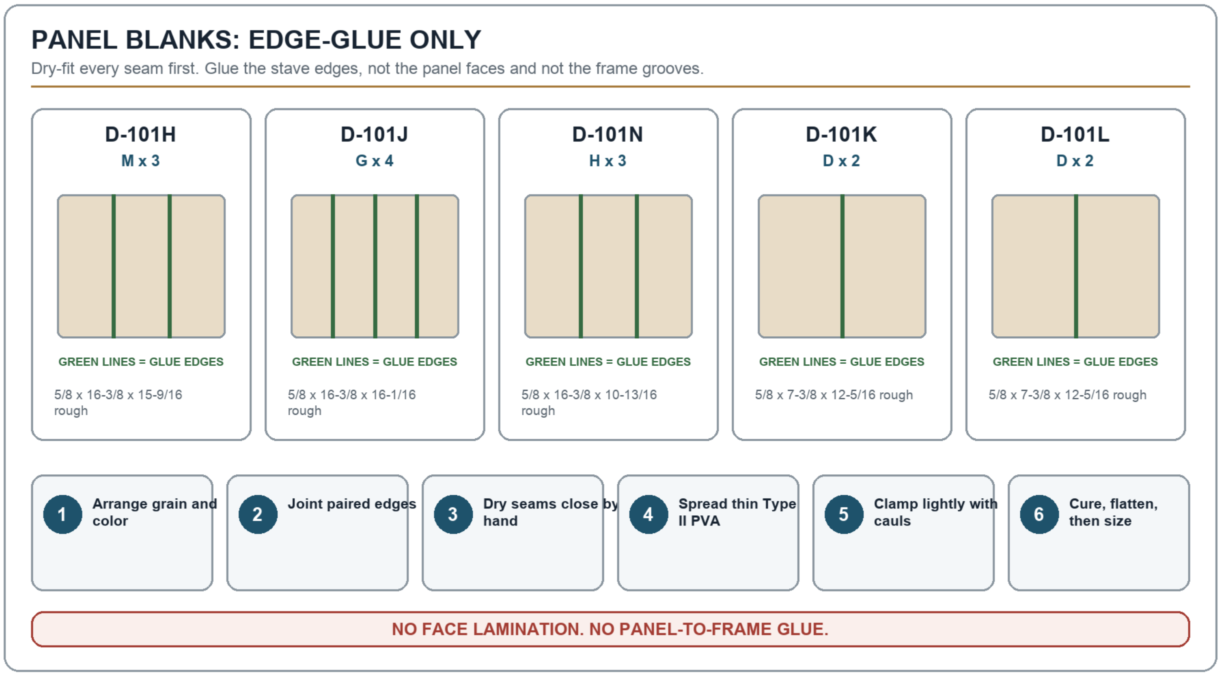
Accept: All grooves are .250–.255 wide and .500–.510 deep; the rail islands remain intact and centered.

STOP: If the guide shifts, retract the router vertically and quarantine the part.

REMAKE: An overrun into 6-1/8–9-1/8, a clipped island corner, or an actual web below minimum cannot be filled or patched.

Prepare and glue solid panel blanks

Arrange, joint, dry-check, edge-glue, clamp lightly, cure, and flatten the five panel blanks.



Step	Action	Check
1	Arrange the labeled staves for grain and color.	Grain runs vertically in door orientation.
2	Joint mating edges as pairs.	Dry seams close by hand.
3	Mark SF and stave order.	The layout cannot be reversed.
4	Glue edges with a thin Type II PVA film.	No dry spots or heavy flooding.
5	Clamp lightly between flat cauls.	Blank stays flat; fine squeeze line appears.
6	Cure fully, then flatten and size.	No open seam and full finished size remains.

STOP: A seam that will not close dry is re-jointed before glue; a skinned glue line is disassembled immediately while safe.

REMAKE: Open seams, offset exceeding available surfacing stock, or a blank that will not flatten to 1/2 inch at full size require replacement.

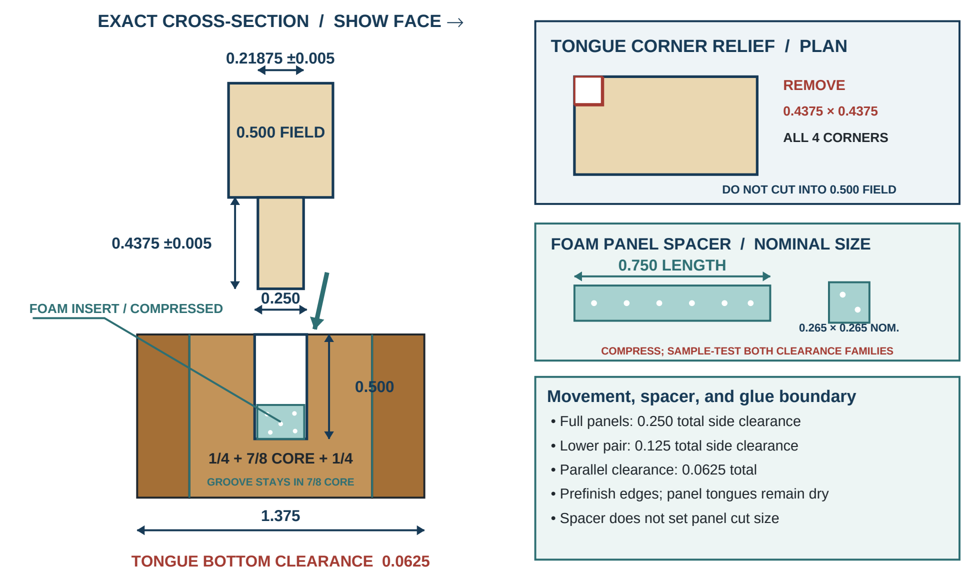
Size, tongue, and relieve corners

Form a 7/32 +/-0.005 x 7/16 +/-0.005 tongue with bottom clearance and remove each complete tongue corner square.

4 Panel groove and floating tongue cutaway

CUTAWAY

The 7/32 +/-0.005-inch tongue enters a centered 1/4 x 1/2 groove without glue.



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Panel	Clear frame opening	Finished panel	Across-grain clear.	Parallel clear.
D-101H	15-1/4 x 14-1/4	15-7/8 x 15-1/16	1/4 total	1/16 total
D-101J	15-1/4 x 14-3/4	15-7/8 x 15-9/16	1/4 total	1/16 total
D-101N	15-1/4 x 9-1/2	15-7/8 x 10-5/16	1/4 total	1/16 total
D-101K	6-1/8 x 11	6-7/8 x 11-13/16	1/8 total	1/16 total
D-101L	6-1/8 x 11	6-7/8 x 11-13/16	1/8 total	1/16 total

TONGUE	SPACER	RELEASE
7/32 +/-0.005 thick x 7/16 projection +/-0.005; relieve the complete 7/16 corner square.	3/4-in longitudinal length; two per vertical groove after the two-family Page 25 release.	Every dry panel moves laterally by hand.

1. Flatten cured blanks and plane to 1/2 inch with SF identified. Square one end and one long edge, then size overall: H 15-7/8 x 15-1/16; J 15-7/8 x 15-9/16; N 15-7/8 x 10-5/16; K/L 6-7/8 x 11-13/16.
2. Verify all panel grain runs vertically in door orientation. Mark the two vertical side edges that will receive spacers.
3. On TP-G, set the tongue operation to leave a **7/32-inch-thick tongue +/-0.005** centered in the 1/2-inch field. Preserve equal shoulders unless the approved sample specifically records an offset.
4. Form the tongue on all four edges to a **7/16-inch projection +/-0.005**. Make several light, supported passes and use backing at exits to prevent cross-grain breakout.
5. At all four corners remove the **complete 7/16 x 7/16 tongue square** exactly to the panel field corner. Do not leave a triangular nib and do not cut into the 1/2-inch field.
6. Test every edge in the retained groove sample. A seated tongue retains about 1/16-inch nominal bottom clearance and slides by hand. The groove stays 0.250 +0.005/-0.000 inch; check PB265's published 0.220-0.250 range at Page 25 and never widen a groove to fit foam.

STOP: Do not tune a panel by widening a production frame groove.

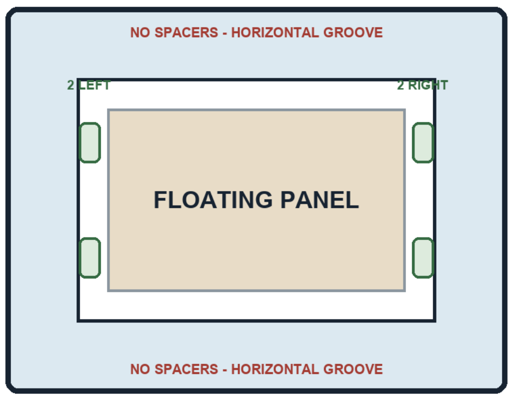
REMAKE: Undersize overall panel dimensions, tongue thickness outside .21375–.22375, projection outside tolerance, or relief cut into the field requires a new panel.

Spacer compression and edge prefinish

Use four spacers per panel in vertical grooves only; test H/J/N separately from K/L and keep every panel movable.

FOAM SPACERS: FOUR PER PANEL

Two spacers in each vertical side groove. No spacers in horizontal grooves and no glue on any panel tongue.



FULL-WIDTH FAMILY

H / J / N

Total side clearance: 1/4"

Centered tongue-end gap: 3/16"

Test as TP-SP-FW

LOWER PAIR FAMILY

K / L

Total side clearance: 1/8"

Centered tongue-end gap: 1/8"

Test separately as TP-SP-L

Panels must still move sideways by hand.

Step	Action	Check
1	Measure every spacer-zone groove.	Candidate foam matches the actual groove range.
2	Test H/J/N and K/L as separate families.	Both full-size samples pass independently.
3	Place two spacers in each vertical side groove.	Four spacers per panel; none in horizontal grooves.
4	Prefinish panel edges and tongues only.	No finish enters frame glue surfaces.
5	Close the sample and move the panel sideways.	Shoulders close; no rattle, stress, or binding.

Accept: Both family records identify the released product; finished tongues move freely; 3/4-inch-long foam spacers prevent rattle and recover without crushing; shoulders close without joint stress; and no finish contamination is present.

STOP: Frame finishing remains prohibited until jamb fitting and all hardware dry machining are complete; this panel-prefinish gate is the controlled exception.

REMAKE: A panel bonded by finish bridging, swollen beyond fit, deprived of residual movement, or loaded by spacer-induced joint stress is stripped only if the approved process preserves dimensions; otherwise replace it. Reject the candidate/family combination if rattle, contamination, inadequate recovery, or excessive/insufficient compression is the cause.

Map the physical lock before J-06

Overlay the actual lock and preserve 1/4 inch of clear wood to J-06 and other structural joinery.

5B J-06 joinery and physical lock-envelope gate

The measured lock machining envelope must remain 0.250 inch clear of J-06 structural joinery.

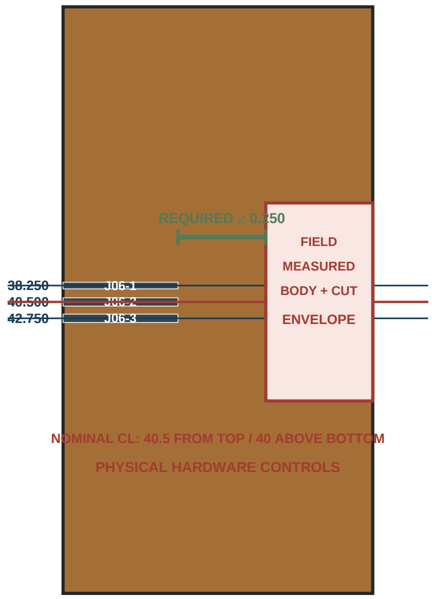
FIELD VERIFY

D-101B EDGE ELEVATION / HARDWARE OVERLAY

HARDWARE MEASUREMENTS

- MANUFACTURER / MODEL
- BACKSET
- BODY HEIGHT
- BODY DEPTH
- BODY THICKNESS
- FACEPLATE H × W
- SPINDLE / KEY LOC.

LEAVE BLANK UNTIL MEASURED



RELEASE LOGIC

1 Map actual lock + allowance.

2 Overlay J-06 mortise volume.

3 Check ≥1/16 to face glue lines.

4 Add 0.250 buffer toward J-06.

5 PASS: envelopes remain separate.

6 FAIL: re-engineer J-06.

7 Never shift the lock by assumption.

NO HARDWARE = J-06 HOLD

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1. Place the actual mortise lock, faceplate, spindle, key/privacy mechanism, strike, knobs/levers, escutcheons, hinges, and screws on the bench. Record manufacturer, model, and every physical dimension.
2. On a full-size side elevation of D-101B, plot lock-body height/thickness, faceplate depth/width, spindle and key/privacy bores, backset, machining tolerance, and screw paths at the nominal lock center: 40 inches above the finished slab bottom, or 40-1/2 inches down from the finished top.
3. Overlay J-06—the D-101D-to-lock-stile joint—with its three planned mortises and 25 mm plunges. Offset the physical hardware machining envelope toward J-06 and any other structural joinery to preserve **0.250 inch minimum clear wood** between the two systems.
4. In edge and face views, check that structural buffer against J-06, adjacent structural mortises, the panel groove, and the rail shoulder. In the cross-thickness view, keep the lock-body machining allowance at least **1/16 inch from each face glue line**. Separately verify that intentional faceplate, spindle, key/privacy, and screw openings suit the stile faces/edge, escutcheons, and actual hardware.
5. If conflict exists, stop and obtain an engineered relocation or joint revision that moves corresponding rail/stile centers together and preserves all groove-web and structural requirements. Update the register and story stick before tests.
6. Trace the released envelope on D-101B waste surfaces and retain the full-size overlay with the traveler through lock installation.

STOP: No J-06 production mortise may be cut before the physical envelope preserves the 1/4-inch structural buffer. Catalog drawings alone are insufficient.

REMAKE: A stile or lock rail mortised inside the released clear-wood envelope is structurally rejected; do not shorten the lock body or hide the conflict.

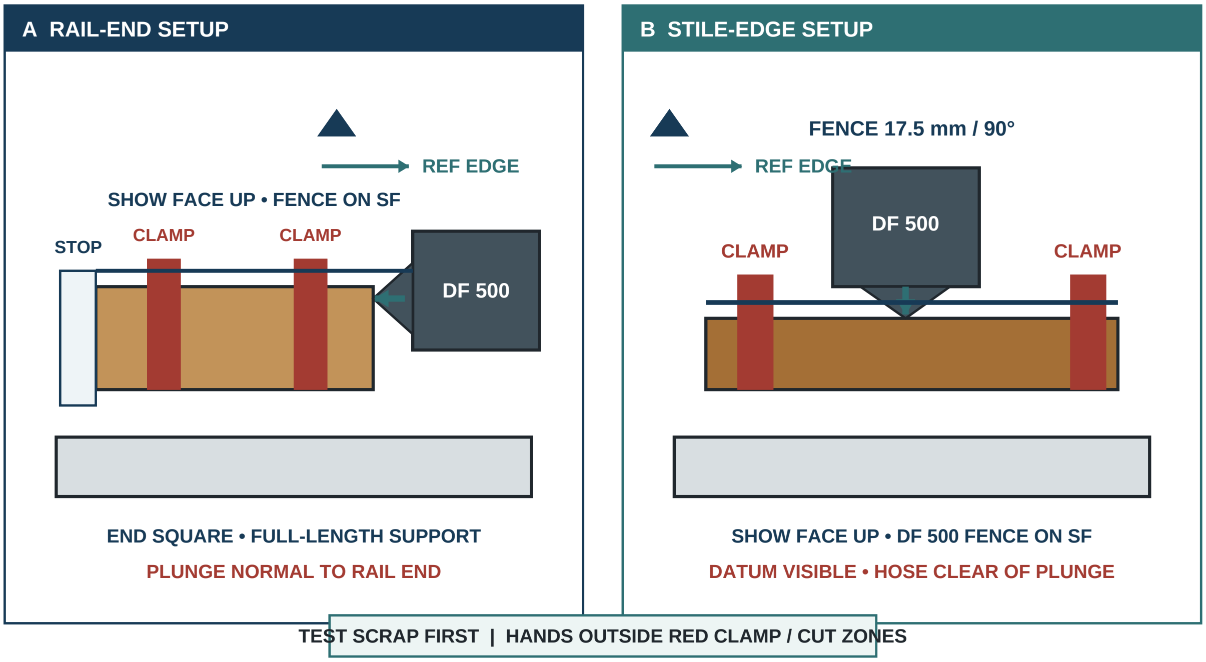
Cutter, fence, width, and workholding

Prove cutter, plunge, 17.5 mm fence height, 90-degree fence, datum face, support, and extraction.

5A DF 500 workholding and reference orientation

Clamp and fully support the test piece exactly as the matching production member will be machined.

WORKHOLDING



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1. Disconnect the DF 500 and install the verified 10 mm cutter for frame joints or 6 mm cutter for J-11/J-12. Check cutter seating, extraction, base, fence pivots, and registration pins.
2. Set fence height to 17.5 mm and fence angle to 90 degrees. Set 25 mm plunge for 10 x 50 tenons and 20 mm plunge for 6 x 40 tenons; verify each with a measured scrap mortise.
3. On the sectioned same-layup coupon, verify the 10 mm and 6 mm cutter bands remain wholly in the 7/8-inch core with at least **0.200 inch actual core wood** to each face glue line. Use the listed rail-local and vertical stile patterns; Rev. G horizontal rail shoulders, muntin footprints, and receiver coordinates control.
4. Place the member SF up on a flat bench. Clamp it against a straight back fence with the end or edge being mortised fully supported and unobstructed.
5. Set the machine fence firmly on SF; use a support platform when the DF 500 base would otherwise hang below a narrow edge. The machine must not pitch, roll, or bridge a clamp.
6. Align the engraved centerline to the knife mark and start the tool before contact. If the next mortise requires a different width, move the selector only while the motor is running and the cutter is clear—never during a plunge—then verify the detent. Seat the fence/base and plunge straight at a controlled rate.
7. Withdraw fully before lifting the fence. Vacuum chips and confirm center, depth, width setting, face reference, and seam clearance after the first mortise of every member.

STOP: A rocking machine, bowed workpiece, obstructed plunge, or uncertain selector position requires reset.

REMAKE: Face breakout, wrong-diameter cut, excessive plunge, or a mortise referenced from the wrong face rejects the member.

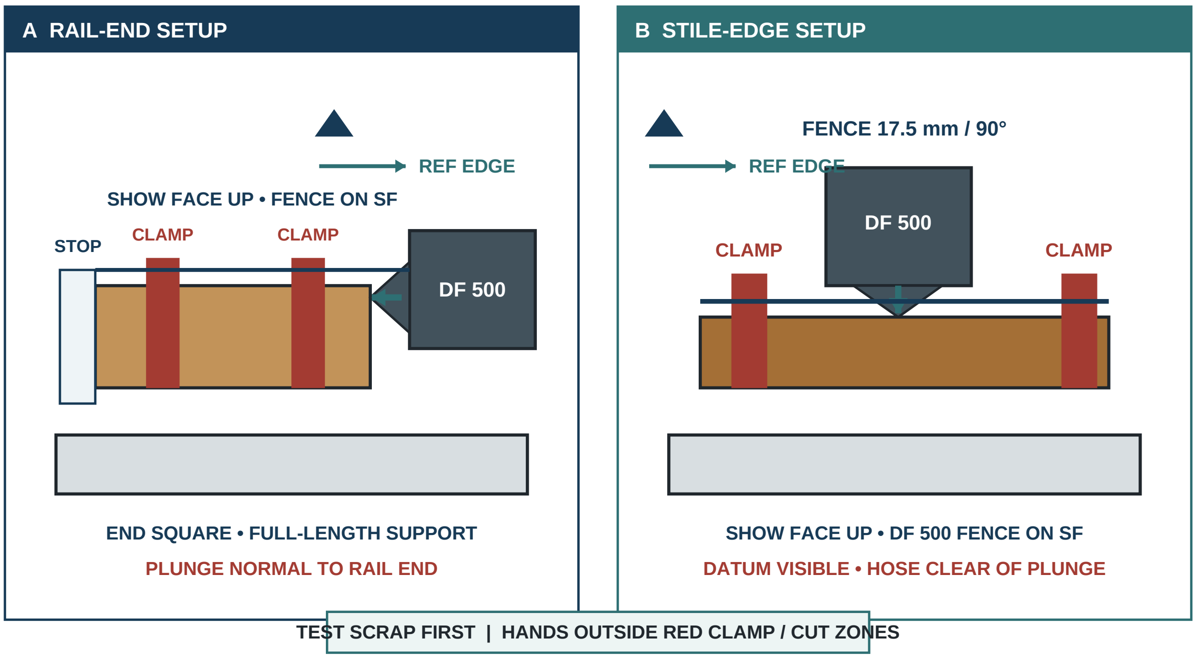
Production samples and relieved assembly tenons

Retain full-thickness joint samples; reserve untouched factory tenons for final adhesive stages.

5A DF 500 workholding and reference orientation

Clamp and fully support the test piece exactly as the matching production member will be machined.

WORKHOLDING



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1. Prepare production-thickness samples with the same 1/4 + 7/8 + 1/4 section, RF/SF, width, groove, seam relationship, and end geometry as each joint family. Label TP-01 through TP-12 before mortising.
2. Cut both mating mortises at their specified centers, widths, cutter, and plunge. First close each paired production-thickness sample with untouched, full-length factory tenons to verify seating and shoulder closure; never hammer.
3. Section representative 10 mm rail samples perpendicular to the mortise/groove relationship. Measure actual wood between **every relevant tight and medium mortise** and its adjacent groove at 0.150 inch minimum.
4. Section TP-11 and TP-12 through both G and rail receivers. Verify G's two tight mortises, the tight/medium rail pair, continuous mortise walls, and at least 0.200 inch actual core wood from each cutter envelope to either face glue line. Measure G groove-to-mortise webs at 0.100 inch minimum, G inter-mortise webs at 0.220 inch minimum, and each rail-receiver inter-mortise web at 0.100 inch minimum.
5. From non-production inventory, create clearly relieved rehearsal tenons by lightly easing the ribbed faces/ends only enough for repeated hand removal. Paint both ends red and mark **DRY ONLY—NO GLUE**.
6. Store the 26 untouched 10 x 50 and 4 untouched 6 x 40 production tenons separately. Count both bags before and after each rehearsal.

Accept: Sample shoulders close to <=.005 inch and adjacent faces finish within .010 inch with moderate hand pressure.

STOP: Any sample requiring hammering or heavy clamp pressure is a setup failure.

REMAKE: A production tenon altered for rehearsal, or a test piece with inadequate actual web, cannot be released for glue-up.

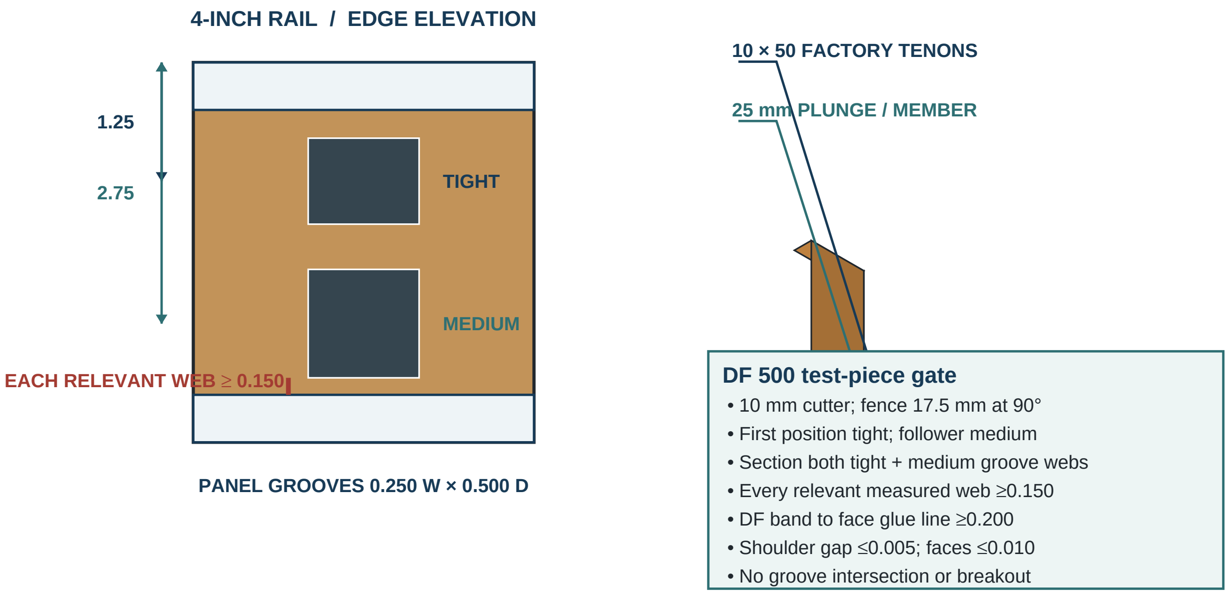
Four-inch rail-to-stile joints

Use revised 1.250 and 2.750-inch centers, one tight locator, and one medium follower.

5 DF 500 rail-to-stile joint

The locator is tight and the follower medium; section and measure every relevant mortise-to-groove web.

JOINT



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1. Assign J-01/J-02 to D-101C, J-03/J-04 to D-101M, and J-07/J-08 to D-101E at hinge/lock stiles respectively.
2. From each rail’s marked top edge, knife centers at **1.250 and 2.750 inches**. On A/B use the matching absolute centers: C 1.250/2.750; M 19.500/21.000; E 54.750/56.250 from the final slab top.
3. Clamp each rail SF up with its end square and supported. Cut the 1.250 mortise tight and the 2.750 follower medium, 10 mm x 25 mm deep.
4. Clamp each stile SF up and cut the matching upper mortise tight and lower follower medium. Keep the 17.5 mm fence firmly on SF.
5. Dry-pair each end with the labeled relieved tenons; identify joint number on both members and verify the shoulder line returns to the story stick.
6. Measure the actual web between **every relevant mortise, including the tight locator and medium follower**, and its adjacent groove on the sectioned family sample; minimum is 0.150 inch.

Accept: Centerline +/-0.010, shoulder <=.005, face flush <=.010, and intact stopped stile-groove zones.

STOP: Do not widen a tight locating mortise to compensate for a transfer error.

REMAKE: A merged mortise/groove, misplaced absolute stile center, or web below 0.150 inch rejects the affected member.

J-05 and hardware-gated J-06

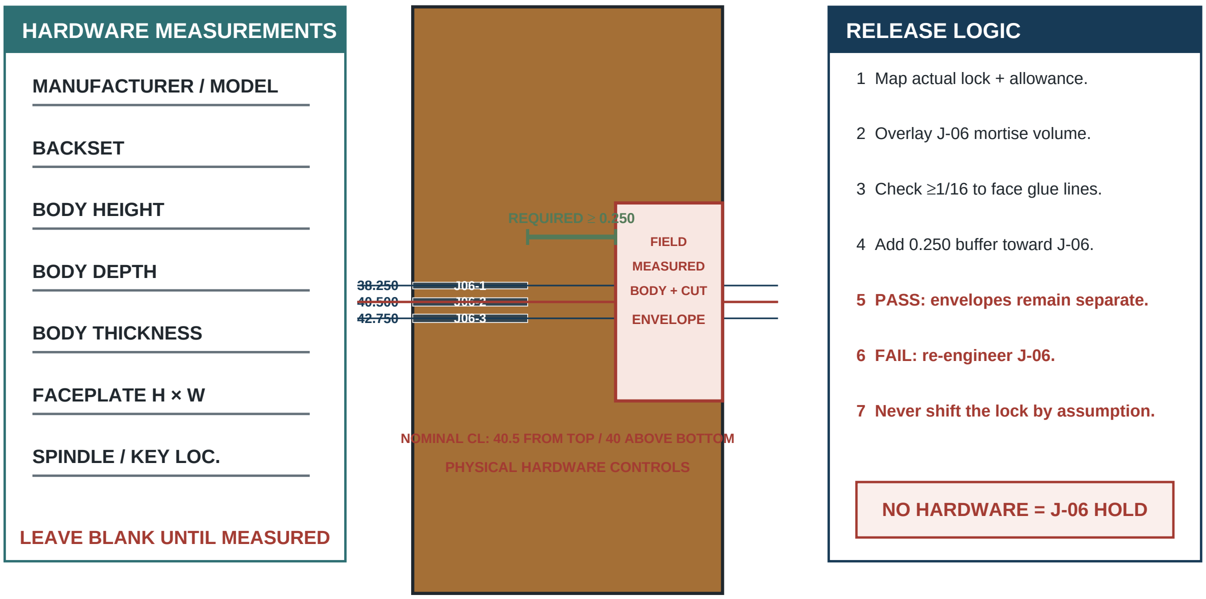
Use 1.250, 3.500, and 5.750-inch centers only after the physical lock envelope passes.

5B J-06 joinery and physical lock-envelope gate

The measured lock machining envelope must remain 0.250 inch clear of J-06 structural joinery.

FIELD VERIFY

D-101B EDGE ELEVATION / HARDWARE OVERLAY



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1. Confirm Page 26 is signed and the physical lock envelope plus 0.250-inch clear-wood buffer does not intersect J-06. Keep the released overlay visible at the DF 500 station.
2. Verify D-101D's core/show/back seam witnesses at **2.250, 4.375, and 4.875 inches from the finished top**, each within +/-1/32 inch and clear of its released mortise planform. Do not move a mortise center to accommodate a seam.
3. From D-101D's top edge, mark centers **1.250, 3.500, and 5.750 inches**. Mark matching absolute stile centers **38.250, 40.500, and 42.750 inches** from the final slab top.
4. On each rail end cut the 1.250 mortise tight and the 3.500/5.750 mortises medium, using the 10 mm cutter and 25 mm plunge.
5. On D-101A and D-101B cut the corresponding upper mortise tight and two followers medium. Recheck the lock-stile overlay immediately before each J-06 plunge.
6. Dry-pair J-05 and J-06 with red relieved tenons. Verify lock-rail shoulders at 37 and 44, face flush, unobstructed lock machining volume, and no mortise entering a stave seam.
7. Section the approved family sample and measure every mortise-to-groove web at 0.150 inch minimum and every DF cutter envelope to face glue line at 0.200 inch minimum.

STOP: A physical hardware change reopens this gate even if J-05 is complete.

REMAKE: Any J-06 cut inside the buffered lock envelope, wrong-width locating mortise, or inadequate web requires replacement or documented re-engineering before more work.

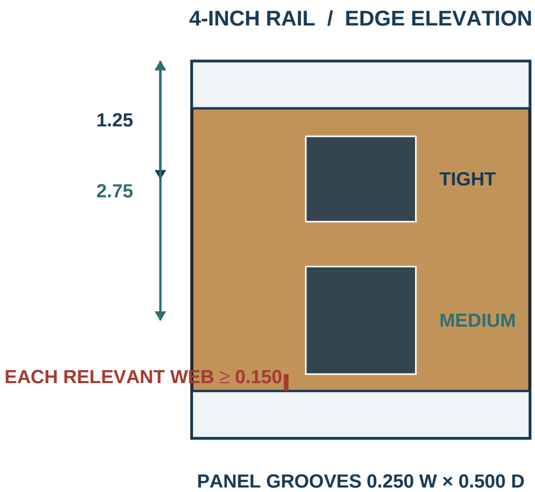
J-09 and J-10

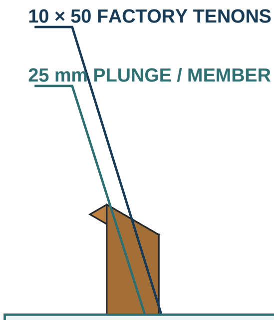
Use four tenons at the revised centers while preserving shoulder, face, and edge integrity.

5 DF 500 rail-to-stile joint

The locator is tight and the follower medium; section and measure every relevant mortise-to-groove web.

JOINT





DF 500 test-piece gate

- 10 mm cutter; fence 17.5 mm at 90°
- First position tight; follower medium
- Section both tight + medium groove webs
- Every relevant measured web ≥0.150
- DF band to face glue line ≥0.200
- Shoulder gap ≤0.005; faces ≤0.010
- No groove intersection or breakout

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1. Verify D-101F’s core/show/back seam witnesses at **5.750, 5.250, and 6.250 inches from the finished top**, each within +/-1/32 inch and clear of its released mortise planform.
2. From D-101F’s top edge mark **1.250, 4.000, 7.500, and 10.750 inches** on both ends. Keep the hinge-side shoulder identity visible so the segmented groove is not mirrored.
3. Mark matching absolute centers on D-101A/B at **69.750, 72.500, 76.000, and 79.250 inches** from the final slab top.
4. Clamp F with its 12-inch width fully supported. Cut 1.250 tight; cut 4.000, 7.500, and 10.750 medium using the 10 mm cutter and 25 mm plunge.
5. Reconfigure workholding for the stiles without changing SF reference. Cut the matching top locating mortise tight and remaining three medium.
6. Dry-pair J-09 and J-10 with relieved tenons. Verify the F top shoulder is at 68-1/2, slab bottom at 80-1/2, the 0–6-1/8 and 9-1/8–15-1/4 groove segments face upward, and no mortise enters a stave seam.
7. Check the sectioned test piece for complete walls, centerline, at least 0.150 inch actual web between mortises and the segmented panel groove, and at least 0.200 inch actual core wood to each face glue line.

STOP: The DF 500 may not be balanced on the unsupported middle of the wide rail; extend the machine support platform.

REMAKE: Mirrored groove orientation, a center outside +/-0.010, or web below 0.150 inch rejects D-101F or its affected stile.

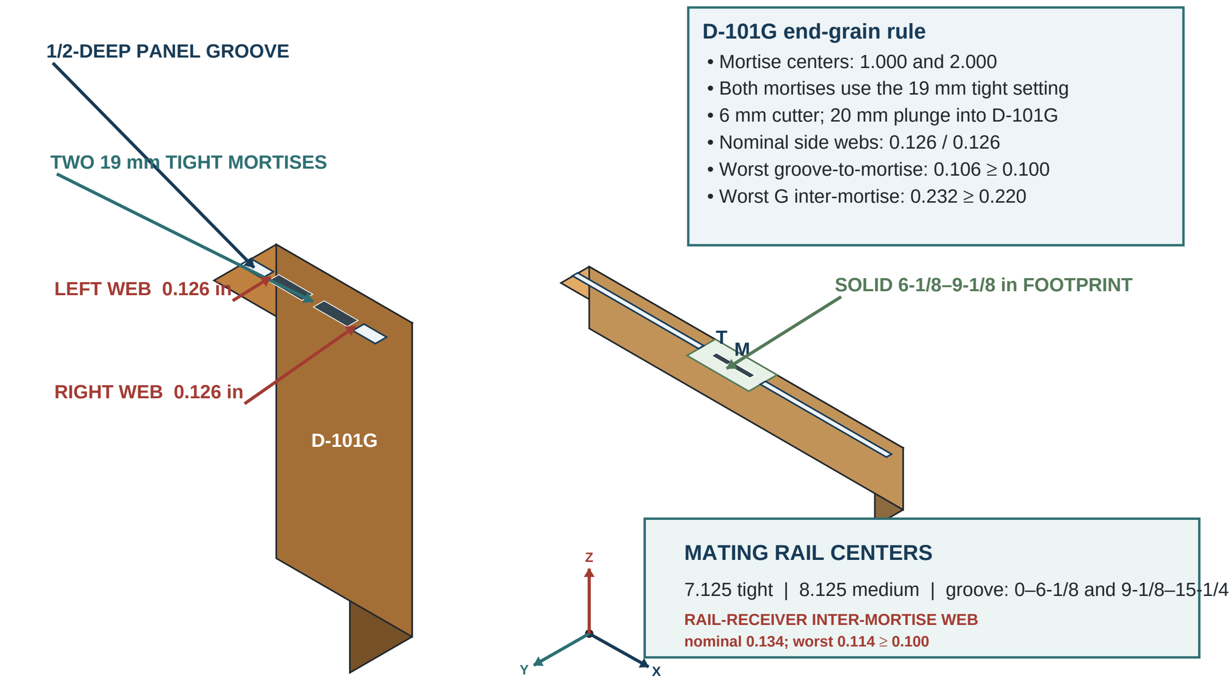
Muntin J-11 and J-12

Use D-101G centers 1 and 2 inches, mating-rail centers 7-1/8 and 8-1/8 inches, and actual web gates.

6A Muntin mortise and groove clearance

The corrected J-11/J-12 pattern preserves solid wood beside both panel grooves.

CRITICAL JOINT



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1. Fit the 6 mm cutter, 20 mm plunge, 17.5 mm fence, and tight-width setting. Reprove the setup on TP-11/TP-12 before touching D-101G.
2. On each G end, mark centers **1.000 and 2.000 inches from its left reference edge**. Cut **both G mortises tight**; do not apply the standard follower-medium rule.
3. On D-101E bottom and D-101F top, mark receiver centers **7-1/8 and 8-1/8 inches from the hinge-side shoulder** inside the solid 6-1/8–9-1/8-inch footprint.
4. Cut each rail receiver's 7-1/8 mortise tight and 8-1/8 follower medium. Keep the mortises square to the edge and centered from SF.
5. Dry-fit each joint with red 6 x 40 relieved tenons. Verify G stands square, panel grooves align, and the two bottom openings remain 6-1/8 inches each.
6. Section the production-equivalent samples. Measure D-101G groove-to-mortise webs at **0.100 inch minimum**, D-101G inter-mortise webs at **0.220 inch minimum**, and E/F rail-receiver inter-mortise webs at **0.100 inch minimum**, all with complete mortise walls.

Accept: J-11/J-12 shoulders <=.005, faces <=.010, centers +/-0.010, G both tight, rails tight/medium.

STOP: Do not widen either G mortise to ease assembly.

REMAKE: Groove intersection, broken wall, G groove-to-mortise web below 0.100 inch, G inter-mortise web below 0.220 inch, or rail-receiver inter-mortise web below 0.100 inch rejects the affected rail or muntin.

Complete setup register and section tests

Check all twelve joint families, member-specific datums, mortise widths, and measured web before assembly.

Joints	Part A to Part B	Tenon	A centers + datum	B centers + datum	Width A / B
J-01/J-02	D-101C to D-101A	10 x 50 mm	1.2500 in; 2.7500 in; top/left marked reference	1.2500 in; 2.7500 in; finished top of stile	first tight; remaining medium / first tight; remaining medium
J-03/J-04	D-101M to D-101A	10 x 50 mm	1.2500 in; 2.7500 in; top/left marked reference	19.5000 in; 21.0000 in; finished top of stile	first tight; remaining medium / first tight; remaining medium
J-05/J-06	D-101D to D-101A	10 x 50 mm	1.2500 in; 3.5000 in; 5.7500 in; top/left marked reference	38.2500 in; 40.5000 in; 42.7500 in; finished top of stile	first tight; remaining medium / first tight; remaining medium
J-07/J-08	D-101E to D-101A	10 x 50 mm	1.2500 in; 2.7500 in; top/left marked reference	54.7500 in; 56.2500 in; finished top of stile	first tight; remaining medium / first tight; remaining medium
J-09/J-10	D-101F to D-101A	10 x 50 mm	1.2500 in; 4.0000 in; 7.5000 in; 10.7500 in; top/left marked reference	69.7500 in; 72.5000 in; 76.0000 in; 79.2500 in; finished top of stile	first tight; remaining medium / first tight; remaining medium
J-11/J-12	D-101G to D-101E	6 x 40 mm	1.0000 in; 2.0000 in; top/left marked reference	7.1250 in; 8.1250 in; hinge-side rail shoulder	both tight / first tight; second medium

REFERENCE	FRAME	WEB GATES
Register the DF 500 fence on the same marked show face for both members; never flip a part.	10 mm cutter; 25 mm plunge in both members; first tight, followers medium.	Frame rail to groove >=0.150; G to groove >=0.100; G between mortises >=0.220; receiver between mortises >=0.100.

1. Reconcile the register to 26 production 10 x 50 tenons and 4 production 6 x 40 tenons. Check every joint ID on both mating members.
2. Review sectioned samples under strong light. Record actual centers, plunge, mortise width setting, **every relevant standard-rail tight/medium mortise-to-groove web >=.150**, DF cutter-to-face-glue-line core clearance >=.200, G groove-to-mortise web >=.100, G inter-mortise web >=.220, and rail-receiver inter-mortise web >=.100.
3. Dry-close every individual joint with relieved tenons before the complete assembly. Shoulders must close <=.005 and faces align <=.010.

STOP: A missing measurement, unsectioned critical sample, or undocumented relocated J-06 blocks assembly.

REMAKE: Breakthrough, merged features, subminimum web, or an unpaired coordinate cannot be waived by glue-up success.

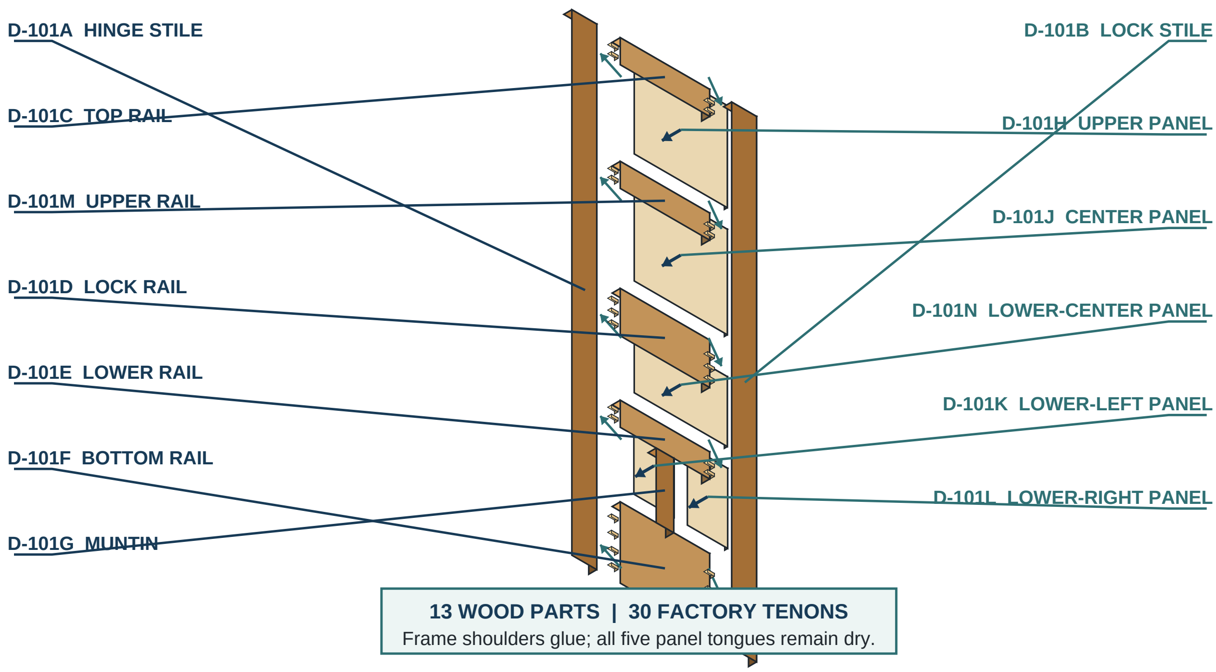
Exploded five-panel door

Reconcile thirteen wood parts, thirty final tenons, five panels, spacers, clamps, and measurement tools.

2 Exploded five-panel door assembly

Every part retains its project identifier; panels move forward and stiles move outward.

13 PARTS



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1. Lay a clean padded assembly platform at least 30 x 84 inches and verify it flat with straightedges and winding sticks. Mark top, hinge, lock, and rail centerlines on removable bench tape.
2. Reconcile 13 finished wood parts: A/B stiles; C/M/D/E/F rails; G muntin; H/J/N/K/L panels. Place all SF marks upward in final orientation.
3. Reconcile five panel bags totaling 20 compression-tested, 3/4-inch-long foam production spacers plus a separate bag of 10 retained test/spare pieces, the red relieved dry tenons, and the sealed production bags containing 26 large and 4 small untouched tenons.
4. Arrange clamps at C, M, D, E, and F rail lines; add padded cauls, two diagonal tapes, winding sticks, squares, dead-blow block for caul adjustment only, timer, brushes, and cleanup tools.
5. Place H between C/M, J between M/D, N between D/E, K/L between E/F around G. Confirm vertical grain and the intended K/L left-right order.
6. Check every groove and tongue for debris, every corner relief for clearance, and every joint label against the register before inserting a dry tenon.

STOP: A missing part, spacer, test result, measuring tool, helper, or clamp postpones the rehearsal.

REMAKE: A panel or frame part that has been damaged in storage returns to its dimensional gate; visual orientation errors are corrected before assembly.

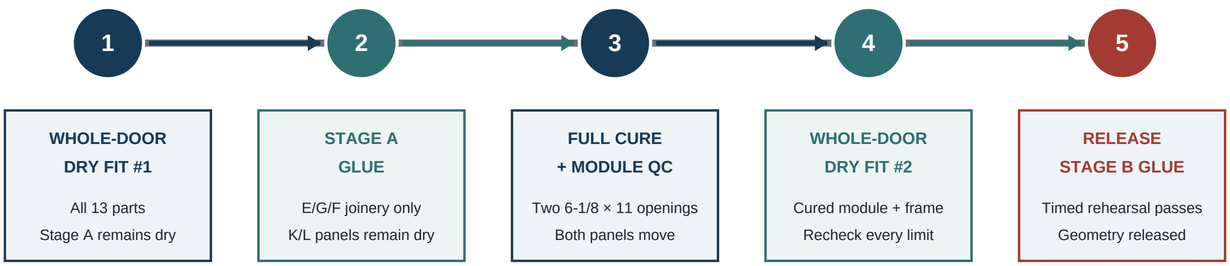
Prove the whole door before Stage A

Keep D-101E/G/F unglued and prove the complete slab, all panels, sequence, square, and twist.

7A Two whole-door dry fits bracket Stage A

The complete door is dry assembled once before lower-module glue and again after that module cures.

TWO GATES



EACH WHOLE-DOOR DRY FIT

- Diagonals differ ≤0.0625
- Twist ≤0.03125
- Shoulder gaps ≤0.005
- All five panels move by hand

TIMING RELEASE

REHEARSED STAGE B TIME

MINUTES ____________

FINAL REHEARSAL GATE
≤ 0.75 × CURRENT PUBLISHED OPEN

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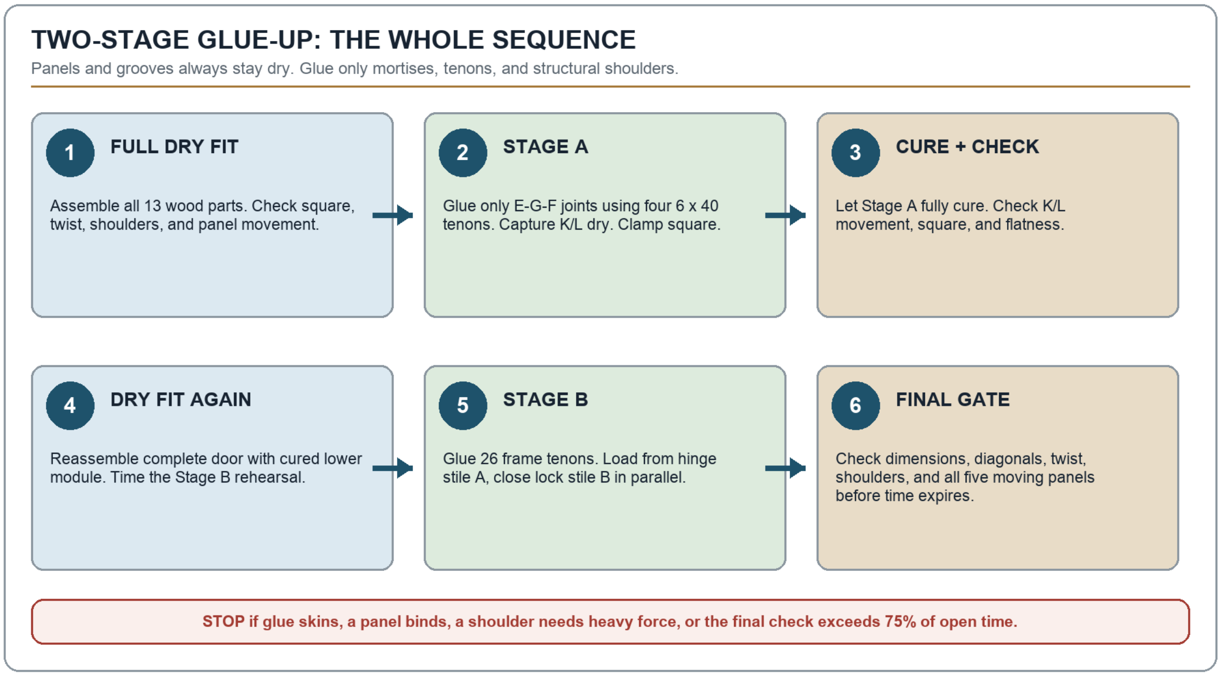
1. Keep E/G/F and K/L completely unglued. Before either panel enters G, retain two approved 3/4-inch-long foam spacers near the upper/lower thirds of each G-side groove using the approved dry method; assemble the lower module with red 6 x 40 dry tenons, capture K/L, and keep the E/F rail ends and K/L outer tongues accessible.
2. Place D-101A SF up and insert red 10 x 50 dry tenons at the C, M, D, E, and F positions. Before any panel tongue enters A, retain two approved 3/4-inch-long foam spacers by the same approved dry method in each A-side groove segment for H, J, N, and K.
3. Seat C, M, and D at their story-stick shoulders. From the open B side, slide H through the C/M grooves and J through the M/D grooves until their outer tongues enter A and compress the staged spacers.
4. Seat the dry E/G/F module into A as one unit so the E/F tenons and K outer tongue enter together; do not seat F a second time. Then slide N through the D/E grooves until its outer tongue enters A.
5. Retain two approved 3/4-inch-long foam spacers in each B-side groove segment for H, J, N, and L before bringing B parallel over all five rail ends and four exposed outer panel tongues. Apply only light rail-line clamp pressure.
6. Verify the 23-7/8 x 80-5/8 x 1-3/8 prefit perimeter, the 80-1/2-inch final story-stick stack inside it, diagonal difference <=1/16, twist <=1/32, shoulders <=.005, and faces <=.010. Confirm the scribed final rectangle will not exceed 23-3/4 x 80-1/2. Move all five panels laterally by hand and confirm the complete door can be disassembled in the marked reverse order.

STOP: Any failure is corrected and the entire first dry fit repeated before Stage A glue.

REMAKE: A panel that needs clamp pressure, a joint that closes only by bowing a stile, or a structural dimension outside tolerance is not “worked in.”

Two-stage glue-up at a glance

Dry-fit first, glue the lower module, cure and dry-fit again, then glue the full frame while every panel remains dry.



Step	Action	Check
1	Dry-fit the complete unglued door.	Square, twist, shoulders, and panels pass.
2	Glue Stage A: E/G/F only.	Use four 6 x 40 tenons; K/L remain dry.
3	Cure and inspect the lower module.	K/L still move; module is square and flat.
4	Dry-fit the full door again and time it.	Finish inside 75% of open time, keeping a 25 percent buffer.
5	Glue Stage B frame joints.	Use 26 large tenons; all panel grooves stay dry.
6	Complete the final timed geometry check.	All five panels move before the time gate.

STOP: If the timed rehearsal exceeds 75 percent of current published OPEN time, divide labor, improve staging, or select an approved adhesive/process; do not simply work faster with glue.

REMAKE: Contaminated mortises or production tenons used in rehearsal are replaced before Stage A.

Time and assemble the lower module

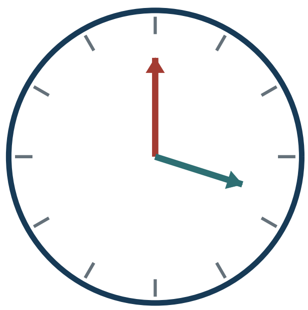
Glue only E/G/F joinery, capture K/L dry, close shoulders, square, and record elapsed time.

8A Stage A glue clock and bench staging

Record the adhesive limit; preload four 3/4-in-long foam panel spacers; keep K/L dry.

STAGE A

A SIGNED TIME GATE



CURRENT PUBLISHED OPEN TIME ___ min

REHEARSAL FINAL GATE ___ min

LIVE FINAL GATE ___ min

BOTH FINAL GATES ≤ 0.75 × PUBLISHED

CLAMP COMPLETE = CHECKPOINT; TIMER CONTINUES

B BENCH MAP / STAGE A

BLUE = PANEL + GROOVE DRY

GREEN = J-11 / J-12 ADHESIVE

PRESET CAULS + G CLAMP • REHEARSE

START TIMER AT FIRST GLUE APPLICATION

STOP ONLY AFTER SQUARE + K/L MOVEMENT PASS

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1. Before starting the timer, retain two approved 3/4-inch-long foam spacers near the upper/lower thirds of each dry G-side groove. Confirm all four are secure, recover freely, and remain outside both J-11/J-12 glue zones.
2. Start the timer at first adhesive contact. Brush a controlled continuous film on both J-11 mortise walls and its two production tenon faces, then seat G into E by hand; avoid hydraulic overfill and keep the 6-1/8–9-1/8-inch footprint centered.
3. Slide K/L from the open F side with grain vertical so their tongues enter the E groove and the preloaded G grooves together. Leave both outer panel edges unspaced until the stiles are installed in Stage B.
4. Glue J-12 mortise walls and the remaining two production tenons, then lower F squarely over G and both panel tongues. Keep all panel tongues and horizontal grooves dry.
5. Apply padded cauls and the pre-positioned clamp at G only until both shoulders close. Record **clamp complete as a checkpoint only**; the live timer continues. Check the E/F shoulders remain 15-1/4 inches long and G occupies 6-1/8–9-1/8, leaving two 6-1/8-inch clear openings.
6. Complete the square/opening check and confirm K/L still shift across grain, then record the **final geometry/panel-movement time**. Stage A passes only when that final time is <=75 percent of the current published open time. Clean accessible squeeze-out without washing oak pores after the gate is recorded.

STOP: Glue skinning, expired open time, a dropped/dirty tenon, or panel bonding requires immediate safe disassembly before set.

REMAKE: A cured joint forced closed after the time limit, or adhesive in a panel groove that cannot be fully removed, fails Stage A.

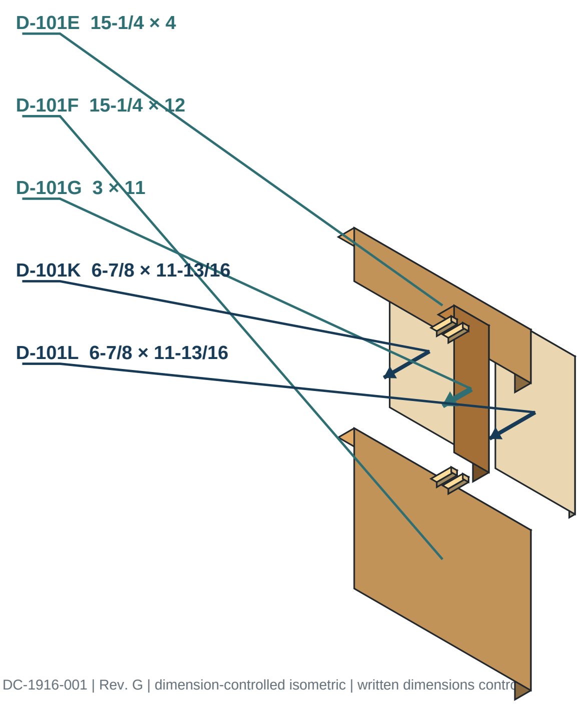
Cure and inspect the lower module

Verify size, square, flatness, web integrity, panel travel, and full cure before reuse.

6 Lower rail-and-muntin module

D-101E, D-101G, and D-101F form the only cured subassembly before the full frame glue-up.

STAGE A



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Stage A / controlled sequence

- Four 6 × 40 mm factory tenons
- Glue E/G/F mortises and tenons only
- Preload 4 foam spacers - 3/4 in long; K/L dry
- Clamp two clear 6-1/8 × 11 openings
- Cure fully; recheck panel travel

VERTICAL STACK

BOTTOM RAIL 12.000
CLEAR OPENING 11.000
LOWER RAIL 4.000
TOTAL 27.000

1. Leave Stage A clamped, flat, and undisturbed for at least the recorded minimum clamp time; then support it flat through the full stated cure before load testing or surfacing.
2. After clamp removal, inspect J-11/J-12 in raking light. Measure shoulder gap <=.005, adjacent face flush <=.010, G square to E/F, and equal 6-1/8-inch bottom openings.
3. Measure both module diagonals from rail-shoulder corners and check twist with short winding sticks. Record actuals rather than forcing the module against the bench.
4. Move K/L across grain. Confirm each has two spacers at G, remains unbonded, and will accept its outer two spacers during full-door assembly.
5. Remove hardened squeeze-out only from accessible frame surfaces. Do not scrape inside grooves in a way that enlarges them or cuts the prefinished panel tongues.
6. Mark the cured module **STAGE A ACCEPTED**, date, adhesive lot, cure conditions, and checker initials.

STOP: Keep the module on hold if full cure is uncertain or if a panel does not move.

CORRECT: A surface-only glue bead may be carefully removed without changing geometry.

REMAKE: Open structural shoulders, module twist that cannot meet the full-door limit, a bonded panel, or a displaced G requires rebuilding the module.

Repeat with the cured lower module

Reassemble the complete door and release the exact Stage B loading and clamp sequence.

7A Two whole-door dry fits bracket Stage A

The complete door is dry assembled once before lower-module glue and again after that module cures.

TWO GATES

1. **WHOLE-DOOR DRY FIT #1**
   All 13 parts
   Stage A remains dry
2. **STAGE A GLUE**
   E/G/F joinery only
   K/L panels remain dry
3. **FULL CURE + MODULE QC**
   Two 6-1/8 × 11 openings
   Both panels move
4. **WHOLE-DOOR DRY FIT #2**
   Cured module + frame
   Recheck every limit
5. **RELEASE STAGE B GLUE**
   Timed rehearsal passes
   Geometry released

EACH WHOLE-DOOR DRY FIT

- Diagonals differ ≤0.0625
- Twist ≤0.03125
- Shoulder gaps ≤0.005
- All five panels move by hand

TIMING RELEASE

REHEARSED STAGE B TIME

MINUTES ____________

FINAL REHEARSAL GATE
≤ 0.75 × CURRENT PUBLISHED OPEN

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1. Re-stage the flat platform and inspect the cured E/G/F module. Use red relieved 10 x 50 tenons only; all 26 production tenons stay sealed.

2. Before loading, retain two approved 3/4-inch-long foam spacers in each A-side groove segment for H, J, N, and K. Seat C, M, and D into A, then slide H and J through their paired rail grooves until their outer tongues enter A.

3. Seat the cured E/G/F module into A so E/F tenons and K's outer tongue enter together; then slide N through D/E until its outer tongue enters A.

4. Retain two approved 3/4-inch-long foam spacers in each B-side groove segment for H, J, N, and L, then bring B over every rail end and exposed panel tongue parallel to A. Close lightly at all five rail lines.

5. Repeat every full-door check: 23-7/8 x 80-5/8 prefit perimeter; 80-1/2-inch final story-stick stack; 23-3/4 x 80-1/2 maximum final scribe rectangle; diagonals <=1/16 difference; twist <=1/32; shoulders <=.005; faces <=.010. Confirm all five panels retain residual lateral movement.

6. Time a complete simulated Stage B sequence from first glue location through the final clamp, square, twist, and five-panel movement check. It must finish within 75 percent of the adhesive's current published open time.

7. Mark final loading order, brush ownership, clamp order, helper positions, preloaded spacer locations, and safe disassembly plan.

STOP: Any failure repeats this complete dry fit after correction.

REMAKE: Do not plane a cured Stage A shoulder, thin a panel tongue, or widen a mortise merely to make the full frame close.

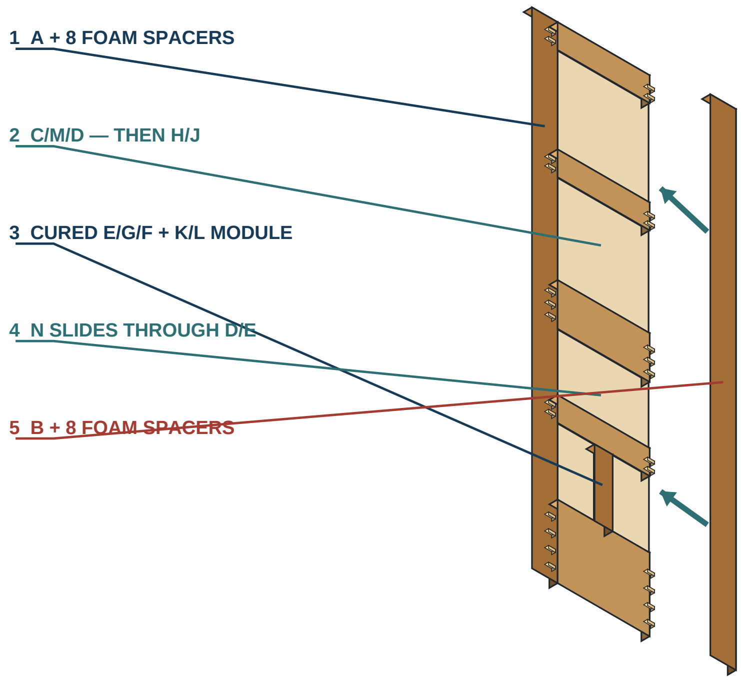
Timed adhesive application and loading

Load from the hinge stile, keep every panel dry, and close D-101B parallel across all joints.

7 Load the frame from the hinge stile

Stage 3/4-in-long foam panel spacers before loading parts; close D-101B evenly.

STAGE B



CONTROLLED CLOSURE

- Stage 3/4-in-long foam spacers first.
- Seat the cured module as one unit.
- Start every lock-side tenon and tongue.
- Keep A and B parallel.
- No clamp force to start a joint.

SECOND WHOLE-DOOR DRY FIT

Required after Stage A has fully cured.

Release only after panels move by hand.

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1. Confirm the timed rehearsal passed, the platform is flat, and fresh adhesive/open-time conditions match the recorded plan. Count 26 untouched 10 x 50 production tenons. Before timing, retain two approved 3/4-inch-long foam spacers in each A-side H/J/N/K groove and each B-side H/J/N/L groove, clear of all structural mortises.
2. Start the timer at first adhesive contact. Brush complete films on the A-side rail/stile mortise walls and corresponding tenon faces; avoid mortise-bottom overfill and every panel groove.
3. Insert the production tenons, then seat C, M, and D at their marked A shoulders. Slide H and J through their paired rail grooves until their outer tongues enter A; keep each rail square and SF aligned.
4. Seat the cured E/G/F module into A so the E/F tenons and K outer tongue enter together. Slide N through D/E into A and verify every tongue enters without shaving or glue contact.
5. Apply adhesive to B-side structural mortises/tenons, keeping the preloaded B spacers dry. Bring B parallel over all five rail ends and the exposed H/J/N/L tongues.
6. Seat all rail lines progressively by hand and light clamp pressure in the rehearsed order. Record clamp complete as a checkpoint only; do not stop the live timer before final geometry and five-panel movement are verified.
7. Count unused/rejected tenons and verify exactly 26 large tenons entered the ten specified frame joints.

STOP: A skinned glue film, expired time, contaminated tenon, misloaded panel, or joint requiring heavy force triggers immediate safe disassembly while possible.

REMAKE: Never inject glue later into a starved joint or keep an assembly that closed only after the published time.

Clamp, square, and remove twist

Work from light pressure, correct the long diagonal, and verify winding sticks before final pressure.

8 Clamp, square, and remove twist

Keep the Stage B timer live through final geometry and five-panel movement; clamp complete is only a checkpoint.

GEOMETRY

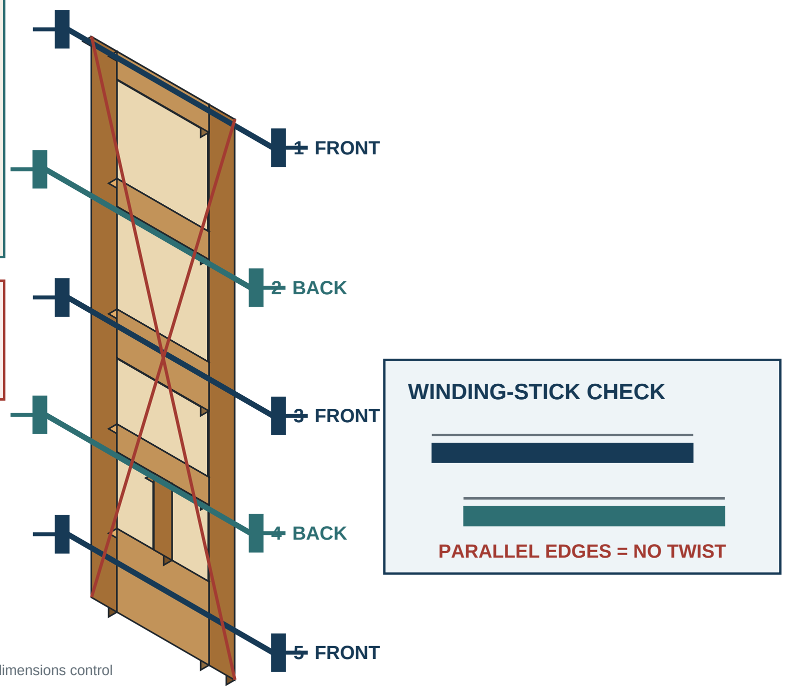
Live final gate

- Diagonal A - B <= 1/16
- Twist <= 1/32 with winding sticks
- Shoulder gaps <= 0.005
- All five panels move laterally
- Clamp complete = checkpoint only
- Final gate <=75% of published OPEN time

DIAGONAL MAP

A TOP HINGE TO BOTTOM LOCK

B TOP LOCK TO BOTTOM HINGE



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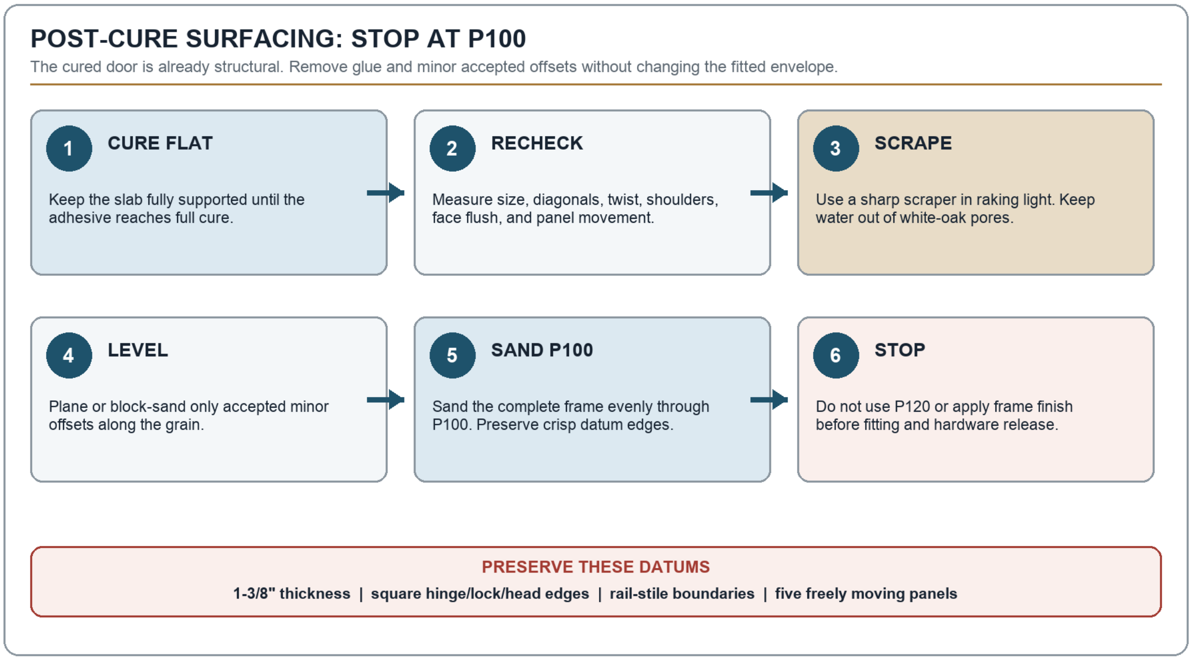
1. Place one padded clamp at each C, M, D, E, and F rail line. Alternate clamp bars above/below the slab or use balanced cauls so force passes through the door's mid-thickness.
2. Bring every clamp only to shoulder contact. Measure width at top/middle/bottom; correct parallelism before adding pressure.
3. Measure both diagonals. If one is longer, apply a controlled diagonal pull between that diagonal's corners or shift the lightly clamped frame until the difference is <=1/16; recheck shoulders after every adjustment.
4. Set winding sticks near top/bottom and across intermediate rail lines. Correct a high corner with a padded caul to the verified flat platform; never twist by crushing a panel or overloading one joint.
5. Confirm the 23-7/8 x 80-5/8 prefit perimeter, complete 80-1/2-inch final story-stick stack, twist <=1/32, shoulders <=.005, and adjacent faces <=.010 while adhesive remains workable. Verify the final 23-3/4 x 80-1/2 rectangle remains available inside the prefit perimeter.
6. Move every panel laterally and make the final time-stamped geometry/panel-movement record. Stage B passes only when this final timestamp is <=75 percent of the manufacturer's current published open time. Then clean accessible squeeze-out; do not disturb accepted clamps until minimum clamp time passes.

STOP: If the final geometry/panel-movement check cannot finish within 75 percent of current published open time, the run fails this release gate; follow the adhesive maker's safe disassembly limits and do not keep tightening.

REMAKE: A cured door beyond diagonal/twist limits or with a bonded panel fails even if its outer size is correct.

Scrape, flush, and sand through P100

Recheck geometry, remove glue, level accepted offsets, and stop at P100 until fitting and hardware machining pass.



Step	Action	Check
1	Allow full cure on a flat support.	Temperature and humidity are recorded.
2	Remove clamps and recheck geometry.	Size, diagonals, twist, shoulders, and panels pass.
3	Scrape glue in raking light.	Do not flood white oak with water.
4	Level only accepted minor offsets.	No hollow or rounded datum edge.
5	Sand uniformly through P100 and stop.	P120 waits for fitting and hardware release.

STOP: Do not sand to P120 or apply frame finish before completed-jamb fitting and all hardware dry machining/release.

REMAKE: Structural shoulder opening, hidden glue-starved joint, excessive flush correction, or irreversible hollow at an edge requires formal rejection/rebuild.

Calculate the installed slab

Move from the 24 x 81 jamb to the prefit slab and then the 23-3/4 x 80-1/2 fitted slab without mixing the three states.

FROM JAM OPENING TO FINISHED DOOR

Keep the jamb, prefit slab, and fitted slab as three separate measurements.

1. **REMEASURE** Confirm the finished jamb is still 24 x 81 at all required locations.
2. **CALCULATE** Width: 24 - 1/8 - 1/8 = 23-3/4. Height: 81 - 1/8 - 3/8 = 80-1/2.
3. **COMPARE** Overlay the fitted rectangle on the 23-7/8 x 80-5/8 cured prefit slab.
4. **MARK** Record hinge, head, lock, and bottom removal. No single edge may require more than 1/16.
5. **FIT IN ORDER** Hinge edge -> head -> lock edge/bevel -> bottom.

Fitted-slab control	Confirmed / shop input	Calculation / transfer	Acceptance
Width	24-in clear jamb; 1/8 each side	24 - 1/8 - 1/8 = 23-3/4	Maximum rectangle; even reveals
Height	81-in clear jamb; 1/8 head; 3/8 bottom	81 - 1/8 - 3/8 = 80-1/2	Head and bottom clearances pass
Prefit rectangle	Cured assembly 23-7/8 x 80-5/8	Scribe from final top and hinge datums	1/16 removable stock per edge
Hinge edge	Jamb plumb/twist	Fit first; preserve reference face	Full leaf bearing
Lock edge	Handing and swing	Apply bevel inside 23-3/4 maximum width	1/8 reveal through swing

ORDER	ALLOWANCE	REDESIGN
Hinge edge, head, lock edge, then bottom.	Remove no more than the controlled 1/16 inch per prefit edge.	A larger change requires geometry revalidation.

STOP: Excess removal, unclear operating gaps, moving jambs, or a floor change requires revised approval.

REMAKE: An incorrectly calculated edge cannot be restored with veneer, filler, or a wider stop.

Fit the hinge edge and head

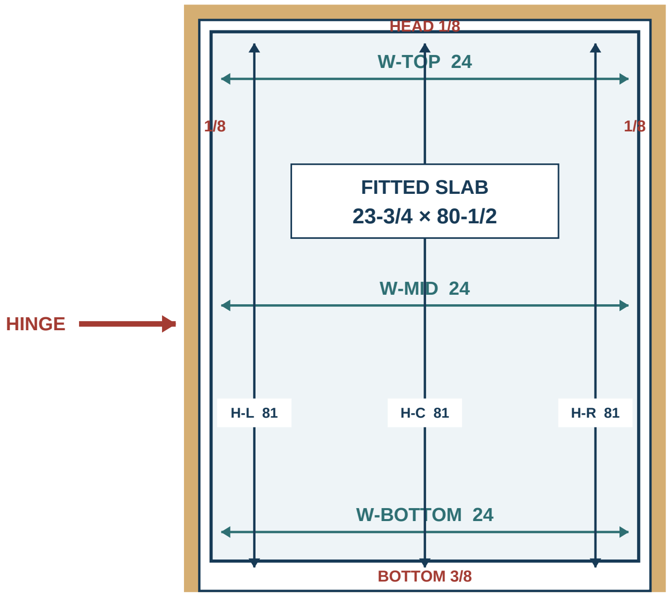
Establish the hinge datum first, then transfer the head condition and test frequently.

9A Confirmed jamb and fitted-slab gap map

The 24 x 81 clear jamb is fixed; the 23-3/4 x 80-1/2 slab resolves the written operating gaps.

CONTROLLED FIT

CONFIRMED INSTALLED JAMB / FRONT ELEVATION



CONTROLLED FIT + ORDER

1. VERIFY 24 × 81 AT 3 STATIONS
2. HINGE EDGE / PLUMB DATUM
3. HOLD 1/8 HEAD + BOTH SIDES
4. HOLD 3/8 ABOVE FINISHED FLOOR
5. REHANG + FULL-SWING TEST

DO NOT REWORK AN OBSOLETE SLAB
BUILD THE REV. G FRAME GEOMETRY
23-3/4 × 80-1/2 MAX ENVELOPE

GAP GRAPHICS EXAGGERATED / NTS

FINISHED FLOOR DATUM | 3/8 BOTTOM GAP | WRITTEN DIMENSIONS CONTROL

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1. Support the slab vertically in a padded door cradle with the hinge edge up. Use a long straightedge and the signed fit story stick to mark the finished hinge line.
2. Plane the hinge edge straight in full controlled passes from the appropriate end to avoid breakout. Check width at top/middle/bottom after each pass; keep the edge square unless the released hinge geometry requires otherwise.
3. Place the slab in the opening on temporary **3/8-inch floor blocks**. Register the fitted hinge edge at a measured **1/8-inch operating clearance** and transfer the head condition at both corners for a **1/8-inch head clearance**.
4. Remove and support the slab with the head edge accessible. Connect only the verified transfer points; preserve a field-required taper rather than forcing the head square to an out-of-square jamb.
5. Plane to the line with backing at the lock corner. Return the slab to the opening frequently and gauge the head and hinge clearances at **1/8 inch each** through the swing start.
6. Mark the accepted hinge/head dimensions and protect those edges from later “cleanup” passes.

STOP: A hollow greater than the released clearance over any 12-inch edge length or more than 1/16 removal requires review.

REMAKE: Cutting past the installed envelope, tearing through a hinge-gain zone, or rounding the hinge datum requires engineered repair or slab rejection.

Fit lock edge, bevel, and undercut

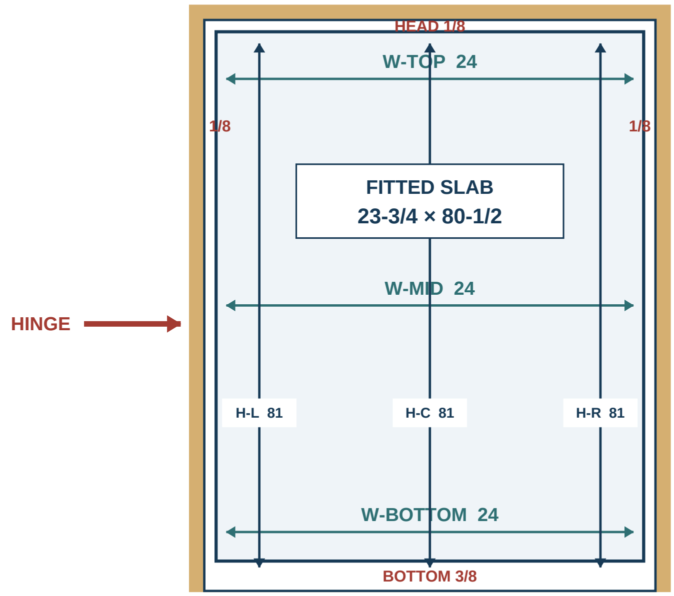
Carry the measured closing bevel and floor datum while staying inside the released removal allowance.

9A Confirmed jamb and fitted-slab gap map

The 24 x 81 clear jamb is fixed; the 23-3/4 x 80-1/2 slab resolves the written operating gaps.

CONTROLLED FIT

CONFIRMED INSTALLED JAMB / FRONT ELEVATION



CONTROLLED FIT + ORDER

1 VERIFY 24 × 81 AT 3 STATIONS

2 HINGE EDGE / PLUMB DATUM

3 HOLD 1/8 HEAD + BOTH SIDES

4 HOLD 3/8 ABOVE FINISHED FLOOR

5 REHANG + FULL-SWING TEST

DO NOT REWORK AN OBSOLETE SLAB
BUILD THE REV. G FRAME GEOMETRY
23-3/4 × 80-1/2 MAX ENVELOPE

GAP GRAPHICS EXAGGERATED / NTS

FINISHED FLOOR DATUM | 3/8 BOTTOM GAP | WRITTEN DIMENSIONS CONTROL

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1. With hinge edge/head accepted, hold the slab on temporary hinges or positioning blocks and transfer the **1/8-inch lock-jamb clearance** at top, lock, and bottom. Mark the latch location and retained J-06/lock buffer.

2. Use a bevel gauge or full-size template to establish only the bevel needed for the actual swing and jamb geometry. Mark both face limits so the faceplate remains fully supported.

3. Clamp the slab in the cradle and plane the lock edge in long passes, maintaining the released bevel without a mid-edge hollow. Check with the bevel gauge after every pass.

4. Set the slab on **3/8-inch floor blocks** and scribe the bottom from the finished floor/threshold through the complete swing path. The released bottom operating gap is **3/8 inch**; do not substitute a carpet estimate.

5. Support both bottom corners and cut/plane to the line without breakout. Preserve stile width and the lower J-09/J-10 structure.

6. Trial the slab in the opening and record installed width at top/middle/bottom and height at both sides.

STOP: Binding local to jamb twist is diagnosed before more wood is removed.

REMAKE: Excessive bevel that exposes lock machining, undercut beyond approval, or fitting into a Domino/clear-wood buffer requires structural disposition, not filler.

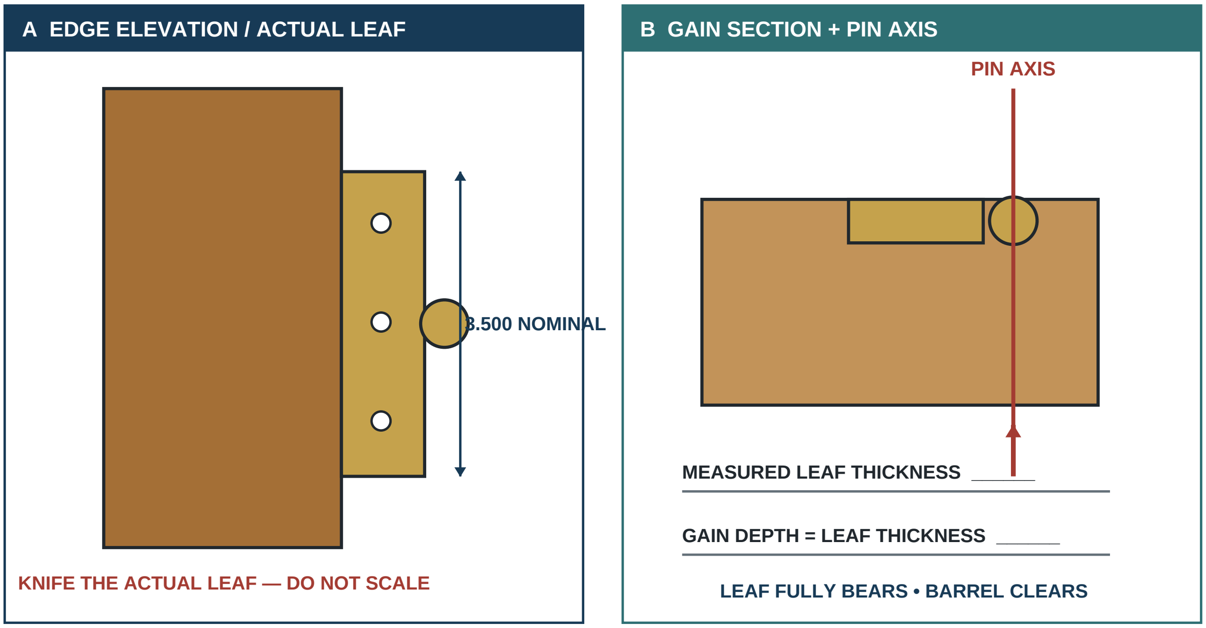
Lay out and fit the actual leaves

Knife physical leaves, rout slightly shallow, pare to flush, drill centered pilots, and use steel setup screws.

9B Hinge-gain transfer from the physical leaf

The measured brass leaf controls gain depth, screw pattern, pin axis, and barrel projection.

HINGE DETAIL



STEEL SETUP SCREWS FIRST | BRASS FINAL SCREWS AFTER FIT + FINISH

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1. Use the physical 3-1/2-inch hinges. Record each leaf outline, thickness, barrel offset, screw root diameter, screw length, and any variation among leaves.
2. Transfer released hinge locations from sound existing jamb gains or the approved hardware plan to the fitted slab hinge edge. Knife directly around the assigned leaf; keep each barrel on the common axis.
3. Clamp the slab in a stable edge-up cradle. Rout or chisel each gain in controlled shallow passes to actual leaf thickness; square corners only to match the physical leaf.
4. Fit the leaf flush within .005 inch with full bearing and no rocking. Align all barrels with a straightedge/string before drilling.
5. Use a self-centering guide and depth stop for pilot holes sized to the measured steel setup-screw root. Wax if compatible and install matching steel screws by hand for dry hanging.
6. Fit the jamb leaves by the same physical transfer and depth method. Do not use brass screws until final installation.

STOP: Split fibers, wandering pilots, loose jamb substrate, or barrel-axis disagreement requires correction before hanging.

REMAKE: An oversized/deep gain receives a fitted matching-oak Dutchman and complete remachining; if full screw purchase or stile structure cannot be restored, reject the slab.

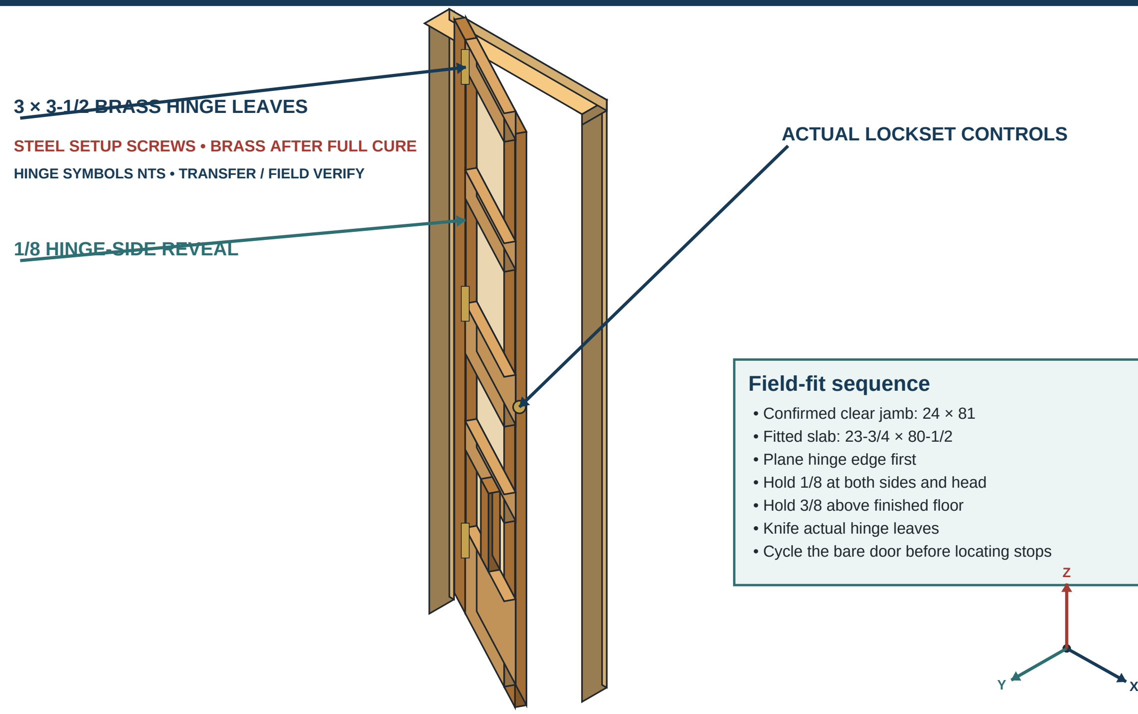
Diagnose swing and reveals

Cycle the unfinished slab through the full arc and correct hinge-axis or edge geometry by cause.

9 Fit, hinge, and dry-cycle the bare slab

The completed jamb controls prefinish edge fit and hardware transfer; use steel setup screws.

HARDWARE



Field-fit sequence

- Confirmed clear jamb: 24 × 81
- Fitted slab: 23-3/4 × 80-1/2
- Plane hinge edge first
- Hold 1/8 at both sides and head
- Hold 3/8 above finished floor
- Knife actual hinge leaves
- Cycle the bare door before locating stops

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1. With a helper and protected floor, hang the slab on steel setup screws. Support its weight until every screw has adequate thread engagement.
2. Remove floor blocks and measure operating clearances at the hinge edge, head, lock edge, and bottom. Require **1/8 inch at the hinge edge, head, and lock edge within +/-1/32 inch**, and **3/8 inch at the bottom within +/-1/16 inch**. Record at least top/middle/bottom; these are operating gaps, not the 3/16-inch casing reveal.
3. Swing slowly through the full arc. Mark contact with removable chalk or paper, not by guessing from a light line.
4. Diagnose before correcting: uniform head taper suggests fit; changing hinge gap suggests barrel-axis/gain depth; lock-edge rub only during swing suggests bevel; bottom contact follows the actual floor path.
5. Check whether the door remains where placed. A slab that falls open/closed indicates hinge-axis or jamb-plumb conditions to document and correct where authorized.
6. Make only the signed corrective pass, rehang, and repeat the complete measurement/swing check. Preserve hardware and joint no-cut zones.

STOP: Do not move the strike, stop, or casing to mask a hanging error.

REMAKE: Repeated planing without a diagnosed contact point, stripped screw purchase, or a hinge repair that cannot hold axis requires engineered correction before lock machining.

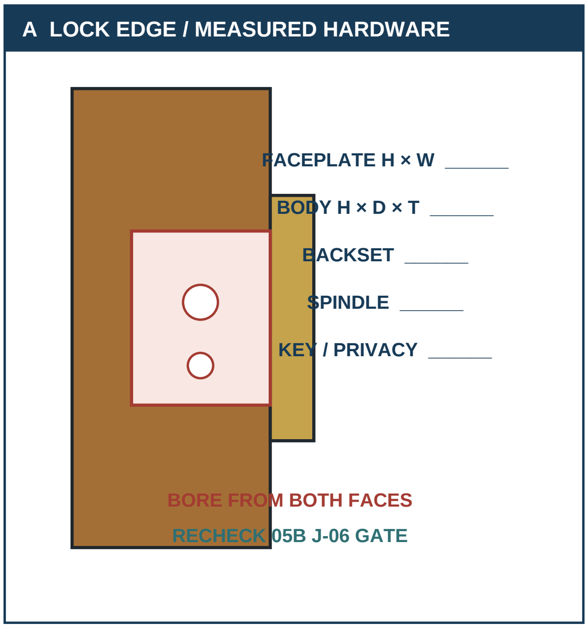
Cut body mortise and faceplate

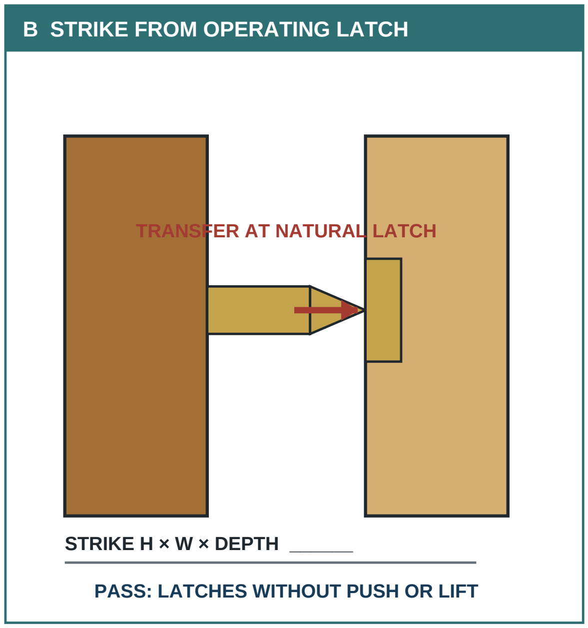
Machine incrementally from the physical lock, protect J-06 clearance, and fit the faceplate flush.

9C Mortise lock and strike transfer

Machine from measured hardware, hang the bare door, then locate the strike from the operating latch.

LOCK + STRIKE





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1. Remove the temporarily released slab and clamp it in a rigid edge-up cradle. Reapply the Page 26 physical lock overlay and confirm the 0.250-inch clear-wood buffer to J-06 remains after fitting.
2. Transfer the actual lock-body height, thickness, backset, depth, and faceplate outline from the hardware—not a generic template. Mark a centerline on the edge and both faces.
3. Use a guided drilling jig, hollow-chisel setup, or plunge-router method appropriate to the physical body. Set positive depth stops and remove waste in shallow, overlapping increments without crossing the envelope.
4. Square and clean the body mortise with sharp hand tools. Test-fit frequently; the body must enter by hand, seat at full depth, and remain aligned without side pressure.
5. Knife the physical faceplate and cut its gain to the actual thickness. Accept it flush within .005 inch with full edge bearing and screw holes centered in sound wood.
6. Remove the lock and inspect the mortise with light and depth gauge. Reconfirm remaining clear wood to every J-06 feature before proceeding.

STOP: Unseen metal, unexpected void, approaching J-06, or a changed lock blocks deeper cutting.

REMAKE: Breakthrough into joinery, loss of the 1/4-inch buffer, or an oversized cavity that cannot retain the body square requires a structural repair design or slab rejection.

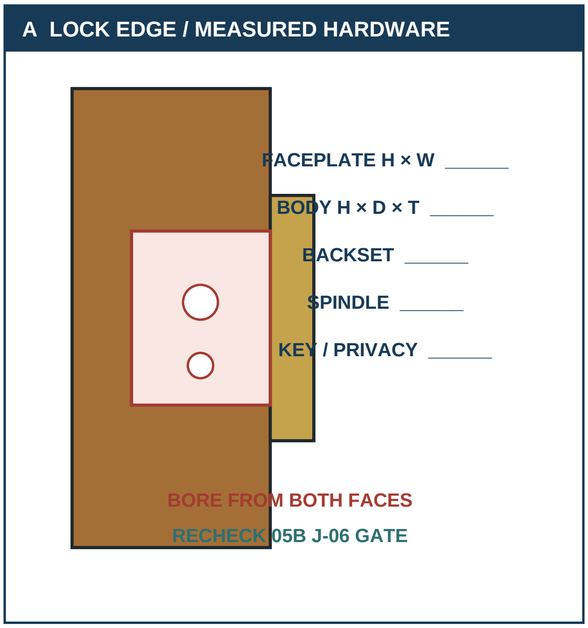
Bore from both faces and transfer the latch

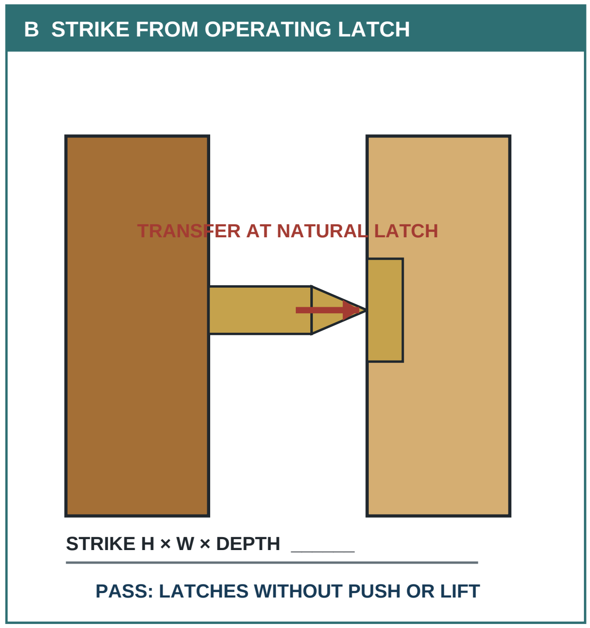
Prevent breakout, locate the strike from the operating latch, and correct sag before enlarging anything.

9C Mortise lock and strike transfer

Machine from measured hardware, hang the bare door, then locate the strike from the operating latch.

LOCK + STRIKE





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1. Temporarily seat the actual lock body and faceplate with steel setup screws. Transfer the spindle and key/privacy centers mechanically from the body to both slab faces; confirm backset independently.
2. Clamp a sacrificial backer to the exit face. Bore each hole perpendicular to its local face from one side only past the centerline, then complete from the opposite transferred center to prevent breakout.
3. Fit spindle, key/privacy parts, escutcheons, and knobs/levers dry. Confirm free movement, full fastener cover, and no hole edge visible outside the physical trim.
4. Rehang the slab on steel screws and install the lock/latch temporarily. Apply removable transfer medium to the latch nose or use a fitted center marker.
5. Close the door naturally without lifting, pushing, or preloading it. Transfer the actual latch center to the strike jamb; knife the physical strike outline and measure its required recess.
6. Cut the strike pocket and plate gain in shallow controlled increments. Establish centered pilots with steel screws, then cycle the latch at least 20 times from both sides.

Accept: Latch enters and releases without lifting/pushing; strike and faceplate are flush within .005; spindle turns without bind.

STOP: Do not enlarge the strike until hinge sag, barrel axis, bevel, and latch projection have been diagnosed.

REMAKE: Breakout beyond escutcheon/strike cover, a hole intersecting the buffered J-06 envelope, or a strike pocket without sound screw purchase requires a fitted wood repair and complete remachining or rejection.

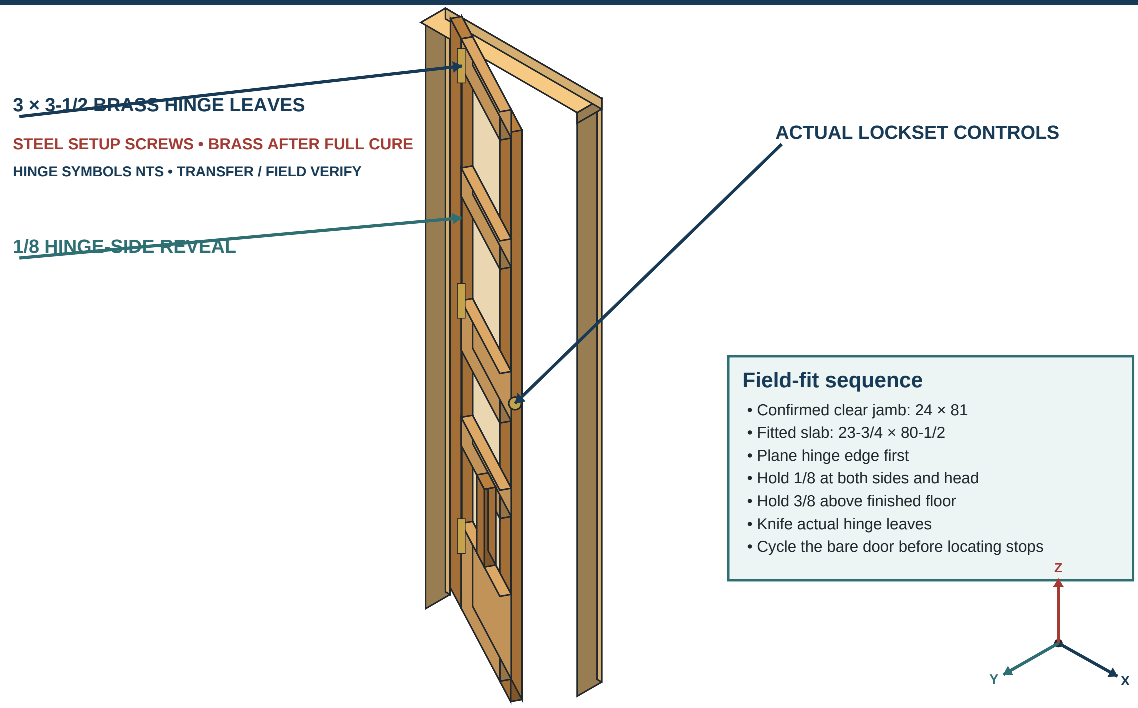
Release the operating door before finish

Confirm full swing, clearances, latch, strike, hardware fit, and panel freedom; then remove hardware.

9 Fit, hinge, and dry-cycle the bare slab

The completed jamb controls prefinish edge fit and hardware transfer; use steel setup screws.

HARDWARE



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1. Install all hinges, lock parts, spindle, escutcheons, knobs/levers, and strike with steel setup screws. Do not install final brass screws or apply frame finish.
2. Hang the slab and measure hinge/head/lock/bottom operating clearances at all recorded locations. Verify **1/8 inch at the hinge edge, head, and lock edge within +/-1/32 inch**, and **3/8 inch at the bottom within +/-1/16 inch**, consistent with Page 43’s released 23-3/4 x 80-1/2-inch envelope.
3. Swing the door through the full arc 20 times, pause it at several angles, and operate the latch/lock 20 times from both sides.
4. Confirm hinge leaves and faceplates are flush <=.005, barrels share one axis, the latch enters without force, the slab neither scrapes nor springs, and the bottom clears the complete floor path.
5. Confirm H/J/N/K/L still move laterally and do not rattle. Inspect all shoulders and the lock rail after hanging load.
6. Photograph accepted reveals, hinge axis, lock/strike, bottom clearance, and panel positions. Record actual results and sign **OPERATING DOOR RELEASED FOR FINISH**.
7. Remove, label, and bag hardware by location; protect machined gains/holes. Any subsequent fitting change voids this release and requires another dry hang.

STOP: No frame, trim, or hardware finish proceeds until this gate passes.

REMAKE: Finish is not a correction for rub, sag, misalignment, rattle, or exposed machining.

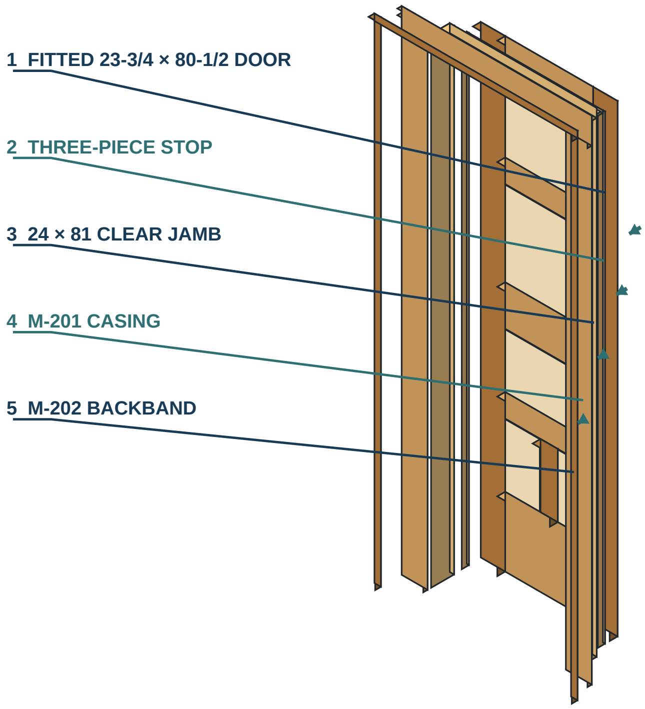
Profiles and layer relationships

Separate the one stop set from face-dependent casing, backband, cap, plinth, and base transitions.

10 Installed door and millwork layer relationship

The spaced layers show fit order and substrate relationships, not a literal exploded installation.

MILLWORK

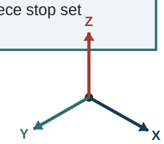


Layer anatomy / inside to out

- 1. Fitted slab: 23-3/4 × 80-1/2
- 2. Three-piece stop set
- 3. Clear jamb: 24 × 81
- 4. M-201 casing at 3/16 reveal
- 5. M-202 backband, 1/4 proud
- Fit gaps: 1/8 sides/head; 3/8 bottom

Face-count rule

- One face: 2 legs + head + 3 bands
- Two faces: double casing and band
- Do not double the three-piece stop set



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1. Reconfirm the released trim face count and head scheme separately for Face A and Face B. Two faces double casing/backband; they do not double the one three-piece stop set.
2. Mill M-201 casing to 3/4 x 4-1/2 inches with a flat face and approximately 1/32-inch easing on approved visible long arrises.
3. Mill M-202 backband to 1 x 1-1/8 inches. Its back shares the wall plane and its face stands 1/4 inch proud of the casing.
4. Mill M-203 stop to 3/8 x 1-1/4 inches; ease only the exposed door-side arris approximately 1/16 inch after the line is proven.
5. Mill M-204 cap only when approved, 7/8 x 1-1/2 inches. Define **P = 3/4-inch end projection beyond the completed M-202C head backband**; cap finished length is the actual accepted M-202C outside length + 2P. M-204 bears on M-202C and never substitutes for it. Mill M-205 plinth only from a field-approved profile; 7/8 inch is a provisional thickness, not design authority.
6. Produce and label a full-size sample of every used profile and corner treatment. Compare projection, easing, ray figure, and head language with historic evidence.
7. Use standard ripping/crosscutting blades and controlled router/hand-tool relief. The default moulding requires **no dado blade**.

STOP: A second face, cap, or plinth without written release is scope creep, not extra yield.

REMAKE: A production run that differs from the approved profile sample in section, easing, or show-face orientation is rejected.

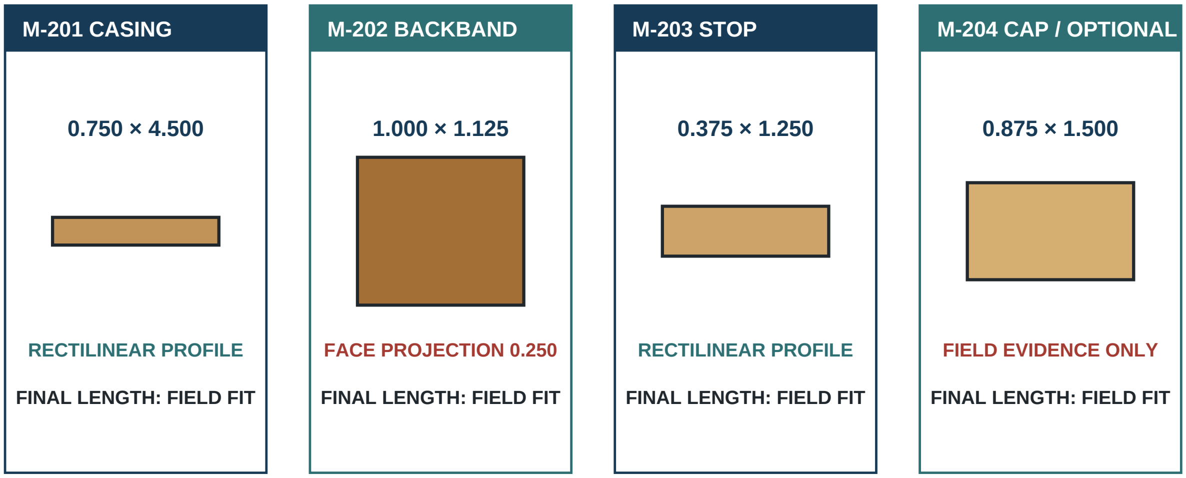
Mill matched moulding families

Machine casing, backband, stop, and optional parts as labeled families with full-size approved samples.

10A Custom moulding stock and milling sequence

Mill each profile family to its controlled rectangular section; final lengths remain field transfers.

PROFILE MILLING



SAFE MACHINING SPLIT

Rip / crosscut with standard blades; plane broad mother blanks before ripping narrow stop strips.

No dado blade is required for default moulding profiles. Local back relief uses router or hand tools.

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1. Allocate straight quarter-sawn stock by face and visible family. Mark RF, RE, show face, wall edge, jamb edge, top, and left/right before milling.
2. Joint one face/edge, plane to thickness, rest stickered stock, then final-plane all members in a family together. Keep paired casing legs within .005 inch thickness.
3. Rip casing and backband from stable wide blanks with continuous support. For M-203, plane and edge-profile a wide mother blank before ripping individual 1-1/4-inch strips.
4. Crosscut only one square reference end initially. Preserve the field formula plus 2-inch rough allowance at the other end until the member is transferred at the opening.
5. Ease profiles with a finely set hand plane, sanding block, or guarded router setup proven on the profile sample. Use backers at end grain and preserve square glue/joint lands.
6. Check section at both ends and center, straightness against a long reference, face quality in raking light, and grain pairing in installed order.
7. Bundle by Face A/Face B and layer M-201/M-202/M-203. Store flat with formula cards attached.

STOP: Do not pull bowed casing flat with nails or mill a narrow stop freehand.

REMAKE: A member below section, with unsafe short grain at a miter, or unable to bear flat after authorized relief is replaced.

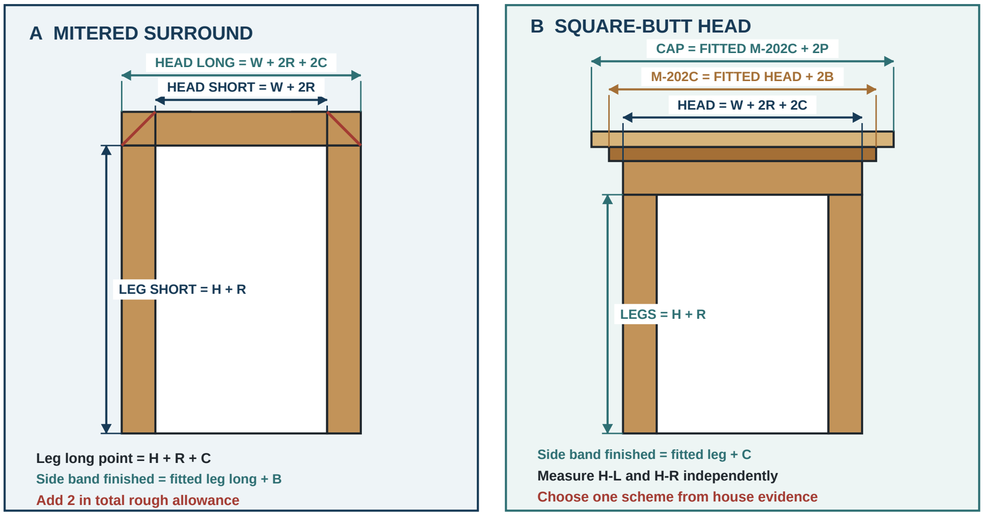
Face-specific formula register

Calculate each face from its own W, H-L, H-R, reveal, casing width, joint scheme, and datum.

4 Transfer formulas

Arithmetic sets rough stock; the fitted opening controls every final cut.

FORMULAS



R = 3/16 reveal | C = 4-1/2 casing | B = 1-1/8 backband | P = 3/4 cap overhang

DC-1916-001 | Rev. G | quarter-sawn white oak | written dimensions control

Part	Production rough-length formula
M-201A/B	longest required point + 2; mitered long point = H side + R + C + 2
M-201C	W + 2R + 2C + 2
M-202A/B	mitered: fitted side-casing outer long + B + 2; square: fitted side-casing length + C + 2
M-202C	fitted head-casing outside width + 2B + 2
M-203A-C	jamb member + 2; fit head first and sides beneath
M-204	actual fitted head-backband outside width + 2P + 2
M-205A/B	field dimension + 0.25

FACE A	FACE B	TRANSFER
Use its own W, H-L, H-R, joint, cap, plinth, and floor datum.	Recalculate independently; do not copy Face A leg lengths.	Final cuts come from installed work after each preceding layer is fitted.

1. For each face define **W** = released clear jamb width at head; **R** = 3/16 casing reveal; **C** = 4-1/2 casing width; **H-L/H-R** = that face's floor or approved plinth-top datum to underside of head jamb.
2. For a mitered surround calculate head inside short point = **W + 2R** and head outside long point = **W + 2R + 2C**.
3. Calculate mitered left-leg outside long point = **H-L + R + C** and right-leg outside long point = **H-R + R + C**.
4. For a square-butt head calculate left leg = **H-L + R**, right leg = **H-R + R**, and head overall = **W + 2R + 2C**.
5. Add the standard **2-inch total rough-length allowance** only after selecting the formula. Mark the calculated target and rough cut separately on each member.
6. Use formulas to select stock and establish safe rough cuts. Transfer final joint lines from the fitted jamb/casing because wall projection, scribe, and out-of-square conditions remain field-specific.

Inspection: Recalculate independently and compare units, face, bottom datum, and scheme before cutting.

STOP: Do not mix H-L/H-R across faces or mix mitered and square formulas on one face.

REMAKE: A formula-cut member shorter than the physical transfer line is replaced; joint filler is not added length.

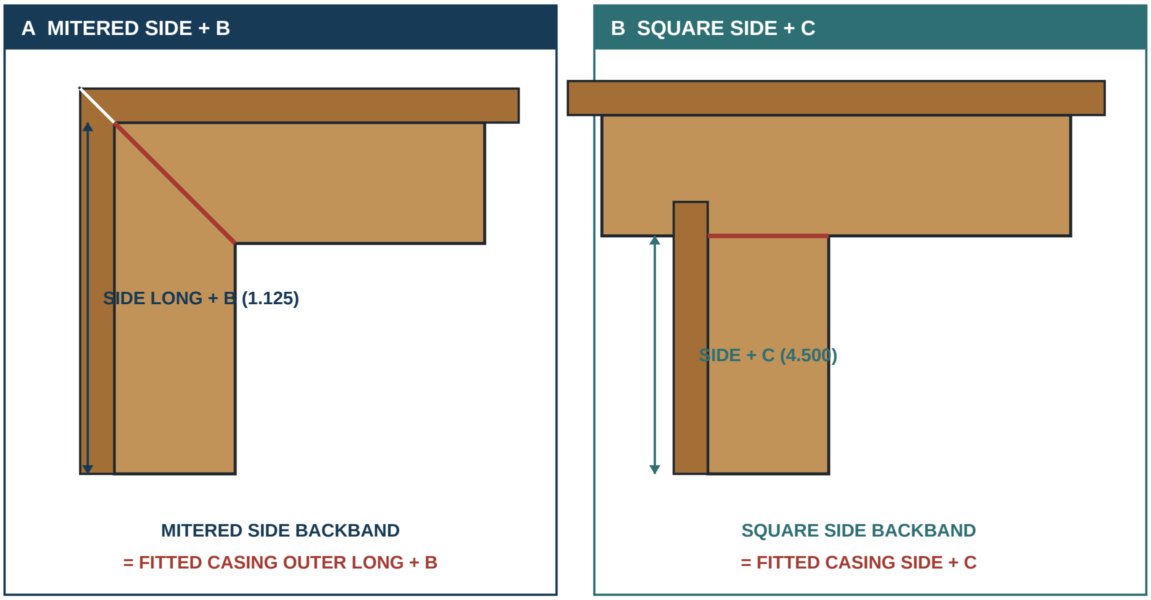
Scheme-specific side and head formulas

Mitered sides add B; square sides add C; the head adds 2B—then each rough blank receives allowance.

11A Backband corner formulas by head scheme

A mitered side gains one band width B; a square-butt side gains one casing width C.

CORNER LOGIC



DC-1916-001 | Rev. G | dimension-controlled isometric | written dimensions control

Backband member	Finished target	Production rough length
M-202A/B, mitered	Fitted side-casing outside long + B	Target + 2 inches
M-202A/B, square	Fitted square side-casing length + C	Target + 2 inches
M-202C head	Fitted head-casing outside width + 2B	Target + 2 inches

DO NOT COPY	FINAL TRANSFER	DATUMS
Mitered and square side-band lengths are not interchangeable.	Fit the casing first; transfer every backband joint from installed work.	Use B = 1-1/8 and C = 4-1/2 only after profile samples pass.

1. Define **B = 1-1/8 inches** backband width and retain **C = 4-1/2 inches** casing width. Measure only after the complete M-201 surround is fitted and accepted.
2. For a **mitered** face, calculate each side-backband rough length as **actual fitted side-casing outer long point + B + 2 inches**.
3. For a **square-butt** face, calculate each side-backband rough length as **actual fitted side-casing length + C + 2 inches**.
4. For either scheme, keep M-202C under its dedicated rule: **rough length = actual fitted head-casing outside width + 2B + 2 inches**. Its finished outside target is fitted head outside + 2B.
5. Record A/B/C identities on the back: M-202A hinge side, M-202B lock side, M-202C head. Mark room face and selected corner language.
6. Dry-position the rough members over the fitted casing and transfer actual intersections, wall scribes, and return lines. Do not final-cut from arithmetic alone.

STOP: M-202C is never derived from nominal jamb or door width, and side formulas are not interchangeable between schemes.

REMAKE: A backband shortened by using the former generic “side casing + 2” rule is replaced rather than hidden at a return.

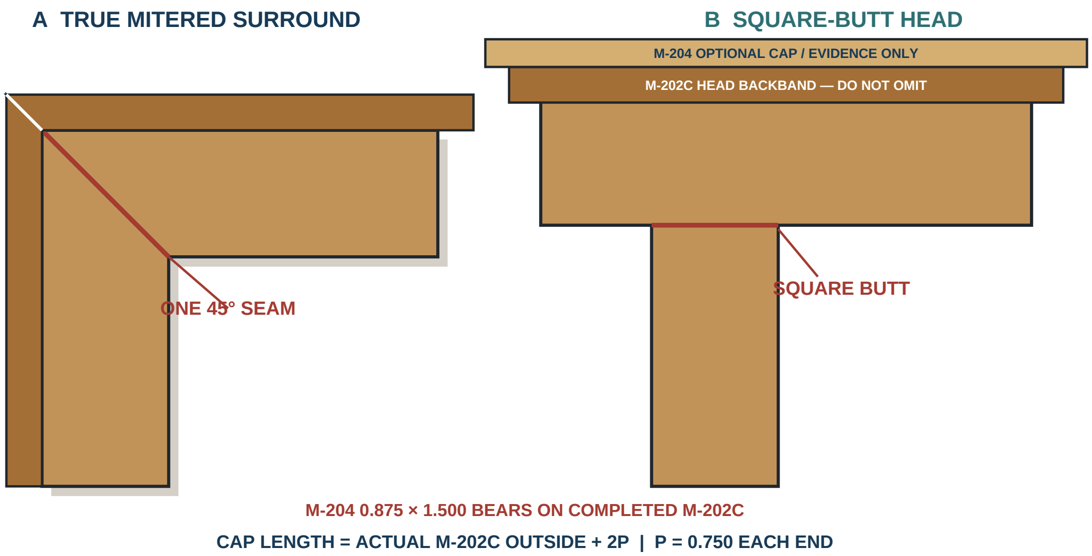
Fit a true mitered surround

Transfer both legs, tune miters on a shooting board, and dry-fit the complete casing and backband corners.

11 Head casing joint options

Match one adjacent historic joint language: mitered surround or square-butt head.

FIELD CHOICE



Length controls

- Mitered head short = W + 2R; long = W + 2R + 2C.
- Square leg = H side + R; square head = W + 2R + 2C.

DC-1916-001 | Rev. G | dimension-controlled isometric | written dimensions control

1. Make a rigid 3/16-inch reveal gauge and mark the two jamb legs and head from the completed jamb edge. Verify the marks meet cleanly at both corners.
2. Fit each M-201 leg independently at its floor/plinth datum. Cut its 45-degree upper miter long, hold the short point on the reveal intersection, and scribe the wall edge without moving the reveal edge.
3. Hold both fitted legs temporarily with non-marring clamps or minimal removable pins. Transfer the actual head short points to an overlength head blank.
4. Cut the first head miter, fit it to its leg, then mark and shoot the second miter in place. Use a shooting board or supported fine-adjustment method to keep the show face crisp.
5. Dry-clamp the three-piece surround on a flat surface and again at the opening. Optional alignment splines/5 x 30 Dominos require a sectioned scrap test proving no face breakout.
6. Accept reveal variation <=1/32, miter gap <=.010, adjacent face flush <=.010, leg plumb <=1/16 overall, and head level <=1/16 overall.

STOP: Fasteners or glue may hold an accurate miter but may not pull an open one closed.

REMAKE: A short point off the reveal, crushed miter, unsafe thin edge, or joint that opens when unclamped is recut within allowance or replaced.

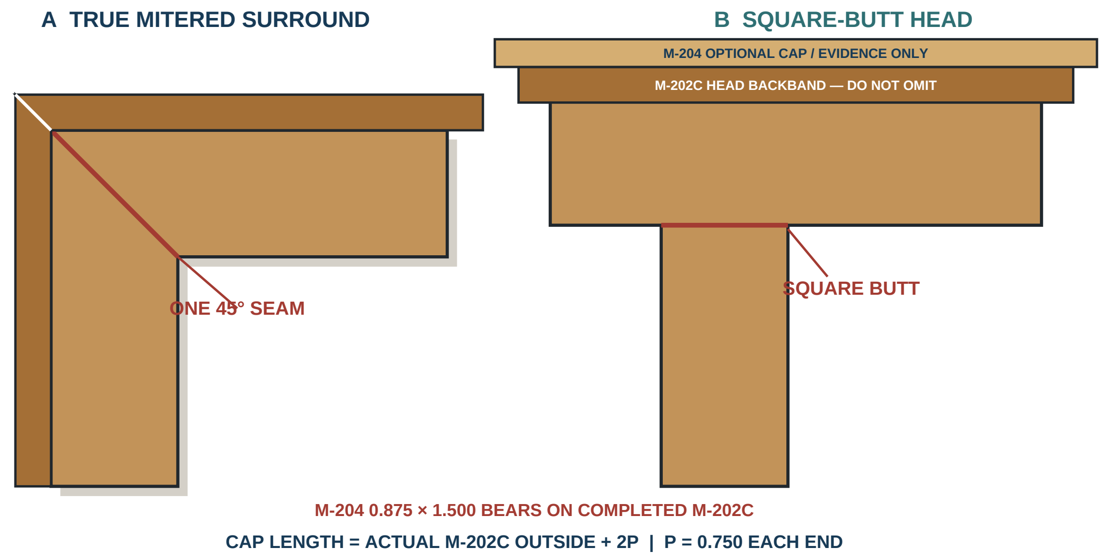
Fit a square-butt surround

Fit independent legs, span them with the head, and add a cap only when field evidence supports it.

11 Head casing joint options

Match one adjacent historic joint language: mitered surround or square-butt head.

FIELD CHOICE



Length controls

- Mitered head short = W + 2R; long = W + 2R + 2C.
- Square leg = H side + R; square head = W + 2R + 2C.

DC-1916-001 | Rev. G | dimension-controlled isometric | written dimensions control

1. Mark the same rigid 3/16-inch reveal on both legs and head. Confirm the face's historic evidence specifically releases a square-butt head.
2. Fit the left and right M-201 legs to **H-L + R** and **H-R + R**, adjusted only by physical transfer at the selected bottom datum. Cut upper ends square and coplanar.
3. Hold the fitted legs on their reveal lines and transfer the actual outside-to-outside width onto the overlength head casing.
4. Fit the head across both leg ends, maintaining the head's inside edge on the reveal and equal intended overhang. Tune one end at a time; preserve square bearing.
5. Where field evidence calls for a cap, preserve the square bearing condition and dry-fit the complete M-201 surround, then M-202C, then M-204. The cap bears on the completed M-202C head backband; M-202C is never omitted.
6. Accept butt gaps <=.010, adjacent faces flush <=.010, reveal variation <=1/32, legs plumb <=1/16 overall, and head level <=1/16 overall.

STOP: Do not miter the backband or add a cap until the face-specific square-head detail is confirmed.

REMAKE: A head cut short, a leg top out of plane, or a butt joint held closed only by nail pressure requires recut/replacement.

Fit the operating-door stop set

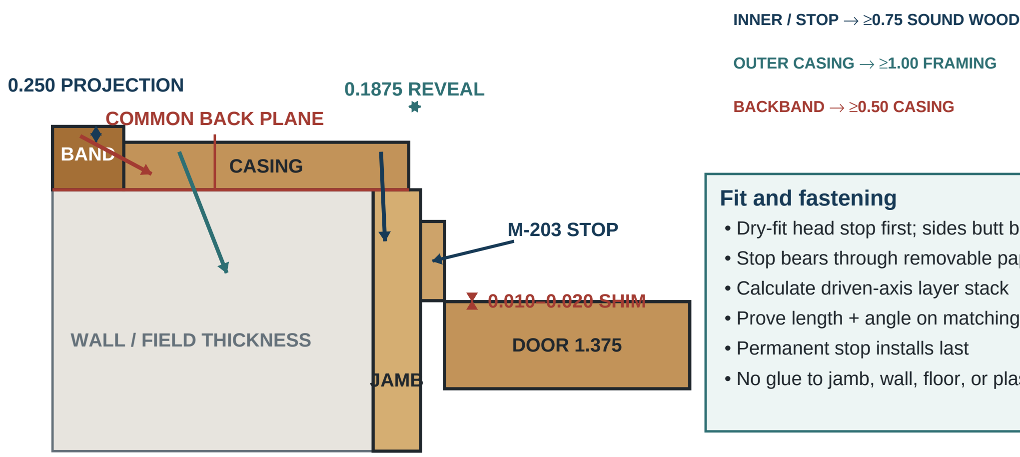
Fit head first, sides beneath, tack through removable shims, cycle, then complete fasteners.

12 Door stop, casing, and backband section

Prove each layer stack and sound-wood embedment on a matching offcut before field fastening.

SECTION

TRUE PLAN SECTION / ROOM FACE AT TOP



Fit and fastening

- Dry-fit head stop first; sides butt below
- Stop bears through removable paper shims
- Calculate driven-axis layer stack
- Prove length + angle on matching offcut
- Permanent stop installs last
- No glue to jamb, wall, floor, or plaster

SUBSTRATE HOLD

Hold for unknown layers, framing, utilities, or hazards.

No field drive until offcut embedment proof passes.

DC-1916-001 | Rev. G | dimension-controlled isometric | written dimensions control

1. Rehang the operating-door-released slab on steel screws. Fit the head M-203 between jamb legs with its visible edge parallel to the naturally latched door face.
2. Place removable 0.010–0.020 paper/card gauges between slab and stop at hinge, center, and latch reference points; do not force the door against the gauges.
3. Fit each side stop from the released floor/threshold condition to the underside of the head stop. Scribe localized jamb irregularity while maintaining full bearing.
4. Tack with the shortest suitable 18-gauge brads at three accessible locations per member. Cycle and latch the door 20 times; listen and feel for preload, rub, spring-back, and rattle.
5. Mark final fastener locations at 10–12 inches on center, offset from hinge/strike, casing nails, jamb splits, and utilities. Remove/label the stops for final P120/finish when the chosen sequence requires shop finishing.
6. After full finish cure and only after any released plinth, casing, backband/returns, and cap are permanently installed, install M-203 last at the accepted witness lines and mapped final pattern. Remove gauges and repeat the complete operating test.

Accept: Quiet continuous contact without slab bow or latch preload.

STOP: A stop may not be moved to conceal a reveal, bevel, hinge, or strike fault.

REMAKE: A bowed, over-scribed, split, or nail-blown show member is replaced.

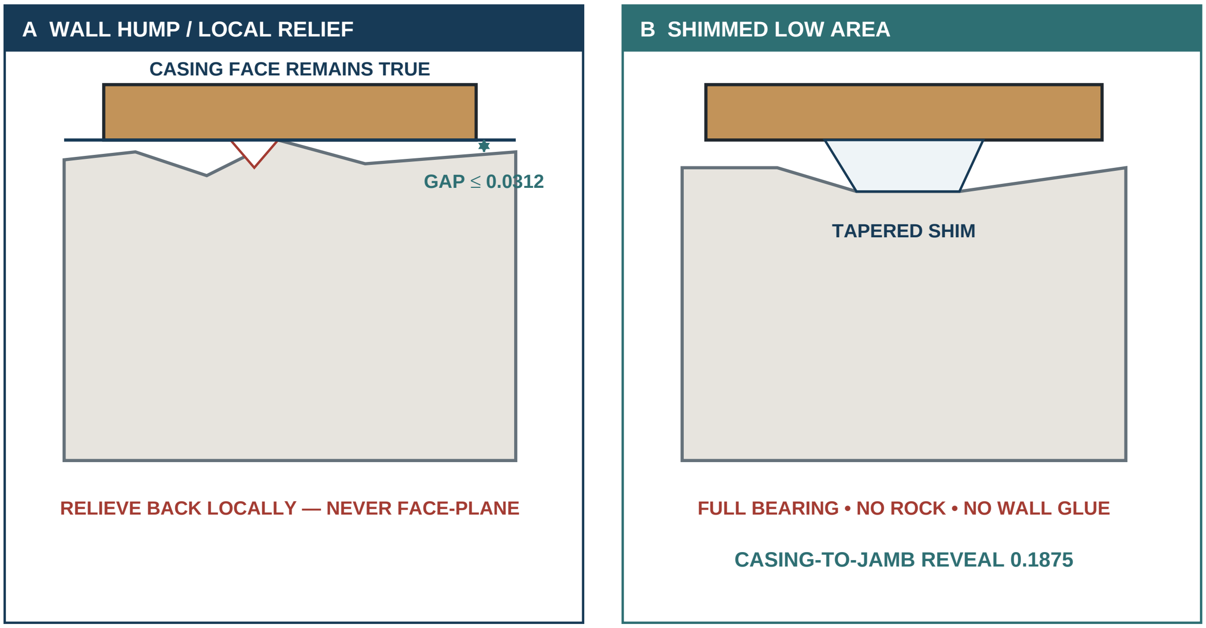
Fit, scribe, and dry-assemble each face

Establish the casing-to-jamb reveal, fit both bottoms independently, transfer the head, and bear flat.

12A Casing wall-scribe and local bearing correction

Preserve the jamb reveal while correcting only the back of the casing enough to bear on sound substrate.

SCRIBE DETAIL



HOLD FOR MOISTURE, LOOSE PLASTER, OR UNSOUND SUBSTRATE

DC-1916-001 | Rev. G | dimension-controlled isometric | written dimensions control

1. Fit the selected mitered or square M-201 surround on the 3/16 jamb reveal before addressing the wall edge. The jamb edge remains the controlling datum.
2. Measure jamb projection/recession behind each member. For a proud jamb, relieve only the casing back with a controlled router/hand-tool cut while preserving at least 1/2 inch of sound casing thickness.
3. For a recessed jamb, fit a continuous wood extension/shim with full bearing; do not bridge a void and bow the casing with nails.
4. Hold the accepted reveal and head joints in position, then scribe the wall-side edge with the smallest compass offset that closes the mapped humps. Plane to the scribe while retaining the visible profile.
5. Dry-pin at three points and check reveal variation <=1/32, wall scribe gap <=1/32, legs plumb <=1/16, head level <=1/16, and full back bearing without nail force.
6. Mark verified jamb/framing fastener zones and concealed utility exclusions on the back. Remove and label parts for finish without losing face/order.

STOP: Projection greater than 1/8 inch, loose plaster, moisture, abrupt humps, or less than 1/2 inch remaining after relief requires substrate correction/design review.

REMAKE: A casing over-scribed past the profile or bowed flat by pins is replaced.

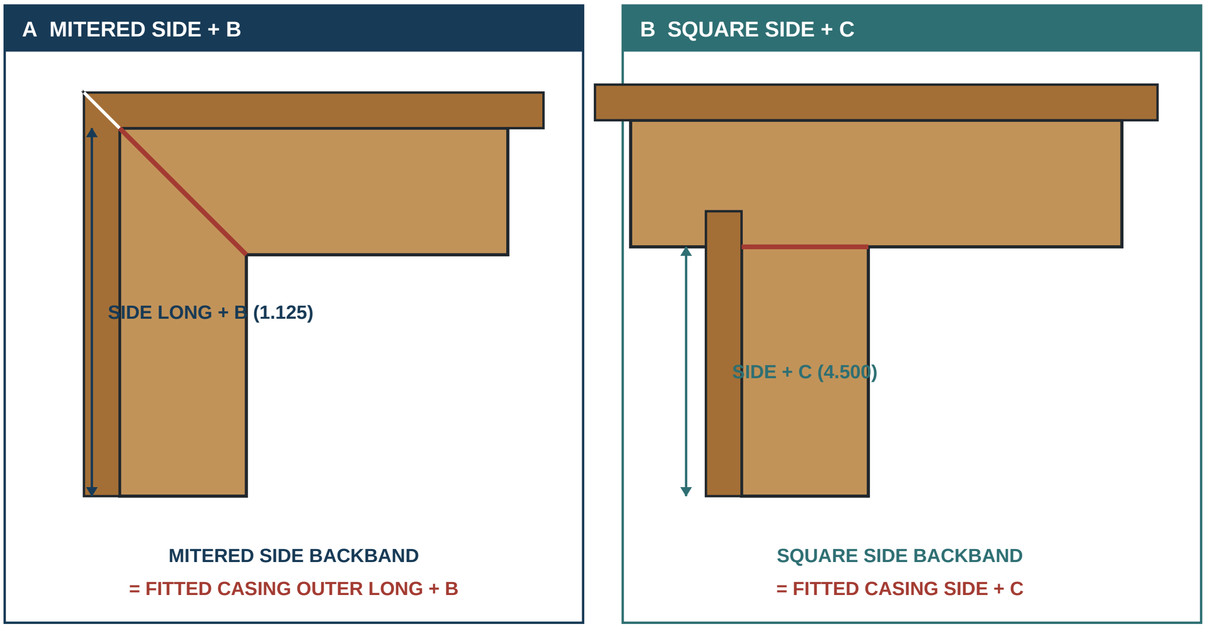
Backband, cap, plinth, and returns

Transfer from fitted casing, preserve the chosen joint language, and close every visible termination.

11A Backband corner formulas by head scheme

A mitered side gains one band width B; a square-butt side gains one casing width C.

CORNER LOGIC



DC-1916-001 | Rev. G | dimension-controlled isometric | written dimensions control

1. Fit M-202C to the actual accepted head-casing outside width, then fit M-202A/B from their scheme-specific rough lengths. Maintain a 1/4-inch face projection within +/-1/32.
2. Carry the selected mitered or square head language through every visible backband corner. Shoot joints until gap <=.010 and adjacent faces are flush <=.010 without fastener pressure.
3. Scribe the wall-side backband edge only after its casing projection is fixed. Do not reduce the visible 1-1/8-inch family unpredictably to chase a wall hump.
4. At an exposed termination, dry-fit a self-return with grain/show-face orientation matched and no exposed end grain at normal view. Label its parent member and orientation, mask both mating glue lands, remove it before finish, and retain it in the same labeled set; **do not glue the return until all parts have fully cured after finish**.
5. Dry-fit M-204 only where released. It must sit on the completed M-202C head backband and span **actual accepted M-202C outside length + 2P**, where P = 3/4 inch at each end. Preserve full bearing and the approved chamfer/profile sample; never omit M-202C beneath the cap.
6. Fit M-205 only to documented adjacent work. Its nominal 7/8-inch thickness remains provisional; field evidence controls final width, height, thickness, baseboard interface, and profile.
7. Dry-fit baseboard to the plinth or directly to casing as released; never notch the casing around an existing baseboard.

STOP: Unsupported cap/plinth design or a return without substrate remains on hold.

REMAKE: Wrong projection, visible end grain, short return, or scheme-mismatched corner requires a new piece.

Fastener paths and glue boundaries

Test depth and angle, verify substrate and utilities, and keep adhesive off jamb, wall, plaster, floor, and slab.

12 Door stop, casing, and backband section

Prove each layer stack and sound-wood embedment on a matching offcut before field fastening.

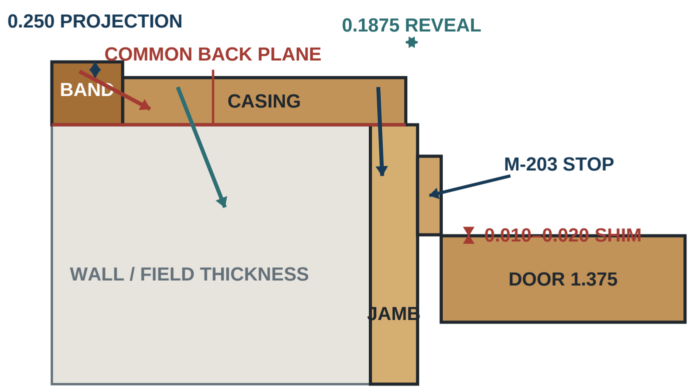
SECTION

TRUE PLAN SECTION / ROOM FACE AT TOP

INNER / STOP → ≥0.75 SOUND WOOD

OUTER CASING → ≥1.00 FRAMING

BACKBAND → ≥0.50 CASING



Fit and fastening

- Dry-fit head stop first; sides butt below
- Stop bears through removable paper shims
- Calculate driven-axis layer stack
- Prove length + angle on matching offcut
- Permanent stop installs last
- No glue to jamb, wall, floor, or plaster

SUBSTRATE HOLD

Hold for unknown layers, framing, utilities, or hazards.

No field drive until offcut embedment proof passes.

DC-1916-001 | Rev. G | dimension-controlled isometric | written dimensions control

1. Map substrate depth, jamb thickness, shims/voids, framing, utilities, hardware screws, and prior nail paths. For every fastener path calculate **sound embedment = fastener length along its driven axis - all trim, shim, and non-sound layers crossed**; select the shortest fastener that passes its numeric gate without entering a hazard.
2. Fasten M-201 inner edges to sound jamb wood with 18-gauge brads at 12–16 inches on center and **>=0.75 inch sound-wood embedment**. Fasten outer casing zones only to verified framing with 15-gauge finish nails about 16 inches on center and **>=1.00 inch framing embedment**.
3. Fasten M-202 into M-201 with angled 18-gauge brads at 8–12 inches on center, offset from casing fasteners, and prove **>=0.50 inch embedment into sound casing**. After its dry-cycle release, fasten M-203 as the **last permanent trim member**, at 10–12 inches on center clear of hardware, with **>=0.75 inch sound-jamb embedment**.
4. Use glue only at casing-to-casing joints, backband-to-casing edges, backband corner/return joints after full finish cure, and approved cap/plinth wood joints. Never glue trim to plaster, drywall, jamb, floor, or slab.
5. Prove each calculated length, drive angle, pressure, set depth, and resulting sound embedment on an offcut layer stack matching the actual trim/shim/substrate assembly before field fastening. Use sequential fire and keep every hand outside the projected path.
6. If a fastener deflects or blows out, stop; disconnect the tool, protect the point, and remove the member when possible. Extract from the back or cut flush only by a safe method, repair the substrate, and reassess the show face.

STOP: Unknown layer thickness, unverified framing, an unlocated utility/hazard, or an offcut proof below the required embedment holds fastening. Never drive a second fastener beside a hidden deflection.

REMAKE: Split trim, visible show-face blowout, weak embedment, or a hole that cannot disappear under the approved fill system requires member replacement.

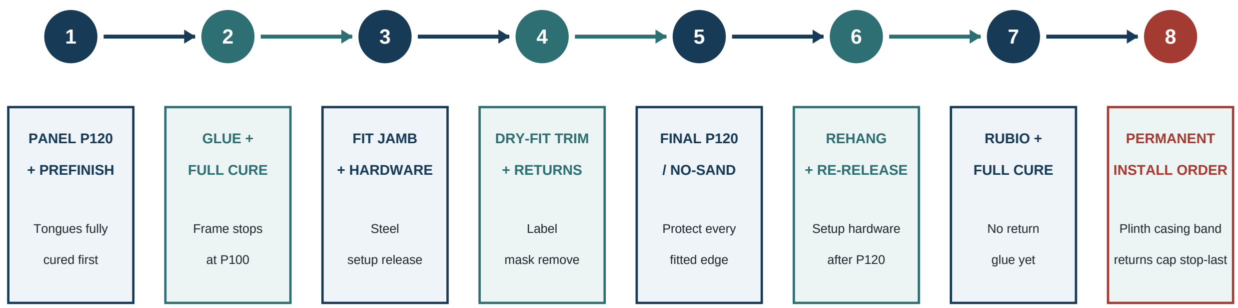
P120 preparation, Rubio application, and cure

After all fitting and dry machining, final-sand, clean, apply the accepted sample system, and control oily waste.

12B Integrated door and moulding finish sequence

Protect every fitted boundary, re-release operation after P120, then finish, fully cure, and install in controlled order.

FINISH



Rubio Monocoat Oil Plus 2C / Pure • CURRENT CONTAINER + TDS CONTROL

NO-SAND / MASK BOUNDARY

HINGE EDGE • LOCK BEVEL • HEAD / BOTTOM

GAINS • FACEPLATE / STRIKE SEATS • GLUE LANDS

REHANG + REPEAT OPERATING RELEASE BEFORE FINISH

FULL CURE / PERMANENT ORDER

A:B = 3:1 • THIN COAT • WIPE EXCESS • FULL CURE

PLINTH → CASING → BAND / RETURNS → CAP

DOOR STOP LAST • THEN FINAL OPERATING CYCLE

DC-1916-001 | Rev. G | dimension-controlled isometric | written dimensions control

1. Confirm Page 50 operating-door release and all stop/casing/backband/cap/plinth/return dry-fit records are signed. Remove steel setup hardware and label every fitted trim part.

2. Progress only the unfitted frame faces from P100 to **P120** with rigid blocks and grain direction. Establish a **NO-SAND boundary** on the fitted hinge edge, lock bevel, head and bottom edges, hinge gains, lock faceplate seat, strike seat, and all other fitted bearing/register surfaces; do not alter them. Protect prefinished floating panels and movement gaps.

3. Prepare moulding to P120 while preserving scribe lines, bearing faces, miters, reveals, return fits, masked return glue lands, and the 1/4-inch backband projection. Ease only approved profile edges.

4. Reinstall the labeled setup hardware after P120, rehang the slab, and repeat every Page 50 operating-door release check and cycle. Finish remains on hold until the repeated release passes; then remove and bag setup hardware again.

5. Approve Rubio Monocoat Oil Plus 2C **Pure** on labeled actual-stock offcuts under normal/raking light. Mask hinge gains, lock/strike seats, screw threads, return and other later glue lands, and critical bearing surfaces.

6. Clean only by the current Rubio-approved process—never mineral spirits. Mix Part A:Part B **3:1** by volume or weight, spread one thin coat, observe the stated reaction period, and wipe all unbonded excess before its stated limit. Record mix time/area, then cure parts flat and separated for the current container/TDS full-cure period; manage oily cloths under Page 09.

7. After full cure, inspect uniform color/sheen, complete wipe-off, clean movement gaps, unchanged hardware/return fit, and all five panel movements. No return or permanent hardware is installed before this gate passes.

STOP: Do not install final brass hardware or stack finished parts before the stated cure permits it.

REMAKE: Pools, tacky residue, contaminated mortises, finish-bridged panels, or sanding through the approved color require a sample-approved correction or replacement.

Brass hardware and complete system

After full cure, install brass hardware by hand, hang, fit trim layers, touch up, and cycle again.

Step	Installed condition	Release before next step
1	Fitted and hardware-machined slab fully cured	Finish, edges, and masked areas pass
2	Final brass hardware installed by hand	Leaf, lock, screws, and faceplate pass
3	Slab rehung and cycled	Full swing, signed clearances, latch, strike pass
4	Head stop fitted and lightly tacked	Paper-shim contact test passes
5	Side stops fitted beneath head	Quiet continuous contact passes
6	M-201 casing fitted and fastened	Reveal, plumb, level, joints pass
7	M-202 and approved optional details fitted	Projection, transitions, returns pass
8	Fill and sample-controlled touch-up complete	Color, sheen, fasteners pass
9	Opening cleaned and cycled again	Complete-system release signed

1. After full permitted cure, install assigned hinge leaves, lock, faceplate, spindle, knobs/levers, escutcheons, and strike. Reuse established threads and drive final brass screws by hand with a fitted driver.
2. Seat brass screw heads uniformly without cam-out or over-torque. A screw that resists is removed; the pilot/thread is corrected with the steel setup screw before brass is retried.
3. Hang the slab with a helper, protect all finished surfaces, and repeat full swing/latch operation before surrounding trim is completed.
4. Install any evidence-supported M-205 plinth first and complete its released baseboard transition. Then install M-201 casing on the 3/16 reveal, closing only accurate casing-to-casing joints with approved glue.
5. Install M-202 backband; glue each labeled, dry-fitted self-return to its masked mating land only now, after full finish cure. Install any released M-204 cap last within this trim group, seated on completed M-202C at **M-202C actual outside + 2P**.
6. Set trim fasteners just below the surface without crushing fibers and fill by the approved color-matched system. Install M-203 door stop **last** at its accepted witness line and final mapped pattern.
7. Remove protection, clean with finish-compatible materials, and perform the final fully layered swing/latch cycle 20 times from both sides after the stop is installed.

STOP: A brass screw is never driven with an impact driver and finished trim is never pulled flat by excessive nail pressure.

REMAKE: Cammed visible hardware, split trim, failed joint, or finish damage outside an approved invisible repair requires replacement.

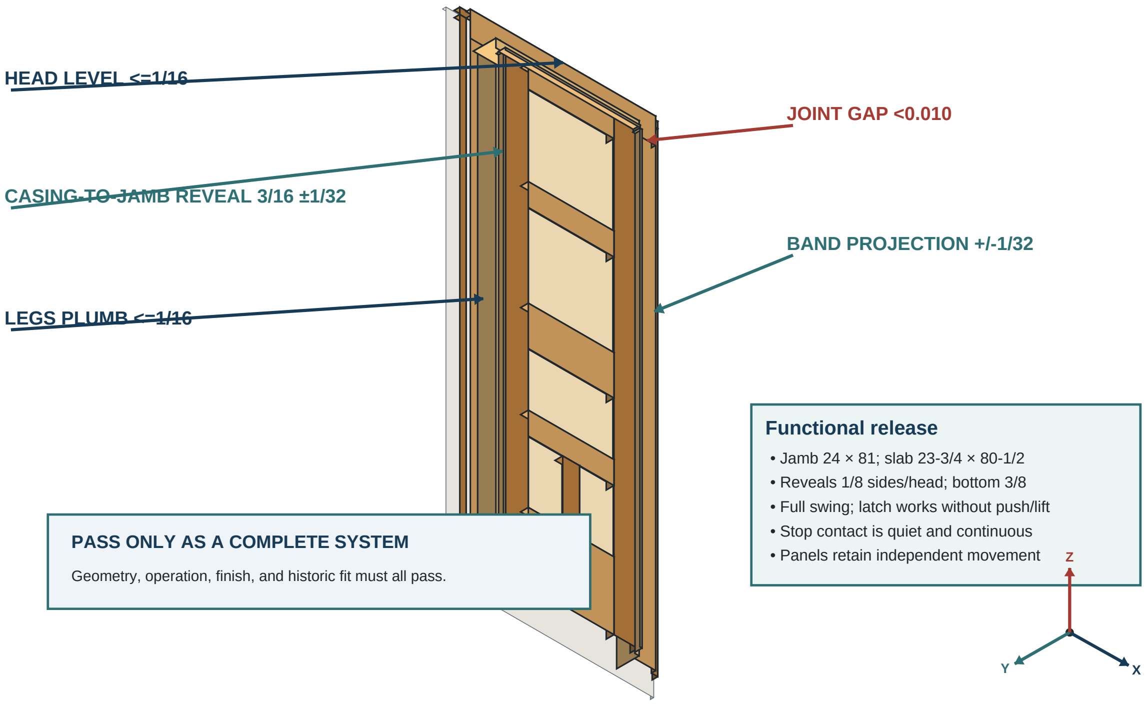
Troubleshooting and numeric acceptance

Diagnose geometry, joinery, panels, hardware, trim, fasteners, and finish against controlled thresholds.

13 Finished opening and quality release

Inspect appearance, geometry, hardware, stop contact, and seasonal panel freedom as one system.

FINAL QC



DC-1916-001 | Rev. G | dimension-controlled isometric | written dimensions control

1. **Shoulder will not close:** remove relieved tenons; check chips, mortise center/width, swollen tenon, groove interference, and part orientation. Correct the cause—never increase clamp force.
2. **Door rubs or will not stay:** map contact; check jamb twist, hinge-gain depth, barrel axis, bevel, floor, and finish buildup before planing.
3. **Latch needs lift/push:** check hinge sag and strike transfer before enlarging the strike. **Panel rattles/sticks:** verify the approved 3/4-inch-long foam spacer and its orientation, then repeat the movement/joint-stress gate or remove finish/glue bridging. Never add rigid packing, widen the groove, or reduce the calculated clearance.
4. **Trim joint opens:** correct cut, bearing, substrate, or scribe. Nails and filler do not convert an open joint to acceptable work.
5. Final numeric door QC: confirmed jamb **24 x 81**; prefit assembly **23-7/8 x 80-5/8**; fitted slab no larger than **23-3/4 x 80-1/2 x 1-3/8**; side/head reveals **1/8**; bottom gap **3/8**; diagonal difference **<=1/16**; twist **<=1/32**; shoulder gap **<=.005**; joint face flush **<=.010**.
6. Feature QC: frame section nominal 1/4 + 7/8 + 1/4; grooves .250–.255 wide and .500–.510 deep inside the core; DF cutter-to-face-glue-line core clearance >=.200; every relevant standard-rail tight/medium mortise-to-groove web >=.150; G groove-to-mortise web >=.100; G inter-mortise web >=.220; rail-receiver inter-mortise web >=.100; tongue 7/32 +/-0.005 thick x 7/16 projection +/-0.005.
7. Trim QC: casing reveal variation <=1/32; joint gap/flush <=.010; leg plumb/head level <=1/16 overall; backband projection variation and wall-scribe gap <=1/32.

STOP: One failed numeric or functional check holds final release.

REMAKE: Use the disposition on the originating page; do not create an undocumented “field tolerance.”

Field release, record, and maintenance

Pass the work only when measured geometry, operation, movement, historic fit, finish, documentation, and owner handoff agree.

Final release	Pass / value	Inspector
Door size and reveals	__________________________	_____________
Swing, latch, and strike	__________________________	_____________
Panel movement	__________________________	_____________
Stop contact	__________________________	_____________
Casing / backband geometry	__________________________	_____________
Finish sample match	__________________________	_____________
Build record complete	__________________________	_____________

RELEASE	RECORD	MAINTENANCE
All gates closed; measurements and test-piece results retained.	Installed slab, face formulas, finish sample, and hardware IDs entered.	Owner advised that floating panels must never be caulked or glued.

1. **Field release record:** enter the confirmed 24 x 81 jamb, measured 23-7/8 x 80-5/8 prefit assembly, final fitted width at top/middle/bottom, final height at both sides, both 1/8-inch side reveals, 1/8-inch head reveal, 3/8-inch bottom gap, diagonal difference, twist, shoulder/flush results, and five panel-movement checks.
2. Record LS-05/RL-02 stock allocations, no-resaw stile-layer results, staged face-planing/rest records, short-member map, same-layup coupon/wood-failure results, finished layer thicknesses, D/F seam positions, groove/tongue samples, every DF-to-glue-line and mortise-to-groove web, all J-01–J-12 results, Stage A/B adhesive lots/final-gate times/cure, actual spacer data, and every accepted repair or approved change.
3. Record hardware manufacturer/model, hinge and screw sizes, lock envelope clearance, dry-hang results, final latch/strike operation, and the final 20-cycle test.
4. For each room face record W, H-L, H-R, bottom datum, head scheme, casing and backband formulas, M-202C actual, stop contact, cap/plinth decision, fastener map, and baseboard return.
5. Record Rubio product/lot, sample ID, preparation grit, application/cure dates and conditions, color/sheen inspection, photographs, builder, checker, and owner acceptance.
6. Give the owner a maintenance sheet: keep panel edges uncaulked/unbonded; control leaks and abrupt humidity; use only finish-compatible cleaners; inspect hinge/lock screws, latch/strike, stop contact, panel movement, and finish seasonally.
7. Release only when the door swings freely, remains safe, latches positively without lift/push, contacts the stop quietly, retains panel movement, and all trim/finish/document controls pass.

Final signature: Builder, checker, owner/representative, date, opening ID, and Rev. G / LS-05 / RL-02.

STOP: An incomplete record is an incomplete build.

REMAKE: Any later material change is logged, re-inspected at its originating gate, and added to the maintenance history; a failed replacement follows the original page disposition.